The CSP Dichotomy Theorem

Abstract

The Constraint Satisfaction Problem over a finite constraint language is solvable in polynomial time if the language admits a weak near-unanimity polymorphism, and is NP-complete otherwise. This document is an exposition of the proof, following Zhuk's simplified argument [14] and supplying the prerequisites it imports.

It states what is being proved and separates the parts that mention a model of computation from the parts that do not (§1); fixes the vocabulary (§§2–5); states the theory of strong subuniverses and proves it (§§6–7); states the main induction and the three theorems the algorithm consumes (§8); and states the algorithm and the hard half (§§9–10).

§11 says, for every statement here, whether it is proved, imported with an exact citation, outlined, stated only, or open, and lists the six items that block the rest. It should be read before any part of this document is relied on: the theory of §7 is largely complete, and §8 is not.

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Introduction

The dichotomy conjecture of Feder and Vardi [7] was settled independently by Bulatov [6] and Zhuk [13] in 2017. Both proofs are long. Zhuk's, in the form published in 2020, is seventy-eight journal pages, and its central induction alternates between global and local information in a way its author later described as forcing "a very complicated induction to prove most of the claims".

In 2024 Zhuk published a simplified proof [14], and it is the argument followed here. The simplification is a redesign of the theory of strong subalgebras so that *every reduction is either strong or global*: for each domain D_x one can build a chain $D_x \supsetneq D_x^{(1)} \supsetneq \cdots \supsetneq \{a\}$ in which every step is either a strong subset or the intersection with a block of an equivalence relation with very strong properties. The types of the intermediate sets need not be tracked — only that such a chain exists, written $C \lhd\!\lhd\!\lhd B$. That is what makes the argument tractable, and it is what this exposition is organised around.

What is written out, and what is not. This is a corrected exposition and not a survey: where the argument as usually presented is wrong, incomplete, or ambiguous, the corrected statement is given here and proved, without marginal notes about what other presentations have. A companion audit document records those departures, with the source's text quoted and the reason for each change, so that the corrections can be checked rather than taken on trust.

It is also not complete. §11 is where the incompleteness is recorded: one statement is *open*, several are proved only in outline, and the spine of §8 — the lemma that produces bridges from an instance — is stated without proof. Read that table before relying on anything here.

Route and sources. [14] is the spine. Several prerequisites are imported rather than proved, chiefly the Absorption Theorem in Brady's form [2] and the strong-subalgebra calculus of [15]; §11.4 lists all of them with their exact hypotheses. [13] supplies the algorithm itself, which [14] does not contain. Section 4 of [14], on XY-symmetric operations, is an independent second result and is not needed for the dichotomy; it is omitted.

Caveats. Marked environments throughout record places where the informal argument has latitude that a careful one does not: a suppressed side condition, a quantifier whose scope is ambiguous, an object asserted to exist without an argument, or a step whose well-foundedness is left to the reader. They are part of the mathematics rather than commentary on it — several of the corrections below were found by writing one down. A sample of what they turn up: that $\mathbf{Z}_p$ lies in $\mathcal{V}_n$ only when $p \mid n-1$ (Caveat 3.3); that σ^* need not be a congruence (Caveat 3.21); that images do not commute with intersection, and that one proof needs them to (Caveat 3.14); that "we cannot weaken forever" needs a measure that is not the obvious one, and that the obvious one points the wrong way (Caveat 5.11); and that the main induction quantifies over the instance *inside* (Caveat 8.2).

1 What is being proved

1.1 The theorem

Fix a finite set A and a finite set Γ of relations on A, the *constraint language*. An instance of $\mathrm{CSP}(\Gamma)$ is a conjunction $R_1(\bar{x}_1) \wedge \cdots \wedge R_s(\bar{x}_s)$ with each $R_i \in \Gamma$; the question is whether it is satisfiable.

Theorem 1.1 (Dichotomy; Bulatov, Zhuk)**.** *If some weak near-unanimity operation preserves* Γ*, then* $\mathrm{CSP}(\Gamma)$ *is solvable in polynomial time. Otherwise* $\mathrm{CSP}(\Gamma)$ *is* NP*-complete.*

Caveat 1.2 (The theorem needs a core, and a Γ closed under constants)**.** Theorem 1.1 is stated for an arbitrary Γ, and in that form the weak near-unanimity criterion is not the right invariant: it is applied after passing to a core of Γ and adjoining every singleton unary relation, which is what makes the polymorphism algebra idempotent. The reduction — finite templates have cores; adjoining all singleton unary relations to a core gives a logspace-equivalent rigid core; algebraically this is passage to the idempotent reduct — is [2], "Cores and Idempotent Reducts". §10 uses it and this document does not prove it.

Everything after §2 therefore assumes that Γ is a rigid core containing all singleton unary relations, so that Convention 2.1 applies to $\mathrm{Pol}(\Gamma)$. Restating Theorem 1.1 under that hypothesis — and connecting the special weak near-unanimity operation of Lemma 6.21, which is a *term* operation of the polymorphism algebra and therefore still preserves every relation of Γ — is work this document has not yet done; see the status table, §11.

1.2 The five layers

Both halves of Theorem 1.1 are conditional on a model of computation, and that is not a formality to wave through. "Solvable in polynomial time" quantifies over an encoding of instances as strings, a machine, and a polynomial bound on its step count; "NP-complete" additionally quantifies over all problems in NP and over polynomial-time many-one reductions. The theorem therefore splits into statements that mention a machine and statements that do not, and the split does not agree with the tractable/hard split.

Write $\mathbf{A} = (A; w)$ for a finite idempotent algebra whose basic operation w is a special weak near-unanimity operation, and $\mathcal{V}_n$ for the class of such algebras with w of arity n (Convention 2.1).

Tractable half. Three layers.

- **(T0)** ***The algebraic core.*** The three theorems of §8 — the existence of a strong subuniverse or a linear congruence, the safety of a reduction to a strong subuniverse, and the codimension-one theorem — together with everything they rest on. These are statements about finite algebras, congruences, and CSP instances regarded as finite combinatorial objects. No computation appears in them.
- **(T1)** ***Correctness of the algorithm.*** Zhuk's algorithm written as a total function $\textsc{Solve}$ from instances to Booleans, and the proposition

$$\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

  This is a statement about a recursive function and is provable from (T0) with no cost model.
- **(T2)** ***The complexity wrapper.*** That $\textsc{Solve}$ runs in polynomial time in the size of the instance, for fixed Γ. This is the one place a cost model is needed on this side. It is stated here and not proved.

Hard half. Two layers.

(H0) ***The algebraic core.*** If no weak near-unanimity operation preserves Γ then Γ primitively positively interprets every finite constraint language — in particular one whose CSP is NP-hard. This is again pure algebra: the Galois connection between polymorphisms and primitive positive definability, the Maróti–McKenzie theorem that a Taylor term yields a weak near-unanimity term, and the Bulatov–Jeavons–Krokhin reduction.

(H1) ***The complexity wrapper.*** That a primitive positive interpretation yields a polynomial-time many-one reduction, and hence NP-hardness. This does need a program, but the program performs a syntactic substitution of formulas whose size is linear in the instance, so it is genuinely small.

(T0), (T1) and (H0) are the mathematics, and are what the literature actually proves. This document is written for them. (T2) and (H1) are stated precisely in §§9–10 and are not proved here; recovering the word *complete* in Theorem 1.1 also needs the standard fact that a fixed finite-template CSP lies in NP, which belongs with them.

Remark 1.3. The separation is not a dodge. Layer (T1) is strictly stronger than " CSP(Γ) is decidable", which is trivial for a finite template: it says that one particular, highly non-obvious algorithm — the one whose existence is the content of the dichotomy — is correct. Everything that makes the dichotomy hard lives in (T0). Conversely, no part of (T0)–(T1) becomes easier if a cost model is available.

2 Standing conventions

The standing hypotheses of this theory are usually carried in prose. Carried that way, a reader cannot audit which results actually consume which hypothesis, and several statements below turn on exactly that. We therefore fix the hypotheses as a labelled convention and record, for each one, where it is consumed.

Convention 2.1 (Standing hypotheses)**.** Throughout, $n \geq 3$ is a fixed integer and:

(a) every algebra is *finite* and has a *nonempty* universe;
(b) every algebra has exactly one basic operation, of arity n, written w;
(c) w is *idempotent*: $w(x, \dots, x) = x$;
(d) w is a *weak near-unanimity* operation: $w(y, x, \dots, x) = w(x, y, x, \dots, x) = \cdots = w(x, \dots, x, y)$ for all x, y;
(e) w is *special*: $w(x, \dots, x, y) = w(x, \dots, x, w(x, \dots, x, y))$ for all x, y.

The class of algebras satisfying a–e is written $\mathcal{V}_n$. Since only finite algebras are considered, $\mathcal{V}_n$ is not a variety.

Remark 2.2 (Why one basic operation)**.** Restricting to a single basic operation is not a loss. The dichotomy hypothesis is that *some* weak near-unanimity operation preserves Γ; one may then discard the original signature and work in the algebra $(A; w)$ whose only basic operation is that one, because every relation of Γ is a subuniverse of a power of $(A; w)$ and that is all the argument uses. Passing from a WNU w to a *special* one is Lemma 6.21, and costs an increase of arity from n to $n^{n!}$.

The gain is that a term is then a finite tree with n-ary branching over a single symbol, so "term operation" needs no signature bookkeeping, and $\mathrm{Clo}(\mathbf{A})$ is the clone generated by one operation.

Caveat 2.3 (Where each hypothesis is consumed)**.** a is used in every induction on $|A|$ and in every minimality argument (a minimal element of a nonempty family of subsets exists); it cannot be dropped anywhere. c is what makes singletons subuniverses and makes Sg of a nonempty set nonempty. d is consumed only through the results it implies — the existence of absorption-theoretic structure — and never directly in §§5–8. e is used where a unary polynomial must be idempotent under iteration, chiefly in the theory of Sg-closures of two-element sets, and it is what forces $p \mid n - 1$ in Proposition 2.7.

Convention 2.4 (The empty set)**.** A *subuniverse* of $\mathbf{A}$ is a subset closed under w; the empty set is a subuniverse under this definition, and we allow it. A *subalgebra* is a nonempty subuniverse; $\mathbf{B} \leq \mathbf{A}$ always carries $B \neq \varnothing$.

The relations $<_T$, $\leq_T$ and $\lhd\lhd\lhd$ of §4 are defined only between *nonempty* sets; in particular $\varnothing \lhd\lhd\lhd A$ never holds. Where a construction can produce the empty set — a projection of an intersection, most often — the dotted variants $\dot{<}_T$, $\dot{\leq}_T$, $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ are used, and they mean "the stated relation holds, or the left-hand side is empty".

Caveat 2.5 (The dot is not decoration)**.** Conflating $\lhd\lhd\lhd$ with $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ is the single most common way to misread §4. Every propagation statement of §6 that projects or intersects produces a dotted conclusion, and every statement that consumes a $\lhd\lhd\lhd$ hypothesis needs the undotted one. The two are different relations, and the case split is part of every use.

Convention 2.6 (Relations, and indices)**.** A relation is a subset of a product $A_1 \times \cdots \times A_m$ of universes, indexed by $[m] = \{1, \dots, m\}$; $\mathbf{R} \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ means additionally that R is a subuniverse of the product algebra. We write $\mathrm{pr}_{i_1,\dots,i_s}(R)$ for the projection onto the listed coordinates and $\mathbf{R} \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ when every $\mathrm{pr}_i(R) = A_i$.

Products are indexed by *arbitrary finite index sets*, not only by initial segments of $\mathbb{N}$. Several constructions in §8 form a product over the variable set of an instance, or over a set of subuniverses, and reindexing those to $[m]$ by hand buys nothing.

2.1 Ten readings fixed

Ten notations and definitions below admit more than one reading. The choice is invisible in prose and decisive in the proofs, so all ten are fixed here, once, and the fixed reading is used everywhere.

(i) **The algebra $\mathbf{Z}_p$.** The two roles of $\mathbf{Z}_p$ are kept apart. The abelian *group* $(\mathbb{Z}_p; +, -)$ supplies the relation $x_1 - x_2 = x_3 - x_4$ in Definition 4.10 and carries no arity constraint. The *algebra* $\mathbf{Z}_p = (\mathbb{Z}_p; x_1 + \cdots + x_n)$ lies in $\mathcal{V}_n$ only when $p \mid n-1$ (Corollary 2.8), and it is the algebra that occurs in Definition 3.34, where $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ is a subuniverse of a product and therefore needs a common signature. Every "for some prime p" below silently carries $p \mid n-1$.

(ii) **σ^* is a tolerance.** It is a reflexive symmetric subuniverse of $\mathbf{A}^2$ and *not* in general a congruence (Proposition 3.22). That σ^* is a congruence is item (b) of the definition of a linear congruence and conclusion (a) of Lemma 7.39; typing it as a congruence would make the first vacuous and collapse the linear/PC distinction.

(iii) **The empty set.** As in Convention 2.4: subuniverses may be empty, subalgebras may not, and the typed relations hold only between nonempty sets. Read syntactically, $\varnothing$ is vacuously absorbing and vacuously central, so two clauses need the word *nonempty* written into them: the clause defining $<_{\mathrm{s}}$ requires its witness D nonempty (Definition 4.6), and Lemma 7.19 carries the explicit alternative "$C = \varnothing$" — without which it is false, as $\mathbf{A} = \mathbf{B} = \mathbf{Z}_p$ with $R = \{(x, x+1)\}$ shows. For the same reason Corollary 7.2 concludes a *proper nonempty* subuniverse: without those words it is vacuous, since every algebra is a binary absorbing subuniverse of itself.

(iv) **The empty family is allowed.** In Definition 3.20 the family $S_1, \ldots, S_k$ may be empty, the empty intersection being A^2. This is not a free choice: the two formulations given there agree *only* under it, and under it an irreducible congruence is automatically proper (Proposition 3.22(a)) and σ^* exists.

(v) **$\lhd\lhd\lhd$ carries its chain.** $C \lhd\lhd\lhd^A B$ is *data*: a list of intermediate sets with their typed steps and witnessing congruences, not merely the proposition that such a list exists. Statements below speak of "the dividing congruences coming from $C \lhd\lhd\lhd^A B$", which is a function of the chain and not of the pair. See Caveat 4.25 for the two places where the choice of chain is load-bearing.

(vi) **"Dimension".** $\prod_i \mathbb{Z}_{q_i}$ with distinct primes is not a vector space over anything and has no dimension. We use the composition length of the finite abelian group, together with its additivity along short exact sequences. Nothing in §§4–6 uses the notion; it appears only in §8.3, where the correct statement is that such a group splits as a product over primes of $\mathbb{F}_q$-vector spaces.

(vii) **"Linked instance".** We use the per-variable condition of Definition 5.8, *not* global connectivity of the value graph. The two are inequivalent — a fragmented instance is per-variable linked in each component — so non-fragmentation is carried as a separate hypothesis wherever it is needed (Caveat 3.12).

(viii) **Quotients of subsets.** B/δ always means the *image* $\{b/\delta : b \in B\}$ in $\mathbf{A}/\delta$, for B any subset of A; never the quotient of B by a congruence of $\mathbf{B}$. Consequently $(C_1 \cap \cdots \cap C_t)/\delta \subseteq \bigcap_i C_i/\delta$ always, with equality false in general (Caveat 3.14). The one place the reverse inclusion is needed is Lemma 6.4(fm), which establishes it from an explicit saturation hypothesis.

(ix) **Four small results, stated rather than assumed.** Each is used repeatedly below and each is easy, which is exactly why it is easy to leave unstated: Lemma 4.11 (the relation S of Definition 4.10(c) is a bridge with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$), Lemma 3.24 ($\sigma$ is irreducible, linear or PC exactly when $0_{\mathbf{A}/\sigma}$ is), Lemma 4.8 (the two formulations of S-freeness agree), and Lemma 3.16 (the parallelogram property implies rectangularity, and the rectangular closure exists).

(x) **The three alternatives of the central-relation lemma.** Lemma 7.19 is stated with three alternatives, the third being a proper nonempty binary absorbing subuniverse. A fourth — a nontrivial *projective* subuniverse — appears in the form of this result that holds without Convention 2.1, and is eliminated here by Lemma 7.18. Projectivity is used nowhere else in this document.

Proposition 2.7 (Where $p \mid n-1$ comes from)**.** *Let $\mathbf{U}$ be a finite algebra with a single basic operation w of arity n that is idempotent, weak near-unanimity and special, and suppose $\mathbf{U}$ is affine over $\mathbb{Z}_p$ with $|U| > 1$. Then $p \mid n-1$, and w acts on $\mathbf{U}$ as $x_1 + \cdots + x_n$.*

Proof. Being affine, $\mathbf{U}$ is polynomially equivalent to a $\mathbb{Z}_p$-module, so w acts as $w(\bar{x}) = \sum_i c_i x_i + d$ for scalars $c_i \in \mathbb{Z}_p$ and a constant d. Idempotence gives $(\sum_i c_i)x + d = x$ for all x, hence $\sum_i c_i = 1$ and $d = 0$.

The weak near-unanimity identities $w(y, x, \ldots, x) = w(x, y, x, \ldots, x) = \cdots = w(x, \ldots, x, y)$ read $c_1 y + (1 - c_1)x = c_2 y + (1 - c_2)x = \ldots$, and comparing coefficients of y gives $c_1 = \cdots = c_n =: c$, so $nc = 1$.

Specialness, $w(x, \ldots, x, y) = w(x, \ldots, x, w(x, \ldots, x, y))$, reads

$$(n-1)c\,x + cy \;=\; (n-1)c\,x + c((n-1)c\,x + cy),$$

and comparing coefficients of y gives $c = c^2$, so $c \in \{0, 1\}$. Now $c = 0$ contradicts $nc = 1$, so $c = 1$; then $n \cdot 1 = 1$ in $\mathbb{Z}_p$, i.e. $p \mid n-1$, and w acts as $x_1 + \cdots + x_n$. □

Corollary 2.8. *$\mathbf{Z}_p = (\mathbb{Z}_p; x_1 + \cdots + x_n)$ lies in $\mathcal{V}_n$ if and only if $p \mid n-1$; and whenever a prime p arises from a linear congruence on some $\mathbf{A} \in \mathcal{V}_n$ — so that some subquotient of $\mathbf{A}$ is affine over $\mathbb{Z}_p$ and nontrivial — one has $p \mid n-1$. So the standing identification of $\mathbf{Z}_p$ as an object of $\mathcal{V}_n$ is legitimate wherever this document makes it.*

Proof. For the first claim, $\mathbf{Z}_p$ is affine over $\mathbb{Z}_p$ with $|\mathbb{Z}_p| > 1$, so Proposition 2.7 gives $p \mid n-1$ if $\mathbf{Z}_p \in \mathcal{V}_n$; conversely $p \mid n-1$ makes $x_1 + \cdots + x_n$ idempotent, and it is then a WNU and special, both identities reducing to $(n-1)x + y = y$. For the second, apply Proposition 2.7 to the nontrivial affine subquotient. □

Caveat 2.9 (The proposition is not decoration)**.** Proposition 2.7 is needed twice, and in both places its absence would leave a statement that does not typecheck. It is what makes $\mathbf{Z}_p$ an object of $\mathcal{V}_n$ at all, hence what makes Definition 3.34 well formed; and it is what upgrades the group isomorphism of Lemma 4.17 to an isomorphism of algebras.

3 Vocabulary

This section fixes the universal-algebraic vocabulary. Nothing in it is new, and a reader who knows the subject should skim to Definition 3.20 (irreducible congruences) and Definition 3.26 (bridges), which are the two places where the conventions genuinely matter.

3.1 Algebras, subuniverses, terms

Definition 3.1 (Algebra)**.** An algebra is a pair $\mathbf{A} = (A; w^{\mathbf{A}})$ with A a finite nonempty set and $w^{\mathbf{A}} \colon A^n \to A$, subject to Convention 2.1. We write A for the universe of $\mathbf{A}$ and drop the superscript on w when no confusion arises.

Definition 3.2 (The algebras $\mathbf{Z}_p$)**.** For a prime p with $p \mid n-1$, $\mathbf{Z}_p$ is the algebra on $\{0, 1, \dots, p-1\}$ with $w^{\mathbf{Z}_p}(x_1, \dots, x_n) = x_1 + \cdots + x_n \bmod p$.

Caveat 3.3 ($\mathbf{Z}_p$ is only sometimes in $\mathcal{V}_n$)**.** The operation $x_1 + \cdots + x_n \bmod p$ is idempotent exactly when $n \equiv 1 \pmod p$: idempotence says $nx \equiv x$ for all x, i.e. $p \mid n-1$. Given that, it is also a WNU (all the identities read $(n-1)x + y = y$) and special (both sides reduce to y).

So $\mathbf{Z}_p \in \mathcal{V}_n$ if and only if $p \mid n-1$, and every statement of the form "$\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for some prime p" (Lemma 4.17, Definition 3.34) silently produces a p dividing $n-1$. The side condition must stay visible, because "for some prime p" otherwise reads as an unconstrained existential; Corollary 2.8 is what discharges it.

Definition 3.4 (Subuniverse, subalgebra)**.** $B \subseteq A$ is a *subuniverse* of $\mathbf{A}$ if $w(b_1, \dots, b_n) \in B$ for all $b_1, \dots, b_n \in B$. If moreover $B \neq \varnothing$ then $\mathbf{B} = (B; w\restriction_{B^n})$ is a *subalgebra*, written $\mathbf{B} \leq \mathbf{A}$.

Definition 3.5 (Term operations)**.** $\mathrm{Clo}(\mathbf{A})$ is the least set of finitary operations on A containing all projections and closed under composition with w. Its m-ary part is written $\mathrm{Clo}_m(\mathbf{A})$. Elements of $\mathrm{Clo}(\mathbf{A})$ are the *term operations* of $\mathbf{A}$.

Caveat 3.6 (Terms versus term operations)**.** A term is a finite tree, and distinct terms may induce the same operation. Every statement below that quantifies over $\mathrm{Clo}(\mathbf{A})$ is a statement about *operations*. The syntactic layer nevertheless cannot be dispensed with, because Sg is characterised through terms (Lemma 3.8) and because the arity bookkeeping in the absorption arguments is syntactic.

Definition 3.7 (Generation)**.** For $X \subseteq A^m$, $\mathrm{Sg}_{\mathbf{A}}(X)$ is the least subuniverse of $\mathbf{A}^m$ containing X.

Lemma 3.8 (Generation by terms)**.** *Let $X = \{x_1, \dots, x_k\} \subseteq A^m$ be nonempty. Then*

$$\mathrm{Sg}_{\mathbf{A}}(X) = \{\, t^{\mathbf{A}^m}(x_1, \dots, x_k) \;:\; t \in \mathrm{Clo}_k(\mathbf{A}) \,\}.$$

The generator list is fixed*: the same list $x_1, \dots, x_k$ in the same order serves every element of the closure, with t varying.*

Caveat 3.9. The "fixed list" phrasing is what makes Lemma 3.8 usable. The weaker reading — for each y in the closure there are *some* generators and *some* term — is also true but useless, because every later application applies one term simultaneously to several closure elements and needs their generator lists to coincide.

3.2 Relations

Definition 3.10 (Relational vocabulary)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) R is *subdirect* if $\mathrm{pr}_i(R) = A_i$ for every i; $\mathbf{R} \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ abbreviates "R is a subdirect subuniverse".
(b) For $R \subseteq A^m$: R is *reflexive* if $(a, \ldots, a) \in R$ for every $a \in A$.
(c) For binary $\delta_1 \subseteq A_1 \times A_2$, $\delta_2 \subseteq A_2 \times A_3$: $\delta_1 \circ \delta_2 = \{(a, c) : \exists b \ (a, b) \in \delta_1 \wedge (b, c) \in \delta_2\}$, and $\delta^{-1} = \{(b, a) : (a, b) \in \delta\}$.
(d) For $B \subseteq A_1$ and $\delta \subseteq A_1 \times A_2$: $B \circ \delta = \{c : \exists b \in B \ (b, c) \in \delta\}$.
(e) A subdirect $\delta \subseteq A \times B$ is *linked* if the bipartite graph on $A \sqcup B$ with edge set δ is connected. It is *bijective* if $|\delta| = |A| = |B|$.
(f) For binary σ and $m \geq 2$, $\sigma^{[m]} = \{(a_1, \ldots, a_m) : (a_i, a_j) \in \sigma \ \forall i, j\}$.

Caveat 3.11 (Composition is diagrammatic)**.** $\delta_1 \circ \delta_2$ applies δ_1 *first*, the opposite of the convention for composition of functions. Every composite of bridges and of congruences below is written left to right, and silently reversing it produces statements that are true-looking and wrong — in Corollary 6.15(l) the composite $\sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$ is asymmetric in k and ℓ and the order is the content.

Caveat 3.12 ("Linked" is used for two different things)**.** The word is used both for a binary relation (Definition 3.10(e)) and for a CSP instance (Definition 5.8). They are related but not the same predicate, and both occur in one proof in §8, where the passage from one to the other is the crux; Caveat 8.15 marks the step.

Definition 3.13 (Quotients of relations)**.** For an equivalence σ on A and $a \in A$, a/σ is the class of a; for $B \subseteq A$, $B/\sigma = \{b/\sigma : b \in B\}$; for $R \subseteq A^m$, $R/\sigma = \{(b_1/\sigma, \ldots, b_m/\sigma) : \bar{b} \in R\}$.

Caveat 3.14 (Images do not commute with intersection)**.** B/σ and R/σ are *images*. Three consequences, all of them easy to pass over:

(a) B/σ is a subset of A/σ, not the quotient of B by $\sigma \cap B^2$. For B a subuniverse and σ a congruence the two are canonically isomorphic as algebras, and the two are used interchangeably below; B/σ always denotes the image, and the isomorphism is what licenses the interchange.
(b) In general $(C_1 \cap C_2)/\sigma \subsetneq C_1/\sigma \cap C_2/\sigma$. Lemma 6.4(fm) needs equality for a family $C_1, \ldots, C_t$, and gets it from the extra hypothesis $(C_i \circ \sigma) \cap B = C_i$, i.e. that each C_i is σ-saturated inside B. That is a content step, not notation.
(c) R/σ need not be stable under anything, and need not be a subuniverse of $(\mathbf{A}/\sigma)^m$ unless σ is a congruence.

3.3 Rectangularity

Definition 3.15 (Parallelogram property, rectangularity)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) The i-th variable of R is *rectangular* if for all $\bar{a}, \bar{b}$,

$$\bar{a}[i \mapsto b_i] \in R, \quad \bar{b}[i \mapsto a_i] \in R, \quad \bar{b} \in R \implies \bar{a} \in R,$$

where $\bar{a}[i \mapsto c]$ is $\bar{a}$ with its i-th entry replaced by c. R is *rectangular* if every variable is.
(b) R *has the parallelogram property* if for every permutation of the coordinates giving R' and every $\ell \in [m-1]$,

$$(a_1, \ldots, a_\ell, b_{\ell+1}, \ldots, b_m) \in R', \quad (b_1, \ldots, b_\ell, a_{\ell+1}, \ldots, a_m) \in R', \quad \bar{b} \in R' \implies \bar{a} \in R'.$$

(c) The *rectangular closure* of R is the least rectangular relation containing R.

Lemma 3.16 (Rectangularity basics)**.** (a) *A relation with the parallelogram property is rectangular.*

(b) *The rectangular closure exists.*

Proof. (a) Apply the parallelogram property to the permutation carrying coordinate i to position 1, with $\ell = 1$. (b) Rectangularity at coordinate i is a Horn implication, so the rectangular relations containing R are closed under arbitrary intersection and the family is nonempty (it contains $A_1 \times \cdots \times A_m$); the intersection of all of them is the least one. $\square$

Caveat 3.17 (Quantifying over permutations)**.** Definition 3.15(b) quantifies over $\mathfrak{S}_m$ and over the split point ℓ. Equivalently, and more usably, it quantifies over all ways of splitting $[m]$ into two nonempty blocks; the second formulation avoids transporting R along a permutation and is the one used below.

3.4 Congruences

Definition 3.18 (Stability)**.** Let $R \subseteq A_1 \times \cdots \times A_m$ and let σ be a binary relation on A_i. The i-th variable of R is *stable under* σ if $\bar{a} \in R$ and $(a_i, b) \in \sigma$ imply $\bar{a}[i \mapsto b] \in R$. R is *stable under* σ if every variable is (each with respect to σ on the corresponding factor, when the factors agree).

Definition 3.19 (Congruence)**.** A *congruence* on $\mathbf{A}$ is an equivalence relation on A that is a subuniverse of $\mathbf{A} \times \mathbf{A}$. It is *nontrivial* if it is neither the equality relation $0_{\mathbf{A}}$ nor A^2. The quotient $\mathbf{A}/\sigma$ is the algebra on A/σ with $w(\bar{a}/\sigma) = w(\bar{a})/\sigma$.

Definition 3.20 (Irreducible congruence and its cover)**.** A congruence σ on $\mathbf{A}$ is *irreducible* if it cannot be written as an intersection of binary subuniverses of $\mathbf{A} \times \mathbf{A}$ each stable under σ and each properly containing σ. Equivalently: there are no $\mathbf{S}_1, \ldots, \mathbf{S}_k \leq \mathbf{A}/\sigma \times \mathbf{A}/\sigma$ with $0_{\mathbf{A}/\sigma} = S_1 \cap \cdots \cap S_k$ and $S_i \neq 0_{\mathbf{A}/\sigma}$ for every i.

For irreducible σ, σ^* denotes the least $\delta \leq \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ.

Caveat 3.21 (Three traps in Definition 3.20)**.** (a) Irreducibility is *not* meet-irreducibility in the congruence lattice. The $\mathbf{S}_i$ range over binary *subuniverses* stable under σ, which need not be equivalence relations. The two notions differ, and every use below is the stated one.

(b) σ^* need not be a congruence. It is a subuniverse of $\mathbf{A}^2$ stable under σ, nothing more; that σ^* *is* a congruence is an extra hypothesis in the definition of a linear congruence (Definition 4.10), which would be redundant otherwise.

(c) Existence and uniqueness of σ^* need an argument, given in Proposition 3.22. Uniqueness is exactly what irreducibility buys; without it there is no "the" cover, and σ^* is meaningless for a congruence not known to be irreducible.

(d) The family $S_1, \ldots, S_k$ may be *empty* (item iv of §2.1). This is not a free choice: the two formulations above agree only under it.

Proposition 3.22 (The cover exists, and is a tolerance)**.** *Let σ be an irreducible congruence on $\mathbf{A}$. Then*

(a) *$\sigma \neq A^2$;*
(b) *the family $\mathcal{F}$ of $\delta \leq \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ is nonempty and closed under intersection, so it has a least element σ^*;*
(c) *σ^* is reflexive and symmetric.*

Proof. (a) The empty family is allowed, and its intersection is A^2; so if $\sigma = A^2$ then σ is an intersection of a family of relations each properly containing it, vacuously, and σ is reducible.

(b) $\mathcal{F} \ni A^2$ by (a). If $\delta_1, \delta_2 \in \mathcal{F}$ then $\delta_1 \cap \delta_2$ is again a subuniverse of $\mathbf{A}^2$ stable under σ and contains σ; it contains σ *properly*, since otherwise $\sigma = \delta_1 \cap \delta_2$ exhibits σ as an intersection of two members of $\mathcal{F}$, contradicting irreducibility. So $\mathcal{F}$ is closed under binary, hence finite, intersection, and $\sigma^* = \bigcap \mathcal{F} \in \mathcal{F}$ is its least element.

(c) $\sigma^* \supseteq \sigma \supseteq 0_{\mathbf{A}}$ is reflexive. For symmetry, $(\sigma^*)^{-1}$ is a subuniverse of $\mathbf{A}^2$, is stable under σ because σ is symmetric, and properly contains $\sigma^{-1} = \sigma$; so $(\sigma^*)^{-1} \in \mathcal{F}$ and therefore $\sigma^* \subseteq (\sigma^*)^{-1}$, whence equality. □

Remark 3.23 (Why not a congruence)**.** Nothing in Proposition 3.22 gives transitivity, and none is available: that σ^* is a congruence is item (b) of Definition 4.10 and conclusion (a) of Lemma 7.39, both of which would be redundant otherwise. Symmetry, by contrast, comes free from minimality, and is what licenses the "switching x_1 and x_2" step in the proof of Lemma 4.15.

Lemma 3.24 (Passing to the quotient)**.** *Let σ be a congruence on $\mathbf{A}$. Then $\delta \mapsto \delta/\sigma$ is an inclusion-preserving bijection, commuting with intersections, from the subuniverses of $\mathbf{A}^2$ that contain σ and are stable under σ onto the subuniverses of $(\mathbf{A}/\sigma)^2$, sending σ to $0_{\mathbf{A}/\sigma}$. Consequently σ is irreducible, linear or PC exactly when $0_{\mathbf{A}/\sigma}$ is, and then $\sigma^*/\sigma = (0_{\mathbf{A}/\sigma})^*$.*

Proof. A subuniverse δ of $\mathbf{A}^2$ containing σ and stable under σ is a union of products of σ-blocks, so it is the preimage of δ/σ; conversely the preimage of a subuniverse of $(\mathbf{A}/\sigma)^2$ is a subuniverse containing σ and stable under σ. The two constructions are mutually inverse and preserve inclusion and intersection, and σ is the preimage of $0_{\mathbf{A}/\sigma}$. The statement of Definition 3.20 is a statement about that family alone, so it transports; and both the least element in Proposition 3.22(b) and the data of Definition 4.10 transport with it. □

Caveat 3.25 (This is the licence for "assume $\sigma = 0$")**.** Three proofs in §7 open by factoring out σ and assuming it is the equality relation. Lemma 3.24 is what licenses that (§2.1, item ix).

3.5 Bridges

Definition 3.26 (Bridge)**.** Let σ_1, σ_2 be congruences on $\mathbf{D}_1, \mathbf{D}_2$. A relation $\delta \leq \mathbf{D}_1^2 \times \mathbf{D}_2^2$ is a *bridge from σ_1 to σ_2* if

(a) the first two variables of δ are stable under σ_1;
(b) the last two variables of δ are stable under σ_2;
(c) $\mathrm{pr}_{1,2}(\delta) \supsetneq \sigma_1$ and $\mathrm{pr}_{3,4}(\delta) \supsetneq \sigma_2$;
(d) $(a_1, a_2, a_3, a_4) \in \delta$ implies $\big((a_1, a_2) \in \sigma_1 \iff (a_3, a_4) \in \sigma_2\big)$.

For a bridge δ put $\widetilde{\delta} = \{(x, y) : (x, x, y, y) \in \delta\}$. A bridge $\delta \subseteq D^4$ is *reflexive* if $(a, a, a, a) \in \delta$ for all $a \in D$.

Remark 3.27 (The motivating example)**.** Let $\mathbf{D}$ be the group $\mathbb{Z}_4$ with any compatible idempotent operation — for instance $x_1 - x_2 + x_3$, extended to arity n by dummy variables — so that every subgroup of a power of $\mathbb{Z}_4$ is a subuniverse. Then $\delta = \{(a_1, a_2, a_3, a_4) : a_1 - a_2 = 2a_3 - 2a_4\}$ is a bridge from $0_{\mathbf{D}}$ to congruence mod 2. Condition (d) is the content: $a_1 = a_2$ exactly when $a_3 \equiv a_4 \pmod 2$. Bridges are the formal shadow of a centralizer relation; see [12].

Definition 3.28 (Composition of bridges)**.** For bridges δ_1 from σ_0 to σ_1 and δ_2 from σ_1 to σ_2, set

$$(\delta_1 * \delta_2)(x_1, x_2, z_1, z_2) = \exists y_1 \exists y_2\ \delta_1(x_1, x_2, y_1, y_2) \wedge \delta_2(y_1, y_2, z_1, z_2).$$

Lemma 3.29 (Bridge composition; [13, Lem. 6.3])**.** *If $\sigma_0, \sigma_1, \sigma_2$ are irreducible then $\delta_1 * \delta_2$ is a bridge from σ_0 to σ_2, and $\widetilde{\delta_1 * \delta_2} = \widetilde{\delta}_1 \circ \widetilde{\delta}_2$.*

Caveat 3.30. Both conclusions of Lemma 3.29 are used: the first in every chain of bridges, the second — that $\widetilde{\cdot}$ commutes with composition — in Lemma 6.23, where a bridge with $\widetilde{\delta}$ linked is powered up until $\widetilde{\delta} = A^2$. The irreducibility hypothesis on *all three* congruences is needed and is easy to drop by accident; Lemma 3.32, which has no such hypothesis, proves its own collapse identity directly rather than citing this one.

Definition 3.31 (Symmetric and pair-reflexive bridges)**.** Let δ be a bridge from σ to σ on $\mathbf{D}$. Call δ *symmetric* if $\delta(x_1, x_2, x_3, x_4) \iff \delta(x_3, x_4, x_1, x_2)$, and *pair-reflexive* if

$$(x_1, x_2) \in \operatorname*{pr}_{1,2}(\delta) \implies (x_1, x_2, x_1, x_2) \in \delta.$$

Write $\delta^{-1}(x_1, x_2, x_3, x_4) = \delta(x_3, x_4, x_1, x_2)$.

Lemma 3.32 (Symmetrisation)**.** *Let δ be a reflexive bridge from σ to σ on $\mathbf{D}$ and put $\delta^\circ = \delta * \delta^{-1}$, that is*

$$\delta^\circ(x_1, x_2, x_3, x_4) = \exists x_5 \exists x_6\ \delta(x_1, x_2, x_5, x_6) \wedge \delta(x_3, x_4, x_5, x_6).$$

Then δ° is a reflexive, symmetric, pair-reflexive bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta^\circ) = \mathrm{pr}_{3,4}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta)$ and $\widetilde{\delta^\circ} = \widetilde{\delta} \circ \widetilde{\delta}^{-1} \supseteq \widetilde{\delta}$. Moreover any pair-reflexive bridge ε satisfies $\varepsilon \subseteq \varepsilon * \varepsilon$.

Proof. Symmetry is visible in the formula. For pair-reflexivity, if $(x_1, x_2) \in \mathrm{pr}_{1,2}(\delta^\circ)$ then $\delta(x_1, x_2, x_5, x_6)$ for some x_5, x_6, and taking the same witnesses twice gives $(x_1, x_2, x_1, x_2) \in \delta^\circ$; the same computation shows $\mathrm{pr}_{1,2}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta)$, the inclusion $\subseteq$ being immediate, and symmetry then gives $\mathrm{pr}_{3,4}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta^\circ)$. Clause (c) of Definition 3.26 follows, and clause (d) because $(x_1, x_2) \in \sigma \iff (x_5, x_6) \in \sigma \iff (x_3, x_4) \in \sigma$. Stability is inherited from δ, and reflexivity from $(a, a, a, a) \in \delta$. The identity $\widetilde{\delta^\circ} = \widetilde{\delta} \circ \widetilde{\delta^{-1}} = \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ is Lemma 3.29; it contains $\widetilde{\delta}$ because $\widetilde{\delta}$ is reflexive, so $(x, y) \in \widetilde{\delta}$ gives $(x, y) \in \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ by taking y as the intermediate point. Finally, if ε is pair-reflexive and $\varepsilon(x_1, x_2, x_3, x_4)$, then taking (x_1, x_2) itself as the intermediate pair — legitimate because $\varepsilon(x_1, x_2, x_1, x_2)$ — gives $(\varepsilon * \varepsilon)(x_1, x_2, x_3, x_4)$. □

Caveat 3.33 (Pair-reflexivity is the hypothesis three lemmas need)**.** Pair-reflexivity is a hypothesis of Lemmas 7.36, 7.39 and 7.42, and it is not optional in any of them: Remark 7.37 exhibits a bridge on $\mathbf{Z}_3$ satisfying every other hypothesis of Lemma 7.36 and none of its conclusion. Lemma 3.32 is what makes the hypothesis costless: every bridge these statements are applied to below is a δ°.

Definition 3.34 (Perfect linear congruence)**.** A congruence σ on $\mathbf{A}$ is a *perfect linear congruence* if it is irreducible and there are a prime p and $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ with $\mathrm{pr}_{1,2}(\zeta) = \sigma^*$ and

$$(a_1, a_2, b) \in \zeta \implies \big((a_1, a_2) \in \sigma \iff b = 0\big).$$

Perfect linear congruences are the payoff of the whole bridge machinery: they let the passage from σ to σ^* be tracked by a single element of $\mathbf{Z}_p$, which is what turns a CSP instance into a system of linear equations in §8.3.

4 The six types of strong subuniverse

This section defines six binary relations $C <_T^A B$ between subsets of a fixed algebra $\mathbf{A}$, one for each type T, and the transitive-closure relation $C \lhd\lhd\lhd^A B$. Everything in §§6–8 is expressed in this language, so the definitions repay care.

The informal reading is: $C <_T^A B$ says that C is a *strong* subset of B — strong enough that the CSP algorithm may reduce a variable's domain from B to C without losing all solutions — and T records *why* it is strong. Three of the reasons are absorption-theoretic (BA, C, S) and three are congruence-theoretic (D, L, PC).

4.1 Absorption

Definition 4.1 (Absorbing subuniverse)**.** A subuniverse B of $\mathbf{A}$ is *absorbing* if there is $t \in \mathrm{Clo}_k(\mathbf{A})$, for some k, such that for every $i \in [k]$

$$t(B, \dots, B, \underbrace{A}_{i}, B, \dots, B) \subseteq B.$$

We then say B *absorbs* $\mathbf{A}$ *with the term* t. If t may be chosen binary, B is a *binary absorbing* subuniverse (BA); if ternary, a *ternary absorbing* subuniverse.

Caveat 4.2 (The tuple form, and $k \geq 1$)**.** Definition 4.1 is a condition on *tuples constrained coordinatewise*: for every $\bar{a} \in A^k$ with $a_j \in B$ for all $j \neq i$, one has $t(\bar{a}) \in B$. The equivalent-looking phrasing "for all $b_1, \dots, b_k \in B$ and $a \in A$, $t(b_1, \dots, b_{i-1}, a, b_{i+1}, \dots, b_k) \in B$" invites the variant in which b_i is quantified and then discarded, which is vacuously true at $k = 1$, where the intended content is not vacuous. The arity is required to satisfy $k \geq 1$.

Definition 4.3 (Central subuniverse)**.** A subuniverse C of $\mathbf{A}$ is *central* (C) if it is absorbing and for every $a \in A \setminus C$,

$$(a, a) \notin \mathrm{Sg}_{\mathbf{A}}((\{a\} \times C) \cup (C \times \{a\})).$$

Lemma 4.4 ([15, Cor. 6.11.1])**.** *A central subuniverse is a ternary absorbing subuniverse.*

Remark 4.5 (Zhuk's centre theorem)**.** Lemma 4.4 is the deepest import of this section. It is proved from the definitions via the Barto–Kazda relational description of absorption (Lemma 7.5) and a doubling argument, and it is the step that converts absorption at some arity into absorption at arity three.

Definition 4.6 (BA and centre free; S)**.** $\mathbf{A}$ is *BA and centre free* if it has no proper nonempty binary absorbing subuniverse and no proper nonempty central subuniverse.

For $\varnothing \neq C \lneq B \leq A$ write $C <_{\mathrm{S}}^A B$ if there is a *nonempty* subuniverse $D \subseteq C$ of $\mathbf{B}$ that is simultaneously binary absorbing and central in $\mathbf{B}$. $\mathbf{A}$ is S-*free* if there is no $C <_{\mathrm{S}} \mathbf{A}$.

Caveat 4.7 (The witness must be required nonempty)**.** $\varnothing$ is vacuously binary absorbing and vacuously central (§2.1, item iii), so without the word "nonempty" the clause defining $<_{\mathrm{S}}$ is universally true and the type S collapses. The global convention that subalgebras are nonempty does not repair it on its own, because D is introduced by an existential inside the clause rather than as a subalgebra of the ambient discourse.

Lemma 4.8 (The two readings of S-freeness agree)**.** $\mathbf{A}$ *is* S*-free if and only if no proper nonempty* $D \leq A$ *is simultaneously binary absorbing and central in* $\mathbf{A}$.

Proof. If such a D exists then $D <_{\mathrm{S}} \mathbf{A}$, taking $C = D$. Conversely if $C <_{\mathrm{S}} \mathbf{A}$ then by definition some nonempty $D \subseteq C \lneq A$ is binary absorbing and central in $\mathbf{A}$, and D is proper. □

Caveat 4.9 (S is not the conjunction of BA and C)**.** $C <_{\text{S}} B$ does not say that C is binary absorbing and central in $\mathbf{B}$; it says that C *contains* some D that is. The type S exists precisely so that it is closed under enlarging C, which BA and C are not. Note also the asymmetry with S-freeness, which *is* stated as a conjunction. Getting this backwards makes Lemma 6.4(ft) unprovable.

Distinct from all of these is the notation $C <^{A}_{\text{BA,C}} B$, which is a *conjunction*: it says that $C <^{A}_{\text{BA}} B$ and $C <^{A}_{\text{C}} B$ both hold. It is not a seventh type, and it is not the disjunction.

4.2 Linear and PC congruences

Definition 4.10 (Linear congruence)**.** A congruence σ on $\mathbf{A}$ is *linear* if

(a) σ is irreducible;
(b) σ^* is a congruence;
(c) there are a prime p and $S \leq (\sigma^*)^{[4]}$ such that for every block B of σ^* there is $m \geq 0$ with
$$(B/\sigma;\ S \cap (B/\sigma)^4) \ \cong\ (\mathbb{Z}_p^m;\ x_1 - x_2 = x_3 - x_4).$$

An irreducible congruence that is not linear is a *PC congruence.*

Here $(\sigma^*)^{[4]}$ denotes the set of 4-tuples whose entries lie in a single block of σ^*, which is meaningful because (b) makes σ^* a congruence. The following is used repeatedly below (§2.1, item ix).

Lemma 4.11 (The witness of (c) is a bridge)**.** *Let σ be a linear congruence and S as in Definition 4.10(c). Then S is a reflexive pair-reflexive bridge from σ to σ with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$.*

Proof. Work inside a block B of σ^* and identify B/σ with $\mathbb{Z}_p^m$ so that $S \cap (B/\sigma)^4$ becomes $x_1 - x_2 = x_3 - x_4$; tuples of S meet no other block. Each variable is stable under σ, since the defining equation is an equation between σ-classes. Clause (d) of Definition 3.26 holds because $x_1 - x_2 = 0$ if and only if $x_3 - x_4 = 0$, which says $(x_1, x_2) \in \sigma \iff (x_3, x_4) \in \sigma$. For the projections, any (x_1, x_2) inside a common block satisfies $(x_1, x_2, x_1, x_2) \in S$, so $\mathrm{pr}_{1,2}(S) = \sigma^*$ — which is at once clause (c), since $\sigma^* \supsetneq \sigma$, and pair-reflexivity; symmetry of the defining equation in the two pairs gives $\mathrm{pr}_{3,4}(S) = \sigma^*$. Finally $(x, x, y, y) \in S$ reads $0 = 0$, so $\widetilde{S}$ is the whole of σ^*, and $(a, a, a, a) \in S$ gives reflexivity. ☐

This is the useful reformulation:

Lemma 4.12 ([14])**.** *For an irreducible congruence σ the following are equivalent:*

(a) *σ is linear;*
(b) *there is a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$.*

Caveat 4.13 (Definition (c) versus criterion (b))**.** Definition 4.10(c) is a per-block isomorphism statement with an existential m *depending on the block*, and it quantifies over blocks of σ^*, not of σ. Lemma 4.12 replaces all of that by one bridge, and the criterion is what every consumer below actually uses; the block description is better read as a structure theorem than as a definition.

The dichotomy "linear or PC" is *by definition* exhaustive on irreducible congruences. Note that PC-ness is a negative condition: every lemma about PC congruences is proved by contradiction against Lemma 4.12(b).

Lemma 4.14 (No bridge crosses the types)**.** *If σ_1 is linear, σ_2 is irreducible, and there is a bridge from σ_1 to σ_2, then σ_2 is linear.*

Lemma 4.15 (Bridges from PC congruences are trivial)**.** *Let σ be a PC congruence on A and δ a reflexive bridge from σ to σ with* $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta) = \sigma^*$*. Then*

$$\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4) \quad \text{or} \quad \delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_4) \wedge \sigma(x_2, x_3).$$

Lemma 4.16 (Bridges into PC congruences)**.** *Let δ be a bridge from a PC congruence σ_1 on $\mathbf{A}_1$ to an irreducible congruence σ_2 on $\mathbf{A}_2$, with* $\mathrm{pr}_{1,2}(\delta) = \sigma_1^*$ *and* $\mathrm{pr}_{3,4}(\delta) = \sigma_2^*$*. Then*

(a) *σ_2 is a PC congruence;*
(b) *$\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$;*
(c) *$\{(a/\sigma_1, b/\sigma_2) : (a, b) \in \widetilde{\delta}\}$ is bijective;*
(d) *$\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_3) \wedge \widetilde{\delta}(x_2, x_4)$ or $\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_4) \wedge \widetilde{\delta}(x_2, x_3)$.*

Two results connect the new vocabulary to the linear and PC subuniverses of the original proof.

Lemma 4.17. *If σ is a linear congruence on $\mathbf{A} \in \mathcal{V}_n$ with $\sigma^* = A^2$, then $\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for a prime p.*

Lemma 4.18. *If σ is a PC congruence on $\mathbf{A}$ with $\sigma^* = A^2$, then $\mathbf{A}/\sigma$ is polynomially complete.*

Caveat 4.19 (What the two statements on top rest on)**.** Lemma 4.17 is proved in §7.7 from Lemma 7.39(c), and the route matters: with $\sigma^* = A^2$ that lemma has a single block, so $\mathbf{A}/\sigma \cong \mathbb{Z}_p^m$ as a group, and irreducibility then forces $m = 1$. That the resulting bijection is an isomorphism of *algebras* rather than of groups is Proposition 2.7, without which the statement does not typecheck against $\mathbf{Z}_p \in \mathcal{V}_n$.

Lemma 4.18 is *stated only*. Its proof needs two things this document does not establish: that an idempotent algebra is polynomially complete exactly when every *reflexive* subuniverse of a power is a conjunction of equalities — those being precisely the invariants of the clone generated by the basic operations together with all constants — and that $\mathbf{A}/\sigma$ is BA and centre free, without which the step $\{a_i\} <_{\mathrm{PC}} \mathbf{A}/\sigma$ is not licensed by Definition 4.21. Nothing below consumes it; it is retained only to connect the vocabulary with [13].

Caveat 4.20 (Polynomial completeness)**.** An algebra is *polynomially complete* if the clone generated by its basic operations together with all constants is the clone of *all* operations on its universe. Lemma 4.18 is where the name "PC" comes from, and it is the only place below where polynomial completeness means more than a label; proving it needs the Istinger–Kaiser characterization [9]. Everywhere else "PC congruence" may be read as "irreducible and not linear".

4.3 The types

Definition 4.21 (The six types)**.** Let $\varnothing \neq C \lneq B \leq A$. We write:

- $C <^A_{\mathrm{BA}} B$ if C is a binary absorbing subuniverse of $\mathbf{B}$;
- $C <^A_{\mathrm{C}} B$ if C is a central subuniverse of $\mathbf{B}$;
- $C <^A_{\mathrm{D}} B$ if there is an irreducible congruence σ on $\mathbf{A}$ with
  (i) $B^2 \subseteq \sigma^*$,
  (ii) $C = B \cap E$ for some block E of σ,
  (iii) $\mathbf{B}/\sigma$ is BA and centre free;
- $C <^A_{\mathrm{L}} B$ if $C <^A_{\mathrm{D}} B$ with the witnessing σ linear;
- $C <^A_{\mathrm{PC}} B$ if $C <^A_{\mathrm{D}} B$ with the witnessing σ a PC congruence;
- $C <^A_{\mathrm{S}} B$ as in Definition 4.6.

When the witnessing congruence matters we write $C <^A_{T(\sigma)} B$; for $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$, where no congruence is involved, σ is taken to be A^2.

Caveat 4.22 (The superscript A is not decoration)**.** For $T \in \{\textsc{d}, \textsc{l}, \textsc{pc}\}$ the witnessing congruence lives on $\mathbf{A}$, not on $\mathbf{B}$, and condition (i) relates the two: $B^2 \subseteq \sigma^*$ says B sits inside a single block of σ^*. This is why the relation carries the ambient algebra as a superscript. For $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ the superscript is genuinely irrelevant, and it is retained only so that one relation symbol serves all six types.

Note also that $\mathbf{B}/\sigma$ in (iii) means B/σ with the induced algebra structure, which requires B to be a subuniverse and σ restricted to B; since $B^2 \subseteq \sigma^*$ and not necessarily $B^2 \subseteq \sigma$, the quotient B/σ is in general *not* a single point, and its being BA and centre free is a real condition.

Definition 4.23 (Multi-types)**.** For $T \in \{\textsc{l}, \textsc{pc}, \textsc{d}\}$ write $C <^A_{\mathcal{M}T} B$ if $C \neq \varnothing$ and $C = C_1 \cap \cdots \cap C_t$ with $t \geq 1$ and $C_i <^A_T B$ for every $i \in [t]$.

Definition 4.24 (The strong-reduction relation)**.** Write $C \lhd\lhd\lhd^A B$ if there are $B_0, \ldots, B_m \subseteq B$ and $T_1, \ldots, T_m \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$ with

$$C = B_m <^A_{T_m} B_{m-1} <^A_{T_{m-1}} \cdots <^A_{T_1} B_0 = B.$$

Here $m = 0$ is allowed, so $\lhd\lhd\lhd^A$ is reflexive. A congruence *comes from* $C \lhd\lhd\lhd^A B$ if it is one of the witnessing congruences in such a chain. We write $B \lhd\lhd\lhd A$ for $B \lhd\lhd\lhd^A A$, and $C \leq^A_{T(\sigma)} B$ for "$C = B$ or $C <^A_{T(\sigma)} B$".

Caveat 4.25 ($\lhd\lhd\lhd$ is data, not a proposition)**.** It is tempting to read $C \lhd\lhd\lhd^A B$ as the reflexive transitive closure of $\bigcup_T <^A_T$ and to forget the chain. That is wrong, and the error is not cosmetic: Definition 4.24 also defines "the congruences coming from $C \lhd\lhd\lhd^A B$", which is a function of the chain and not of the pair. Two places consume it, and in both the *choice* of chain is what makes the argument work.

(a) In the "moreover" clause of Lemma 7.31(d) the conclusion is $\delta \supseteq \delta_1 \cap \cdots \cap \delta_s$ where the δ_i are the congruences of the chains for $B \lhd\lhd\lhd A$ and $C \lhd\lhd\lhd A$ *fixed at the top of that proof*; the proof establishes exactly $\sigma \cap \omega \subseteq \delta$ for those two intersections.
(b) In the type-$\textsc{d}$ case of Lemma 7.46 the clause is applied to C_2 rather than to B_2, and its output is the intersection of ω with σ_2 — which is correct only because the chain witnessing $C_2 \lhd\lhd\lhd A$ is chosen to be the chain witnessing $B_2 \lhd\lhd\lhd A$ extended by the single step $C_2 <_{\textsc{d}(\sigma_2)} B_2$. A different chain gives a different family and the step fails.

So $\lhd\lhd\lhd^A$ carries, at each step, the intermediate set, its type, and its witnessing congruence, and extension by one step is an operation on that data. The bare existential statement may be used only where no congruence is named, which is most but not all of what follows. What *is* true, and is the central simplification over [13], is that the intermediate *types* may be forgotten: $\textsc{l}$ and $\textsc{pc}$ are absent from the list in Definition 4.24 because $\textsc{d}$ subsumes them.

Convention 4.26 (Dotted variants)**.** $C \dot{<}^A_T B$, $C \dot{\leq}^A_T B$ and $C \;\dot{\lhd}\dot{\lhd}\dot{\lhd}^A\; B$ mean that the undotted relation holds *or* $C = \varnothing$. See Caveat 2.5.

5 Instances

This section is entirely combinatorial: no algebra is used beyond the fact that each domain carries a w and each constraint relation is a subuniverse of a product of domains. It is also where the most representation decisions are forced, because instances are syntactic objects manipulated by operations — weakening, covering, renaming — that are usually described in words.

5.1 Instances and solutions

Definition 5.1 (Instance)**.** Fix a finite set V of variables and, for each $x \in V$, an algebra $\mathbf{D}_x \in \mathcal{V}_n$. A *constraint* is a pair $C = (\bar{x}, R)$ where $\bar{x} = (x_1, \dots, x_m)$ is a tuple of variables and $R \leq \mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$. An *instance* $\mathcal{I}$ is a finite list of constraints; we write $C \in \mathcal{I}$, and $\mathrm{Var}(C)$, $\mathrm{Var}(\mathcal{I})$ for the variables occurring. A *subinstance* is a sublist.

A *solution* is a map s with $s(x) \in D_x$ for all x such that $(s(x_1), \dots, s(x_m)) \in R$ for every constraint. The solution set of $\mathcal{I}$ is *subdirect* if for every x and every $a \in D_x$ there is a solution with $s(x) = a$.

Caveat 5.2 (Lists, not sets; tuples, not sets of variables)**.** Two representation choices are forced by the way the arguments below go, and both differ from the naive one.

Constraints form a list, not a set. Proofs delete "a constraint $C \in \mathcal{I}$" and reason about $\mathcal{I} \setminus \{C\}$, and the same relation may occur twice on different variable tuples — and, in the expanded coverings of §5.5, the same relation on the *same* tuple can legitimately occur twice. Multiset or list semantics is required.

A constraint's scope is a tuple, not a set. Repeated variables in a scope are allowed and occur — $\zeta(x, x', z)$ in §8 arises by splitting one variable into two, and $R(x, x, \dots, x)$ appears explicitly in Definition 5.14. So $\mathrm{Var}(C)$ is the image of the scope, and $|\mathrm{Var}(C)|$ may be smaller than the arity.

Once repeated variables are allowed, every semantic operation below that names a *variable* rather than a *position* — $\mathrm{pr}_z(C)$ in 1-consistency, $\mathrm{pr}_{z_i,z_{i+1}}(C)$ along a path, $\mathrm{Con}_1(C, x)$, adjacency at a common variable, weakening by a set of variable names, and the projections in Definition 5.8(ii) — must be read as pulled back along the equal-variable diagonal first. Otherwise a constraint $R(x, x)$ has a full first-coordinate projection even when R contains no diagonal tuple and the constraint admits no value for x. Fixing this notation uniformly is work this document has not yet done; see §11.

Definition 5.3 (Relations defined by an instance)**.** For $x_1, \dots, x_m \in \mathrm{Var}(\mathcal{I})$, $\mathcal{I}(x_1, \dots, x_m)$ is the set of $(a_1, \dots, a_m)$ such that $\mathcal{I}$ has a solution with $s(x_i) = a_i$ for all i. It is a subuniverse of $\mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$, being defined by a primitive positive formula over the constraint relations.

5.2 Reductions

Definition 5.4 (Reduction)**.** A *reduction* for $\mathcal{I}$ is a map $D^{(\top)}$ assigning to each $x \in \mathrm{Var}(\mathcal{I})$ a subuniverse $D_x^{(\top)} \subseteq D_x$, possibly empty. It is *nonempty* if every $D_x^{(\top)} \neq \varnothing$. The instance $\mathcal{I}^{(\top)}$ is $\mathcal{I}$ with each domain replaced by $D_x^{(\top)}$ and each constraint relation intersected with the corresponding box. For reductions we write $D^{(\perp)} \lhd\lhd\lhd D^{(\top)}$ and $D^{(\perp)} \leq_T D^{(\top)}$ when the corresponding relation holds at every variable.

Caveat 5.5 (Empty reductions)**.** Reductions are allowed to be empty at some or all variables — the maximal 1-consistent reduction below $D^{(B,\top)}$ in the proof of Theorem 8.1 is constructed exactly so that it may come out empty, and the case split on whether it does is the branch point of the whole argument. But $D^{(\perp)} \lhd\lhd\lhd D^{(\top)}$ requires nonemptiness at every variable (Convention 2.4). Both notions are needed and they must be different objects.

5.3 Consistency

Definition 5.6 (Paths)**.** A *path* in $\mathcal{I}$ is an alternating sequence $z_1 - C_1 - z_2 - \cdots - C_{l-1} - z_l$ with $z_i, z_{i+1} \in \mathrm{Var}(C_i)$. It *connects b and c* if there are $a_i \in D_{z_i}$ with $a_1 = b$, $a_l = c$, and $(a_i, a_{i+1}) \in \mathrm{pr}_{z_i,z_{i+1}}(C_i)$ for every i. $\mathcal{I}$ is a *tree-instance* if it has no path with $l \geq 3$, $z_1 = z_l$ and $C_1, \ldots, C_{l-1}$ pairwise distinct.

Definition 5.7 (Consistency)**.** $\mathcal{I}$ is 1-*consistent* if $\mathrm{pr}_z(C) = D_z$ for every constraint C and every $z \in \mathrm{Var}(C)$. A reduction $D^{(\top)}$ is 1-consistent for $\mathcal{I}$ if $\mathcal{I}^{(\top)}$ is 1-consistent. $\mathcal{I}$ is *cycle-consistent* if it is 1-consistent and every path from z to z connects a to a, for every variable z and every $a \in D_z$.

Definition 5.8 (Linked, fragmented, irreducible)**.** $\mathcal{I}$ is *linked* if for every $z \in \mathrm{Var}(\mathcal{I})$ and every $a, b \in D_z$ some path from z to z connects a and b. $\mathcal{I}$ is *fragmented* if $\mathrm{Var}(\mathcal{I})$ splits into two disjoint nonempty sets X_1, X_2 with $\mathrm{Var}(C) \subseteq X_1$ or $\mathrm{Var}(C) \subseteq X_2$ for every C. $\mathcal{I}$ is *irreducible* if there is no instance $\mathcal{I}'$ with

(i) $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$;
(ii) every constraint of $\mathcal{I}'$ is a projection of a constraint of $\mathcal{I}$ onto some of its variables;
(iii) $\mathcal{I}'$ is not fragmented;
(iv) $\mathcal{I}'$ is not linked;
(v) the solution set of $\mathcal{I}'$ is not subdirect.

Caveat 5.9 (Irreducibility is a Π_1 condition over a large family)**.** Conditions (i)–(v) quantify over *all* instances built from projections of constraints of $\mathcal{I}$ — a finite but exponentially large family (any multiset of projections of any constraints onto any subsets of their variables). It is finite, so the predicate is decidable, but it is not something to unfold. Every use in §8 goes through Lemma 8.12 and *never* through the definition.

5.4 Weakening and cruciality

Definition 5.10 (Weakening)**.** A constraint $R_1(y_1, \ldots, y_t)$ is *weaker or equivalent to* $R_2(z_1, \ldots, z_s)$ if $\{y_i\} \subseteq \{z_j\}$ and $R_2(\bar{z})$ implies $R_1(\bar{y})$; it is *weaker* if in addition $R_1(\bar{y})$ does not imply $R_2(\bar{z})$. *Weakening* a constraint C in $\mathcal{I}$ means replacing C by all constraints weaker than it. An instance $\mathcal{I}'$ is a *weakening* of $\mathcal{I}$ if $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$ and every constraint of $\mathcal{I}'$ is weaker or equivalent to a constraint of $\mathcal{I}$.

Caveat 5.11 ("Replace by all weaker constraints", and its termination)**.** The operation is finite but enormous: there are $2^{|D|^t}$ relations weaker than a given one on a given scope, and one must also range over all sub-scopes. No algorithm performs it — it appears only inside proofs, as a device for reaching a crucial instance (Remark 5.13). Weakening is therefore defined above as a *relation between instances*, not as a function producing one.

The termination of Remark 5.13 is a genuine obligation and is *not* discharged here. Two things make it delicate, and both must be faced before "we cannot weaken forever" — which is used three times in §8 — can be relied on.

The obvious measure points the wrong way. A properly weaker constraint admits *more* tuples, so the total number of tuples in all constraint relations *increases.* A measure with the right orientation would be a rank in the finite implication order on constraints, or the number of assignments forbidden by the instance.

Replacing a constraint by all *its strict weakenings need not make progress.* The conjunction of all strict weakenings of a relation can be the relation itself — intersecting all one-tuple extensions of R recovers R. So the construction must choose one proper weakening, or a finite antichain with a proved rank decrease, and the choice must be part of the statement.

See §11.

Definition 5.12 (Crucial)**.** Let $D^{(\top)}$ be a reduction for $\mathcal{I}$. A variable y_i of a constraint $R(y_1,\dots,y_t)$ is *dummy* if R does not depend on its i-th coordinate. A constraint $C \in \mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has no dummy variables, $\mathcal{I}^{(\top)}$ has no solution, and weakening C in $\mathcal{I}$ yields an instance with a solution in $D^{(\top)}$. The instance $\mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has at least one constraint and all of its constraints are crucial in $D^{(\top)}$.

Remark 5.13 (Reaching a crucial instance)**.** If $\mathcal{I}^{(\top)}$ has no solution, then iteratively weakening constraints that are not crucial, dropping dummy variables as they appear, terminates in an instance crucial in $D^{(\top)}$. Every relation so introduced is still a subuniverse of the corresponding product of domains. The termination is Caveat 5.11 and is not proved here.

5.5 Coverings

Definition 5.14 (Expanded covering)**.** $\mathcal{I}'$ is an *expanded covering* of $\mathcal{I}$, written $\mathcal{I}' \in \mathrm{Exp}(\mathcal{I})$, if there is $S\colon \mathrm{Var}(\mathcal{I}') \to \mathrm{Var}(\mathcal{I})$ with

(i) $S(x) = x$ for $x \in \mathrm{Var}(\mathcal{I}) \cap \mathrm{Var}(\mathcal{I}')$;
(ii) $D_x = D_{S(x)}$ for every $x \in \mathrm{Var}(\mathcal{I}')$;
(iii) for every constraint $R(x_1,\dots,x_m)$ of $\mathcal{I}'$, either $S(x_1),\dots,S(x_m)$ are pairwise distinct and $R(S(x_1),\dots,S(x_m))$ is weaker or equivalent to a constraint of $\mathcal{I}$, or $S(x_1) = \cdots = S(x_m)$ and $\{(a,\dots,a) : a \in D_{x_1}\} \subseteq R$.

It is a *covering* if moreover $R(S(x_1),\dots,S(x_m)) \in \mathcal{I}$ for every constraint; a *tree-covering* if it is a covering and a tree-instance. $S(x)$ is the *parent* of x, and x a *child.*

Proposition 5.15 (Basic properties)**.** *Let $\mathcal{I}'$ be an expanded covering of $\mathcal{I}$.*

(a) *Replacing every x by $S(x)$ and deleting the constraints $R(x,\dots,x)$ yields a weakening of $\mathcal{I}$.*
(b) *A weakening is an expanded covering with $S = \mathrm{id}$.*
(c) *Every solution of $\mathcal{I}$ lifts to a solution of $\mathcal{I}'$.*
(d) *If $\mathcal{I}$ is 1-consistent and $\mathcal{I}'$ is a tree-covering, the solution set of $\mathcal{I}'$ is subdirect.*
(e) *The union of two expanded coverings is an expanded covering.*
(f) *An expanded covering of an expanded covering is an expanded covering.*
(g) *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*
(h) *Every reduction of $\mathcal{I}$ extends to $\mathcal{I}'$, and a 1-consistent one extends to a 1-consistent one.*

Caveat 5.16 (Variable renaming carries obligations)**.** The constructions of §8 say “rename the variables so that the only common variable of Υ_x and $\mathcal{J}$ is x”, and form conjunctions $\bigwedge_{\mathcal{J}\in\Omega} \mathcal{J}$ of instances whose variable sets must first be made disjoint. Renaming is not free: each of 1-consistency, cycle-consistency, cruciality and being a covering has to be shown invariant under it. Those four facts are used throughout §8 and are nowhere stated.

5.6 Induced congruences and connectedness

Definition 5.17 (Con_1)**.** For R of arity m and $i \in [m]$, $\mathrm{Con}_1(R,i)$ is the binary relation

$$\{(y,y') \;:\; \exists\, \bar{x}\;\; R(\bar{x}[i \mapsto y]) \wedge R(\bar{x}[i \mapsto y'])\}.$$

For a constraint $C = R(x_1,\dots,x_m)$ we write $\mathrm{Con}_1(C,x_i)$ for $\mathrm{Con}_1(R,i)$. For an instance, $\mathrm{Con}(\mathcal{I},x) = \{\mathrm{Con}_1(C,x) : C \in \mathcal{I}\}$ and $\mathrm{Con}(\mathcal{I}) = \bigcup_x \mathrm{Con}(\mathcal{I},x)$.

Lemma 5.18. *The i-th variable of R is rectangular if and only if R is stable under $\mathrm{Con}_1(R,i)$. If R is subdirect and its i-th variable is rectangular, then $\mathrm{Con}_1(R,i)$ is a congruence.*

Caveat 5.19. $\mathrm{Con}_1(R, i)$ is always reflexive and symmetric but is *not* transitive in general, hence not always a congruence; Lemma 5.18 is what makes it one, and its hypotheses (subdirect *and* rectangular at i) are both needed. The notation invites the reader to assume congruence-hood, and every use below must check the hypotheses. Note also that $\mathrm{Con}_1(C, x_i)$ depends on the *position* i, not on the variable, so it is ill-defined if x occurs twice in the scope (Caveat 5.2).

Definition 5.20 (Types of relations and instances)**.** R is *of PC type* (resp. *linear type*) if R is rectangular and every $\mathrm{Con}_1(R, i)$ is a PC (resp. linear) congruence. An instance is of PC or linear type if all its constraints are.

Definition 5.21 (Adjacency, connectedness)**.** Two congruences σ_1, σ_2 on $\mathbf{D}_x$ are *adjacent* if there is a reflexive bridge from σ_1 to σ_2. Two rectangular constraints C_1, C_2 are *adjacent in a common variable x* if $\mathrm{Con}_1(C_1, x)$ and $\mathrm{Con}_1(C_2, x)$ are adjacent. An instance $\mathcal{I}$ is *connected* if all its constraints are rectangular, all congruences in $\mathrm{Con}(\mathcal{I})$ are irreducible, and the graph on the constraints with edges the adjacent pairs is connected.

Note that every proper congruence is adjacent to itself, via $\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4)$.

6 Properties of strong subuniverses

This section is the *interface* between the algebraic theory and the CSP argument: §8 uses the results below and nothing else from §§4. The proofs are §7.

Its shape is worth naming. Four of the five groups say that the relation $\triangleleft\triangleleft\triangleleft$ and the types T *propagate*: forward and backward along surjective homomorphisms, down to quotients, across products of a congruence, and through subdirect relations. The fifth — Theorem 6.14 — is the one genuinely hard statement, and it is what converts a failure of consistency into a bridge. Everything the CSP proof does with bridges traces back to it.

Convention 6.1. Where a type is not specified, T ranges over $\{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{pc}, \textsc{l}, \textsc{d}\}$. Where a multi-type $\mathcal{M}T$ appears, T ranges over $\{\textsc{pc}, \textsc{l}, \textsc{d}\}$.

6.1 Ubiquity

Lemma 6.2 (Ubiquity)**.** *If $B \triangleleft\triangleleft\triangleleft A$ and $|B| > 1$, then $C <_T^A B$ for some C and some $T \in \{\textsc{ba}, \textsc{c}, \textsc{l}, \textsc{pc}\}$.*

Lemma 6.2 is the formal content of Informal Claim 8.1: every domain of size at least 2 that has been reached by a chain of strong reductions admits another strong reduction. It is what guarantees the outer loop of the algorithm makes progress, and it is the reason the algorithm terminates with singleton domains or a linear instance.

Caveat 6.3. The hypothesis is $B \triangleleft\triangleleft\triangleleft A$, not $B \leq A$. A subalgebra reached by no chain of strong reductions need admit nothing. In the algorithm the hypothesis is maintained as an invariant of the reduction loop, and it is carried rather than re-derived.

6.2 Propagation

Lemma 6.4 (Propagation along a homomorphism)**.** Let $f\colon \mathbf{A} \to \mathbf{A}'$ be a surjective homomorphism. Then

(f) $C \triangleleft\triangleleft\triangleleft^A B \Rightarrow f(C) \triangleleft\triangleleft\triangleleft f(B)$;
(b) $C' \triangleleft\triangleleft\triangleleft^{A'} B' \Rightarrow f^{-1}(C') \triangleleft\triangleleft\triangleleft f^{-1}(B')$;
(ft) $C <_{T(\sigma)}^A B \triangleleft\triangleleft\triangleleft A \Rightarrow (f(C) = f(B)$ or $f(C) <_{\textsc{s}} f(B)$ or $f(C) <_T^{A'} f(B))$;
(bt) $C' <_{T(\sigma)}^{A'} B' \Rightarrow f^{-1}(C') <_{T(f^{-1}(\sigma))}^A f^{-1}(B')$;
(fs) $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ and $C <_T B \Rightarrow f(C) \leq_T f(B)$;
(fm) $C \leq_{\mathcal{M}T}^A B \triangleleft\triangleleft\triangleleft A$ and $f(B)$ s-free $\Rightarrow f(C) \leq_{\mathcal{M}T}^{A'} f(B)$;
(bm) $C' \leq_{\mathcal{M}T}^{A'} B' \triangleleft\triangleleft\triangleleft A' \Rightarrow f^{-1}(C') \leq_{\mathcal{M}T}^A f^{-1}(B')$.

Caveat 6.5 (The three-way disjunction in (ft))**.** Clause (ft) is the characteristic shape of this theory and the reason it is usable. Pushing a strong subset forward along a homomorphism can

- collapse it ($f(C) = f(B)$: the reduction becomes vacuous),
- degrade it ($f(C) <_{\textsc{s}} f(B)$: still strong, but the type is lost),
- or preserve it ($f(C) <_T^{A'} f(B)$: same type).

The type $\textsc{s}$ exists to be the value of the middle case. Every downstream proof that pushes a reduction through a quotient does a three-way case split here, and the middle case is the one that is easy to forget; in [13] it is what forces the "go forward and backward from global to local" induction that the present route eliminates.

Note the asymmetry: the backward clauses (b), (bt), (bm) are clean implications with no disjunction. Preimages behave; images do not.

Corollary 6.6 (Propagation to a quotient)**.** *Let δ be a congruence on $\mathbf{A}$. Then* (f) $C \lhd\lhd\lhd^{A} B \Rightarrow C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta$*;* (t) $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd A \Rightarrow (C/\delta = B/\delta$ *or* $C/\delta <_{\mathrm{S}} B/\delta$ *or* $C/\delta <^{A/\delta}_{T} B/\delta)$*;* (s) *for* $T \in \{\mathrm{BA}, \mathrm{C}, \mathrm{S}\}$*,* $C <_T B \Rightarrow C/\delta \leq_T B/\delta$*;* (m) $C \leq^{A}_{\mathcal{M}T} B \lhd\lhd\lhd A$ *and* B/δ S*-free* $\Rightarrow C/\delta \leq^{A/\delta}_{\mathcal{M}T} B/\delta$*.*

Corollary 6.7 (Reflection from a quotient)**.** *Let δ be a congruence on $\mathbf{A}$ and $B, C \leq \mathbf{A}$. Then $C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta \iff C \circ \delta \lhd\lhd\lhd^{A} B \circ \delta$, and $C/\delta <^{A/\delta}_{T} B/\delta \iff C \circ \delta <^{A}_{T} B \circ \delta$.*

Corollary 6.8 (Multiplying by a congruence)**.** *Let δ be a congruence on $\mathbf{A}$. Then* (f) $C \lhd\lhd\lhd^{A} B \Rightarrow C \circ \delta \lhd\lhd\lhd^{A} B \circ \delta$*;* (t) $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd^{A} A \Rightarrow (C \circ \delta = B \circ \delta$ *or* $C \circ \delta <^{A}_{\mathrm{S}} B \circ \delta$ *or* $C \circ \delta <^{A}_{T} B \circ \delta)$*;* (e) *if* $\delta \subseteq \sigma$ *and* $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd^{A} A$ *then* $C \circ \delta <^{A}_{T} B \circ \delta$*.*

Corollary 6.9 (Propagation to relations)**.** Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and $B_i \lhd\lhd\lhd A_i$. Then

- (r) $R \cap (B_1 \times \cdots \times B_m) \dot{\lhd}\dot{\lhd}\dot{\lhd} R$;
- (r1) $\mathrm{pr}_1(R \cap (B_1 \times \cdots \times B_m)) \dot{\lhd}\dot{\lhd}\dot{\lhd} A_1$;
- (b) if $C_i \lhd\lhd\lhd^{A_i} B_i$ for all i then $R \cap (C_1 \times \cdots \times C_m) \dot{\lhd}\dot{\lhd}\dot{\lhd}^{R} R \cap (B_1 \times \cdots \times B_m)$;
- (b1) under the same hypothesis, $\mathrm{pr}_1(R \cap \prod C_i) \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A_1} \mathrm{pr}_1(R \cap \prod B_i)$;
- (m) if $C_i \leq^{A_i}_{\mathcal{M}T} B_i$ for all i then $R \cap \prod C_i \dot{\leq}^{R}_{\mathcal{M}T} R \cap \prod B_i$;
- (m1) if in addition $\mathrm{pr}_1(R \cap \prod B_i)$ is S-free, then $\mathrm{pr}_1(R \cap \prod C_i) \dot{\leq}^{A_1}_{\mathcal{M}T} \mathrm{pr}_1(R \cap \prod B_i)$.

For the two absorption types the conclusion is stronger — no chain, no dot on the type, and no subdirectness hypothesis:

Lemma 6.10 ([15, Cor. 6.1.2, 6.9.2])**.** *Let $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ with* $\mathrm{pr}_1(R) = A_1$*, and let $C_i \leq_T A_i$ for every i, where $T \in \{\mathrm{BA}, \mathrm{C}\}$ — and, when $T = \mathrm{BA}$, with a* single *binary term t witnessing all m absorptions. Then* $\mathrm{pr}_1(R \cap (C_1 \times \cdots \times C_m)) \dot{\leq}_T A_1$*, again with the same t when $T = \mathrm{BA}$.*

Caveat 6.11 (Both hypotheses of Lemma 6.10 are needed)**.** Lemma 6.10 is the workhorse of case (2) of Theorem 8.1, and neither of its hypotheses may be dropped.

Subdirectness. Without $\mathrm{pr}_1(R) = A_1$ the statement is false. Take $R = \{(0,0)\} \leq \mathbf{Z}_p \times \mathbf{Z}_p$ — a subuniverse, by idempotence — and $C_i = A_i$, which is allowed because $\leq_T$ admits equality. Then $R \cap (C_1 \times C_2) = R$ and $\mathrm{pr}_1(R) = \{0\}$, which is neither empty nor all of $\mathbf{Z}_p$, so the conclusion would assert $\{0\} <_{\mathrm{BA}} \mathbf{Z}_p$, contradicting Lemma 6.25.

The common term. In the BA case the m absorptions must be witnessed by a *single* binary t; that is why $\leq_{\mathrm{BA}(t)}$ carries the term in the notation, and the conclusion returns the same t.

At the call site inside Theorem 8.1(2) neither hypothesis is currently discharged: the theorem records that every coordinate reduction has type BA, and that does not by itself produce a term common to all of them. Closing this needs either a lemma synthesising one common binary term from a finite family of coordinate absorptions, or reduction data that carries a common witness from the start. See §11.

6.3 Intersections

Lemma 6.12 (Intersecting with a strong subset)**.** *If $B \lhd\lhd\lhd A$ and $D \lhd\lhd\lhd A$ then* (i) $B \cap D \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A} D$*, and* (t) $C <^{A}_{T(\sigma)} B \Rightarrow C \cap D \dot{\leq}^{A}_{T(\sigma)} B \cap D$*.*

Corollary 6.13. *Under the same hypotheses, $B \cap D \dot{\lhd}\dot{\lhd}\dot{\lhd} A$.*

The next theorem is the heart of §6. Read it as follows: if a family of strong reductions is *jointly* inconsistent but *minimally* so — dropping any one of them restores consistency —

then the types must agree, and in the linear case one gets bridges between the witnessing congruences. Bridges are produced nowhere else.

Theorem 6.14 (Stable intersection)**.** Suppose

(i) $C_i <^{A}_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A$ with $T_i \in \{\text{BA}, \text{C}, \text{S}, \text{L}, \text{PC}\}$, for $i \in [m]$, $m \geq 2$;
(ii) $\bigcap_{i\in[m]} C_i = \varnothing$;
(iii) $B_j \cap \bigcap_{i\neq j} C_i \neq \varnothing$ for every $j \in [m]$.

Then one of:

(ba) $T_1 = \cdots = T_m = \text{BA}$;
(l) $T_1 = \cdots = T_m = \text{L}$, and for all $k, \ell \in [m]$ there is a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \sigma_\ell$;
(c) $m = 2$ and $T_1 = T_2 = \text{C}$;
(pc) $m = 2$, $T_1 = T_2 = \text{PC}$ and $\sigma_1 = \sigma_2$.

Corollary 6.15 (Stable intersection, relational form)**.** Suppose $R \leq_{\text{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$, $C_i <^{A_i}_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A_i$ with $T_i \in \{\text{BA}, \text{C}, \text{S}, \text{L}, \text{PC}\}$ and $m \geq 2$, $R \cap (C_1 \times \cdots \times C_m) = \varnothing$, and $R \cap (C_1 \times \cdots \times B_j \times \cdots \times C_m) \neq \varnothing$ for every j. Then one of: (ba) all $T_i = \text{BA}$; (l) all $T_i = \text{L}$ and for all k, ℓ a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \text{pr}_{k,\ell}(R) \circ \sigma_\ell$; (c) $m = 2$ and $T_1 = T_2 = \text{C}$; (pc) $m = 2$, $T_1 = T_2 = \text{PC}$, $\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$, and $\{(a/\sigma_1, b/\sigma_2) : (a, b) \in R\}$ is bijective.

Remark 6.16. Several applications need more than one restriction on a single coordinate of R. The statement is kept simple; to apply it, duplicate the coordinate (form $R' = \{(\bar{a}, a_k) : \bar{a} \in R\}$) and restrict the copies separately.

Caveat 6.17 (Minimality is a hypothesis, not a construction)**.** Hypotheses (ii)–(iii) of Theorem 6.14 say the family is *critically* inconsistent. In applications one starts with an inconsistent family and shrinks it to a critical subfamily. The shrinking step is elementary — *if P is a monotone predicate on subsets of a finite set X and $P(X)$ holds, there is $M \subseteq X$ minimal with $P(M)$* — but it is performed at least four times below, each time with a different P, so it is worth naming rather than repeating.

A related step is used with the multi-types: an inclusion-minimal $\mathcal{M}T$ subuniverse of B *containing a given point* b is automatically minimal outright. That is not immediate, and it is what Lemma 8.11 consumes. It holds because $M(b) = B \cap \bigcap_{\sigma\in\Sigma}[b]_\sigma$, over the T-dividing congruences Σ for B, is the *least* $\mathcal{M}T$ subuniverse containing b (each $B \cap [b]_\sigma$ is $<^{A}_{T} B$, and any $\mathcal{M}T$ set containing b is an intersection of blocks each containing b), and because $c \in M(b)$ forces $[c]_\sigma = [b]_\sigma$ for every $\sigma \in \Sigma$ and hence $M(c) = M(b)$. So the image of $b \mapsto M(b)$ consists of pairwise disjoint minimal sets, and a minimal set containing b is $M(b)$, which has no proper $\mathcal{M}T$ subset.

6.4 Multi-types

Lemma 6.18. *If $C \leq^{A}_{\mathcal{M}T} B$ then $C <^{A}_{T} \ldots <^{A}_{T} B$ and $C \lhd\lhd\lhd^{A} B$.*

That is: an intersection of type-T subsets, though not itself obtained by a single reduction of type T, is reachable by a *chain* of type-T reductions. This is what makes $\mathcal{M}T$ compatible with $\lhd\lhd\lhd$, and it is used whenever the algorithm reduces several variables at once.

Lemma 6.19 (Preserving linkedness)**.** *Let $R \leq_{\text{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, $C_i \leq^{A_i}_{\mathcal{M}\text{D}} B_i \lhd\lhd\lhd A_i$ for $i = 1, 2$, let S be the rectangular closure of R, and suppose $R \cap (B_1 \times B_2) \neq \varnothing$ and $S \cap (C_1 \times C_2) \neq \varnothing$. Then $R \cap (C_1 \times C_2) \neq \varnothing$.*

Lemma 6.20 (Maximal multi-type extension)**.** *Let $C_1 <^A_{\mathcal{M}T} B_1 \lhd\lhd\lhd A$, $B_2 \lhd\lhd\lhd A$, $C_1 \cap B_2 = \varnothing$, $B_1 \cap B_2 \neq \varnothing$, and let σ be a maximal congruence on $\mathbf{A}$ with $(C_1 \circ \sigma) \cap B_2 = \varnothing$. Then $\sigma = \omega_1 \cap \cdots \cap \omega_s$ where each ω_i is a congruence of type T on $\mathbf{A}$ with $\omega_i^* \supseteq B_1^2$.*

6.5 Auxiliary imports

The following are used but not proved in [14].

Lemma 6.21 ([10, Lem. 4.7])**.** *If w is an idempotent WNU operation of arity n on a finite set A, then* $\mathrm{Clo}(w)$ *contains a special idempotent WNU operation of arity $n^{n!}$.*

Lemma 6.22 ([13, Cor. 8.17.1])**.** *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta} = A^2$, then σ is a perfect linear congruence.*

Lemma 6.23. *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta}$ linked, then σ is a perfect linear congruence.*

Proof. Let $\delta^{-1}(y_1, y_2, x_1, x_2) = \delta(x_1, x_2, y_1, y_2)$, which is again a bridge from σ to σ. By Caveat 6.24 the relation $\rho = \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ is reflexive and symmetric, and it is connected because $\widetilde{\delta}$ is linked; so $\rho^N = A^2$ for $N \geq |A|$. Composing $2N$ bridges $\delta, \delta^{-1}, \ldots$ by Lemma 3.29 — legitimate, all the congruences involved being σ, which is irreducible — gives a bridge δ' from σ to σ with $\widetilde{\delta'} = \rho^N = A^2$; apply Lemma 6.22. □

Caveat 6.24 (Where the reflexivity comes from)**.** The step $(\widetilde{\delta} \circ \widetilde{\delta}^{-1})^N = A^2$ is the elementary fact that a reflexive symmetric relation ρ on a finite set A with connected graph satisfies $\rho^N = A^2$ for $N \geq |A|$; reflexivity is what makes the powers increase monotonically.

The reflexivity of $\widetilde{\delta}$ itself does *not* follow from $\mathrm{pr}_{1,2}(\delta) \supseteq \sigma$, which gives only some tuple $\delta(a, a, c, d)$ and not $\delta(a, a, a, a)$. It follows instead from clause (d) of Definition 3.26 together with stability: from $\delta(a, a, c, d)$, clause (d) gives $(c, d) \in \sigma$, and stability of the last two variables under σ replaces d by c, giving $\delta(a, a, c, c)$. So $\widetilde{\delta}$ is left-total, and $\widetilde{\delta} \circ \widetilde{\delta}^{-1}$ is reflexive and symmetric — which is the relation the power is taken of.

Lemma 6.25. *If $B \leq \mathbf{Z}_p \times \cdots \times \mathbf{Z}_p$ then there is no $C <_T B$ with $T \in \{\textsc{ba}, \textsc{c}\}$.*

Proof. $\mathbf{B}$ is affine over the product of the $\mathbf{Z}_p$, so every term operation acts as $\tau(\bar{x}) = \sum_i c_i x_i$ with $\sum_i c_i = 1$, and every nonempty subuniverse is an affine coset. Suppose $C <_T B$ with $T \in \{\textsc{ba}, \textsc{c}\}$; a central subuniverse is absorbing (Definition 4.3), so in either case some k-ary τ absorbs at every coordinate. Translating so that C is a subgroup, absorption at coordinate i says τ carries $C^{i-1} \times B \times C^{k-i}$ into C; taking all entries but the i-th to be 0 gives $c_i b \in C$ for every $b \in B$, so c_i annihilates B/C. Doing this at every i makes $\sum_i c_i = 1$ annihilate B/C as well, i.e. $B = C$, contradicting properness. □

7 Proofs of the strong-subalgebra theory

This section proves the statements of §6. Throughout, Convention 2.1 is in force — every algebra is finite, idempotent, and has a WNU term operation — and §2.1 fixes the readings.

7.1 What is imported

The development is self-contained except for the following, which are proved in the literature and used here as black boxes. Two are substantial; the rest are routine. §11.4 lists them again with their exact hypotheses.

Theorem 7.1 (Absorption Theorem)**.** Let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ be a subdirect relation between finite idempotent Taylor algebras. If R is linked then one of the following holds: $R = A \times B$; or $\mathbf{A}$ has a proper nonempty binary absorbing or centrally absorbing subalgebra; or $\mathbf{B}$ has a proper nonempty subalgebra that is both binary absorbing and centrally absorbing.

Corollary 7.2 (a corollary of Theorem 7.1)**.** *Let $R \lneq_{sd} \mathbf{A} \times \mathbf{B}$ be a* proper *subdirect relation between finite idempotent Taylor algebras. If R is linked then $\mathbf{A}$ or $\mathbf{B}$ has a proper nonempty binary absorbing or central subuniverse.*

Proof. Properness excludes the first alternative of Theorem 7.1. The second is the conclusion on $\mathbf{A}$; the third is stronger than the conclusion on $\mathbf{B}$. □

Caveat 7.3 (Both words in "proper nonempty" are load-bearing)**.** Without them Corollary 7.2 is vacuous, since every algebra is a binary absorbing subuniverse of itself and $\varnothing$ is vacuously one (§2.1, item iii); every use below is the proper nonempty one. Theorem 7.1 is the single heaviest import of this section, and it is stated here in the form its source gives rather than in the weaker symmetric form actually used, so that the strengthening is visible.

Lemma 7.4 ([4, Lem. 2.9], [15, Lem. 6.1, Thm. 6.9])**.** *Let $R \leq A^n$ be defined by a pp-formula Φ in which a relation S occurs, and let R' be defined by the formula obtained from Φ by replacing* every *occurrence of S by some $S' <_T S$, where $T \in \{\textsc{ba}, \textsc{c}\}$. Then $R' \leq_T R$.*

Lemma 7.5 ([4, Prop. 2.14], [15, Lem. 3.2])**.** *For $B \leq \mathbf{A}$ and $n \geq 2$: B absorbs $\mathbf{A}$ with an operation of arity n if and only if there is no $S \leq A^n$ with $S \cap B^n = \varnothing$ and $S \cap (B^{i-1} \times A \times B^{n-i}) \neq \varnothing$ for every i.*

Lemma 7.6 ([15, Lem. 6.8])**.** *If $B \leq_T \mathbf{A}$ with $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ and σ is a congruence on $\mathbf{A}$, then $B/\sigma \leq_T \mathbf{A}/\sigma$.*

Caveat 7.7 (Only the $\textsc{c}$ case is imported)**.** For $T = \textsc{ba}$ the statement is one line: if $t(B, A) \subseteq B$ and $t(A, B) \subseteq B$ then the same t works on images. The case $T = \textsc{s}$ follows from the other two by Definition 4.6. Only $\textsc{c}$ is genuinely imported.

Lemma 7.8 ([15, Lem. 6.24])**.** *If R is a nontrivial strong subuniverse of $\mathbf{A}_1 \times \cdots \times \mathbf{A}_n$ of a type $T \neq \textsc{pc}$, then some $\mathbf{A}_i$ has a nontrivial subuniverse of type T. In particular $B <_T \mathbf{A}^n$ with $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ yields some $C <_T \mathbf{A}$.*

Caveat 7.9 (One run of the proof serves all three types)**.** The cited lemma is stated for every type $T \neq \textsc{pc}$, and its proof is *type-uniform*: it takes $\mathrm{pr}_1(R)$ if that is proper and otherwise a fibre $R' = \{(a_2, \ldots, a_n) : (a_1, \ldots, a_n) \in R\}$, and neither construction mentions the type — the type enters only through the facts that projections and fibres preserve $\textsc{ba}$ and preserve $\textsc{c}$. So a single run produces one witness carrying *every* strong type that R has; in particular a subuniverse that is simultaneously binary absorbing and central yields one that is too, which is the case $T = \textsc{s}$.

Theorem 7.10 ([15, Thm. 6.15])**.** Let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ and $C = \{c \in A : \{c\} \times B \subseteq R\}$. Then C is a central subuniverse of $\mathbf{A}$, or $\mathbf{B}$ has a nontrivial binary absorbing subuniverse, or $\mathbf{B}$ has a nontrivial *projective* subuniverse.

Caveat 7.11 (The heaviest import after Theorem 7.1)**.** Theorem 7.10 is taken on faith here. Its proof in [15] is two pages and uses only the Barto–Kazda criterion (Lemma 7.5) and the theory of projective subuniverses; none of that theory is needed anywhere else below, which is why absorbing it has been deferred. The same applies to [15, Lem. 3.4], used once, in the proof of Lemma 7.19.

Lemma 7.12 ([8])**.** *An algebra is Abelian if and only if some congruence of its square has the diagonal as a block; and a finite algebra with a WNU term operation is Abelian if and only if it is affine.*

Lemma 7.13. *Let $|A| > 1$. Then $\mathbf{A}$ is Abelian if and only if there is a bridge δ from $0_{\mathbf{A}}$ to $0_{\mathbf{A}}$ with $\widetilde{\delta} = \mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta) = A^2$.*

Proof. If $\mathbf{A}$ is Abelian, a congruence of $\mathbf{A}^2$ having the diagonal as a block is such a bridge; clause (c) needs $|A| > 1$, since for $|A| = 1$ no relation properly contains $0_{\mathbf{A}} = A^2$.

Conversely, regard δ as a binary relation $\hat{\delta}$ on A^2, $\hat{\delta}((x_1, x_2), (x_3, x_4)) \iff \delta(x_1, x_2, x_3, x_4)$, and put $\rho = \hat{\delta} \circ \hat{\delta}^{-1}$, a reflexive symmetric compatible relation on $\mathbf{A}^2$ — reflexive because $\mathrm{pr}_{1,2}(\delta) = A^2$ makes $\hat{\delta}$ left-total. Its powers increase, so $\theta = \rho^N$ for $N \geq |A|^2$ is reflexive, symmetric and transitive, hence a congruence of $\mathbf{A}^2$. Clause (d) of Definition 3.26 is preserved by each composition, so every pair of θ relates a diagonal element only to diagonal elements: the diagonal is a θ-block. Now apply Lemma 7.12. □

7.2 Absorption preliminaries

Lemma 7.14. *If $B \leq_T \mathbf{A}$ with $T \in \{\text{BA}, \text{C}\}$ and $C \leq \mathbf{A}$, then $B \cap C \dot{\leq}_T C$.*

Proof. C is defined by the pp-formula $A(x) \wedge C(x)$; replacing the occurrence of the full unary relation A by B yields $B \cap C$, so Lemma 7.4 applies. □

Lemma 7.15 (No absorbing diagonal)**.** *Let $\mathbf{C}$ be a finite idempotent algebra with $|C| \geq 2$. Then $0_{\mathbf{C}}$ is not an absorbing subuniverse of $\mathbf{C}^2$, of any arity; in particular $0_{\mathbf{C}} \not\leq_{\text{BA}} \mathbf{C}^2$ and $0_{\mathbf{C}} \not\leq_{\text{C}} \mathbf{C}^2$.*

Proof. Suppose t is k-ary and absorbs at every position. Fix i, take $a_j \in C$ for $j \neq i$ and $b, c \in C$, and apply t coordinatewise to the tuples $(a_j, a_j) \in 0_{\mathbf{C}}$ for $j \neq i$ and $(b, c) \in C^2$ at position i. The result $(t(\bar{a}; b), t(\bar{a}; c))$ lies in $0_{\mathbf{C}}$, so $t(\bar{a}; b) = t(\bar{a}; c)$. As b, c were arbitrary, t does not depend on its i-th argument; and this holds for every i, so t is constant. Idempotence then forces $|C| = 1$. A central subuniverse is absorbing by Definition 4.3, so the central case is included. □

Lemma 7.16 (Absorption preserves linkedness, subrelation form)**.** *Let $W \leq_{\text{sd}} \mathbf{P} \times \mathbf{A}$ be linked and let $\varnothing \neq W' \leq_T W$ with $T \in \{\text{BA}, \text{C}\}$. Put $P' = \mathrm{pr}_1(W')$ and $A' = \mathrm{pr}_2(W')$. Then $W' \leq_{\text{sd}} \mathbf{P}' \times \mathbf{A}'$ is linked.*

Proof. Let $\Phi_k(x, y)$ be the pp-formula in one binary relation symbol obtained by iterating $x \sim y \iff \exists z\, (W(x, z) \wedge W(y, z))$ k times. Because W is subdirect, $W \circ W^{-1}$ is reflexive and symmetric on P, so for $k \geq |P|$ the formula Φ_k interpreted in W defines the transitive closure of $W \circ W^{-1}$, which is P^2 since W is linked. Interpreted in W' and for the same $k \geq |P| \geq |P'|$ it defines $\lambda := \mathrm{LeftLinked}(W')$. By Lemma 7.4, replacing every occurrence of W by W' gives $\lambda \leq_T P^2$. Since $\lambda \subseteq (P')^2 \leq P^2$, Lemma 7.14 gives $\lambda \leq_T (P')^2$.

Now λ is a congruence on $\mathbf{P}'$: reflexive because W' is subdirect over P', symmetric by construction, transitive because the powers stabilise. So $\lambda \times \lambda$ is a congruence on the algebra $(P')^2$, and the image of λ under $(P')^2 \twoheadrightarrow (P'/\lambda)^2$ is exactly $0_{\mathbf{P}'/\lambda}$; by Lemma 7.6, $0_{\mathbf{P}'/\lambda} \leq_T (\mathbf{P}'/\lambda)^2$. Lemma 7.15 forces $|P'/\lambda| = 1$, i.e. $\lambda = (P')^2$, which says W' is linked. □

Caveat 7.17 (The box form is not enough)**.** The familiar form of this principle, [3, Prop. 2.15(i)], restricts by a *box*: if $W \leq_{\text{sd}} \mathbf{A}_1 \times \mathbf{A}_2$ is linked, B_i absorbs $\mathbf{A}_i$, and $W \cap (B_1 \times B_2)$ is subdirect in $B_1 \times B_2$, then $W \cap (B_1 \times B_2)$ is linked. That form cannot reach the restrictions used in Lemma 7.42, and the two really do differ: for the meet-semilattice on $\{0, 1, 2\}$ with $B = \{0, 1\}$, the subalgebra $K = \{(0, 0), (1, 2)\}$ of A^2 meets B^2 and has

$\mathrm{pr}_1(K) \cap B = \{0, 1\}$ while $\mathrm{pr}_1(K \cap B^2) = \{0\}$. The subrelation form also discharges the box form's subdirectness hypothesis for free.

Lemma 7.18 (A WNU forbids essentially unary quotients)**.** *Let* $\mathbf{U}$ *be a finite algebra with a WNU term operation and* $|U| \geq 2$*. Then some term operation of* $\mathbf{U}$ *depends on more than one variable. Consequently, if* $\mathbf{A}$ *has a WNU term operation then no* $\mathbf{U} \in \mathrm{HS}(\mathbf{A})$ *with* $|U| \geq 2$ *is essentially unary.*

Proof. Suppose every term operation of $\mathbf{U}$ has at most one non-dummy variable, and let w be a WNU term operation of arity $k \geq 2$; it is idempotent. If w has no non-dummy coordinate it is constant, and idempotence gives $|U| = 1$. Otherwise $w(\bar{x}) = g(x_i)$ for a unary g, and $g(x) = w(x, \ldots, x) = x$, so w is the i-th projection. The WNU identities equate the k terms in which a single y sits at each position against a background of x's; two of them are w with y at position i, which evaluates to y, and w with y at some position $\neq i$, which evaluates to x. Hence $x = y$ for all x, y and $|U| = 1$. The consequence follows because idempotence and the WNU identities are identities, hence hold in every member of $\mathrm{HS}(\mathbf{A})$. □

Lemma 7.19 (Central relations)**.** Let $\mathbf{A}, \mathbf{B}$ satisfy Convention 2.1, let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$, and put $C = \{c \in A : \{c\} \times B \subseteq R\}$. Then

(a) $C = \varnothing$, or
(b) C is a central subuniverse of $\mathbf{A}$, or
(c) $\mathbf{B}$ has a proper nonempty binary absorbing subuniverse.

Proof. Apply Theorem 7.10. Its first alternative gives (a) or (b) according as C is empty or not, its second gives (c). In its third, $\mathbf{B}$ has a nontrivial projective subuniverse P; by [15, Lem. 3.4], either P is a binary absorbing subuniverse of $\mathbf{B}$, which is (c) since P is nontrivial, or there is an essentially unary $\mathbf{U} \in \mathrm{HS}(\mathbf{B})$ with $|U| \geq 2$, which Lemma 7.18 forbids. □

Caveat 7.20 (Alternative (a) is not optional)**.** Under Convention 2.4 the two-alternative form (b)–(c) is false: $\mathbf{A} = \mathbf{B} = \mathbf{Z}_p$ with $R = \{(x, x+1)\}$ is subdirect, has $C = \varnothing$, and $\mathbf{Z}_p$ has no nontrivial binary absorbing subuniverse at all. It is true in [15] only because $\varnothing$ counts as central there. Every call site below discharges (a) by exhibiting a point of C first, and two further obligations go with each contrapositive use: the centre produced must be shown *proper*, and a strong subuniverse of a *power* must be moved down by Lemma 7.8.

Lemma 7.21. *Let* $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ *with* $\mathbf{A}$ *BA and centre free,* $\mathrm{LeftLinked}(R) = A^2$*, and* $C = \{c \in B : A \times \{c\} \subseteq R\}$*. Then* $C \neq \varnothing$ *and* $C \leq_{\mathrm{BA,C}} \mathbf{B}$*.*

Proof. For $k \geq 2$ put $W_k = \{(a_1, \ldots, a_k) : \exists b \, \forall i \, R(a_i, b)\}$. If $W_{|A|} = A^{|A|}$ then $C \neq \varnothing$, since a witness b for the tuple listing all of A lies in C. Otherwise choose k minimal with $W_k \neq A^k$. Since $\mathrm{LeftLinked}(R) = A^2$ we get $\mathrm{LeftLinked}(W_k) = A^2$, and viewing W_k as a binary relation in $\mathbf{A} \times \mathbf{A}^{k-1}$ — subdirect by minimality of k — Corollary 7.2 and Lemma 7.8 produce a proper nonempty binary absorbing or central subuniverse of $\mathbf{A}$, contrary to hypothesis. So $C \neq \varnothing$.

By Lemma 7.19 applied to R^{-1} — alternative (a) being excluded — either C is central in $\mathbf{B}$ or $\mathbf{A}$ has a proper nonempty binary absorbing subuniverse; the latter is excluded, so C is central. For binary absorption, suppose not; by Lemma 7.5 there is $S \leq B \times B$ with $S \cap (C \times C) = \varnothing$, $S \cap (B \times C) \neq \varnothing$ and $S \cap (C \times B) \neq \varnothing$. Put

$$W = \{(a_1, \ldots, a_{|A|}) : \exists (b, c) \in S \ (c \in C \wedge \forall i \, R(a_i, b))\}.$$

By Lemma 7.4 $W <_{\mathrm{C}} A^{|A|}$, so by Lemma 7.8 $\mathbf{A}$ has a proper nonempty central subuniverse — again a contradiction. □

Lemma 7.22. *If $C \lhd\lhd\lhd^A B \lhd\lhd\lhd A$ and $D <_{\text{BA,C}} B$ then $C \cap D \neq \varnothing$.*

Proof. Suppose not, and choose C'' minimal with $C \lhd\lhd\lhd^A C' <^A_{T(\sigma)} C'' \lhd\lhd\lhd^A B$ and $C'' \cap D \neq \varnothing$. By Lemma 7.14 $C'' \cap D <_{\text{BA,C}} C''$, so by Lemma 7.23 $T = \text{D}$; and then Lemma 7.6 gives $(C'' \cap D)/\sigma <_{\text{BA}} C''/\sigma$, contradicting the requirement in Definition 4.21 that C''/σ be BA and centre free. $\square$

Lemma 7.23 ([15, Lem. 6.25])**.** *If $B <_{T_1} \mathbf{A}$, $C <_{T_2} \mathbf{A}$, $B \cap C = \varnothing$ and $T_1, T_2 \in \{\text{BA}, \text{C}, \text{S}\}$, then $T_1 = T_2 \in \{\text{BA}, \text{C}\}$.*

7.3 The intersection property

The goal is Lemma 7.31: if $B \lhd\lhd\lhd A$, $C \lhd\lhd\lhd A$ meet and δ is a dividing congruence for B, then $(B \cap C)/\delta$ is a single point or all of B/δ. Everything in §7 that produces a bridge goes through it.

Lemma 7.24 (The shift manoeuvre)**.** *Let $k \geq 2$ and let $R \leq A^k$ be totally symmetric and closed under $(a_1, \ldots, a_k) \mapsto (a_1, a_1, a_2, \ldots, a_{k-1})$. Identify a tuple of R with its entry multiset, a k-element multiset over A. If $\bar{a} \in R$ has entry multiset M, then for any $u, v \in M$ —* occupying distinct positions of $\bar{a}$, though possibly with $u = v$ as elements of A — *the multiset $M + \{u\} - \{v\}$ is again the entry multiset of a tuple of R. Consequently every k-element multiset over the set S of distinct entries of $\bar{a}$ is reached, provided it is reached by such steps; in particular, for any $u \in S$, the constant multiset $k \cdot \{u\}$ is reached, in at most $k - 1$ steps.*

Proof. Total symmetry lets us place u first and v last; the shift then duplicates u and deletes v, giving $M + \{u\} - \{v\}$. For the last clause, apply this $k - 1$ times with the same u, deleting a different position each time. $\square$

Caveat 7.25 (The shift deletes the last entry)**.** A single application of the shift to $(a, b_2, \ldots, b_{k-1}, c)$ destroys c, so $(a, \ldots, a, c) \in R$ does not follow from one step; it needs $k - 2$ steps interleaved with permutations, which is what Lemma 7.24 packages. The lemma is used twice below and the one-step reading fails both times.

Lemma 7.26. *Let $k \geq 2$, let $\mathbf{A}$ be BA and centre free, and let $R \leq A^k$ be totally symmetric, closed under the shift, with $\text{pr}_{1,2}(R) = A^2$. Then $R = A^k$.*

Proof. Induct on k; for $k = 2$ the hypothesis is the conclusion. Let $k > 2$.

Step 1: $(a, \ldots, a, c) \in R$ for all a, c. By total symmetry $\text{pr}_{1,k}(R) = A^2$, so there is a tuple of R with first entry a and last entry c. Apply Lemma 7.24 $k - 2$ times with $u = a$, deleting on each step one position other than the last: the entry multiset becomes $(k - 1) \cdot \{a\} + \{c\}$, which by total symmetry gives $(a, \ldots, a, c) \in R$. Deleting c as well gives $(a, \ldots, a) \in R$.

Step 2: the left projection is full. $R' = \text{pr}_{1,\ldots,k-1}(R)$ inherits total symmetry, has $\text{pr}_{1,2}(R') = A^2$, and is shift-closed: from $\bar{a} \in R'$ extend to R, shift there, and project. By the inductive hypothesis $R' = A^{k-1}$.

Step 3. Only now may R be viewed as a subdirect relation $R \leq_{\text{sd}} A^{k-1} \times A$: Step 2 gives the left projection and Step 1 gives the right. It is linked, since by Step 1 every c is adjacent to the single left vertex $(a, \ldots, a)$. If $R \neq A^k$ then Corollary 7.2 gives a proper nonempty binary absorbing or central subuniverse on $\mathbf{A}^{k-1}$ or on $\mathbf{A}$, and Lemma 7.8 moves the first case to $\mathbf{A}$ — contradicting the hypothesis either way. $\square$

Lemma 7.27. *Let σ be a dividing congruence for $B \leq A$ and let $R \leq (\mathbf{A}/\sigma)^k$ be reflexive, totally symmetric and shift-closed. Then $(B/\sigma)^k \subseteq R$, or R is the set of constant tuples.*

Proof. If $\mathrm{pr}_{1,2}(R)$ is the equality relation then, by total symmetry, all coordinates of every tuple of R agree, and reflexivity supplies all constant tuples.

Otherwise $\mathrm{pr}_{1,2}(R)$ is a subuniverse of $(\mathbf{A}/\sigma)^2$ containing $0_{\mathbf{A}/\sigma}$ properly and stable under it, so by minimality of the cover (Proposition 3.22, transported by Lemma 3.24) it contains σ^*/σ, hence $(B/\sigma)^2$ — the last step by Definition 4.21(i). Now apply Lemma 7.26 to $R\cap(B/\sigma)^k$ over the algebra B/σ, which is BA and centre free by Definition 4.21(iii). Its hypothesis $\mathrm{pr}_{1,2}(R\cap(B/\sigma)^k) = (B/\sigma)^2$ is not immediate from $\mathrm{pr}_{1,2}(R) \supseteq (B/\sigma)^2$, since a witnessing tuple may leave B/σ; Lemma 7.24 repairs it, pulling $(x_1, x_2, y_3, \ldots, y_k)$ back to $(x_1, x_2, x_1, \ldots, x_1)$. □

Lemma 7.28 (Full fibre)**.** *Let F be a finite set with $|F| \le N$ and let $W \subseteq F^N \times E$ satisfy $\mathrm{pr}_{1..N}(W) = F^N$ and be closed under coordinate substitution: if $(x_1, \ldots, x_N, e) \in W$ and $g : [N] \to [N]$ then $(x_{g(1)}, \ldots, x_{g(N)}, e) \in W$. Then $F^N \times \{d\} \subseteq W$ for some $d \in E$.*

Proof. Write $F = \{F_1, \ldots, F_t\}$ with $t \le N$ and put $\bar{x} = (F_1, \ldots, F_t, F_t, \ldots, F_t) \in F^N$. By surjectivity pick d with $(\bar{x}, d) \in W$. Every $\bar{y} \in F^N$ is $\bar{x} \circ g$ for some $g : [N] \to [N]$, since every entry of $\bar{y}$ occurs among the entries of $\bar{x}$; closure gives $(\bar{y}, d) \in W$. □

Caveat 7.29 (Why the arity is $|A|$)**.** Lemma 7.28 is used three times below, always in the shape "since $\mathrm{pr}_{1,\ldots,|A|}(R') = (B/\delta)^{|A|}$, there is d with $(B/\delta)^{|A|} \times \{d\} \subseteq R'$". Surjectivity of a projection does not by itself produce a full fibre; what makes the step valid is that each such R' is cut out by a conjunction of unary and binary conditions on its first $|A|$ entries, hence closed under substituting entries for one another, and that $|B/\delta| \le |A|$. **The exponent $|A|$ is load-bearing**: it must be at least the number of blocks, so that one tuple can enumerate them. Normalising the arity away breaks all three proofs.

Lemma 7.30. Let $B_k <^{A}_{T_k(\sigma_k)} \cdots <^{A}_{T_1(\sigma_1)} B_0 = A$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$, let δ be a congruence on $\mathbf{A}$, and let $m \in [k]$ with $T_m = \textsc{d}$. Then $((B_k \circ \delta) \cap B_{m-1})/\sigma_m$ is either B_{m-1}/σ_m or B_m/σ_m; and in the second case $\sigma_m \supseteq \delta \cap \sigma_1 \cap \cdots \cap \sigma_{m-1}$.

Proof outline. Induction on k, splitting on T_k and on whether $k > m$. The absorption cases run through Lemmas 7.4 and 7.6 to produce a strong subuniverse of B_{k-1}/σ_k and contradict Definition 4.21(iii). The $\textsc{d}$ cases build an auxiliary relation R' on $(B_{k-1}/\sigma_k)^{|A|} \times B_{m-1}/\sigma_m$, obtain a full fibre from Lemma 7.28, and finish with Lemma 7.19, discharging at each such finish the two obligations of Caveat 7.20: the centre is *proper*, the witness being the non-containment established one step earlier, and the binary absorbing subuniverse produced lives on a *power* and is moved down by Lemma 7.8. This is an outline and not a proof; see §11. □

Lemma 7.31 (The intersection property)**.** Let $B \lhd\lhd\lhd A$, $C \lhd\lhd\lhd A$ with $B \cap C \neq \varnothing$. Then

- (d) if δ is a dividing congruence for $B \le A$ then $|(B \cap C)/\delta| = 1$ or $(B \cap C)/\delta = B/\delta$; and if $|(B \cap C)/\delta| = 1$ then $\delta \supseteq \delta_1 \cap \cdots \cap \delta_s$, where $\delta_1, \ldots, \delta_s$ are the congruences of the *chosen* chains witnessing $B \lhd\lhd\lhd A$ and $C \lhd\lhd\lhd A$;
- (s) if $G <_{\textsc{ba,c}} B$ then $G \cap C \neq \varnothing$.

Proof outline, with the delicate steps in full. Fix chains $B = B_k < \cdots < B_0 = A$ and $C = C_\ell < \cdots < C_0 = A$, put $\sigma = \bigcap_i \sigma_i$ and $\omega = \bigcap_j \omega_j$, and induct on $k + \ell$. Every appeal to the inductive hypothesis lands at strictly smaller $k+\ell$: (s) uses (s) and (d) at $k+(\ell-1)$, subcase 2B2 uses (s) at $(n-1)+k$, and subcase 2B3 uses (d) at $(n-1)+k$, with $n \le \ell$ throughout.

(s). Split on the type $\mathcal{T}_\ell$ of the last step of C's chain. In the case $\mathcal{T}_\ell = \textsc{d}$ the natural move is to derive $(G \cap C_{\ell-1})/\omega_\ell <_{\textsc{ba,c}} C_{\ell-1}/\omega_\ell$ and contradict Definition 4.21(iii). But Lemmas 7.14 and 7.6 supply only $\le$, so the case must be split: if the containment is an equality then every

ω_ℓ-block meeting $C_{\ell-1}$ meets $G \cap C_{\ell-1}$, in particular the block E with $C_\ell = C_{\ell-1} \cap E$, so $G \cap C_\ell \neq \varnothing$, which is the conclusion; and if it is proper the contradiction stands.

(d). Put $S = \{(a_1/\delta, \dots, a_{|A|}/\delta) : \forall i,j\ (a_i, a_j) \in \sigma \cap \omega\}$, a reflexive, totally symmetric, shift-closed subuniverse of $(\mathbf{A}/\delta)^{|A|}$, and apply Lemma 7.27.

Case 1: S is the diagonal. Then $\sigma \cap \omega \subseteq \delta$, which is the "moreover" clause; and any $a, b \in B \cap C$ satisfy $(a,b) \in \sigma \cap \omega$, so $|(B \cap C)/\delta| = 1$.

Case 2: $(B/\delta)^{|A|} \subseteq S$. Refine S to $S_{m,n}$ by requiring $a_i \in B_m \cap C_n$, and take the least m (subcase 2A) or least n (subcase 2B) at which containment fails, splitting on the type of the corresponding step. Three steps in that split deserve to be written out. *Properness*: in 2A1 and 2B1 the containments $S_{m,0} \cap (B/\delta)^{|A|} <_{T_m} (B/\delta)^{|A|}$ are proper by the minimal choice of m, and nonempty because the constant tuples $(b/\delta, \dots, b/\delta)$ for $b \in B$ lie in $S_{m,0}$, since $B \subseteq B_m$ and $(b,b) \in \sigma \cap \omega$. *Which block*: in 2B3 the inductive hypothesis offers "size 1 or all of $C_{\ell-1}/\omega_n$", and the stronger reading "$= C_n/\omega_n$" used there is correct because $B \cap C \neq \varnothing$ and $C \subseteq C_n$ force $B_k \cap C_n \neq \varnothing$, so the single block is the one defining C_n. *Full fibres*: the appeals in 2A3 and 2B3 to a d with $(B/\delta)^{|A|} \times \{d\} \subseteq R'$ are Lemma 7.28. Subcase 2C survives and gives $(B \cap C)/\delta = B/\delta$. □

Caveat 7.32 (What remains to be expanded)**.** The argument above is written at the level of its case structure, and is therefore *not yet a proof*: the twelve subcases of (d) remain to be expanded to the level of the rest of this section. This is the largest single piece of writing outstanding, and much of §7.4 onwards depends on it. See §11.

One loose end in the statement rather than the proof: minimality of the chains for B and C is hypothesised and no step appears to consume it — a shorter chain only makes the induction measure smaller. Either the use should be identified or the hypothesis dropped.

Corollary 7.33. *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, $B_1 \lhd\!\lhd\!\lhd A_1$, $B_2 \lhd\!\lhd\!\lhd A_2$, and let σ be a dividing congruence for $B_1 \lhd\!\lhd\!\lhd A_1$. Then $\mathrm{pr}_1(R \cap (B_1 \times B_2))/\sigma$ is empty, of size 1, or equal to B_1/σ.*

Proof. Let $f_i : \mathbf{R} \to \mathbf{A}_i$ be the projections. By Lemma 6.4(b), $f_i^{-1}(B_i) \lhd\!\lhd\!\lhd \mathbf{R}$; apply Lemma 7.31(d) inside $\mathbf{R}$ to these two, with the congruence $f_1^{-1}(\sigma)$, and read the conclusion back through f_1. □

Caveat 7.34. The alternative "empty" is present in Corollary 7.33 and absent from Lemma 7.31, which hypothesises $B \cap C \neq \varnothing$. It is a real alternative, and every call site below has to exclude it separately.

Lemma 7.35 (Parallelogram property for D)**.** *Let $B_1, B_2 \lhd\!\lhd\!\lhd A$, $C_1, C_1' <^A_{\mathrm{D}(\sigma_1)} B_1$ and $C_2, C_2' <^A_{\mathrm{D}(\sigma_2)} B_2$ with $C_1' \cap C_2$, $C_1 \cap C_2'$ and $C_1' \cap C_2'$ all nonempty. Then $C_1 \cap C_2 \neq \varnothing$.*

Proof. By Lemma 7.31(d), $(B_1 \cap B_2)/\sigma_1$ is of size 1 or all of B_1/σ_1. In the first case the single block is the one defining C_1, because $C_1 \cap C_2' \neq \varnothing$ puts a point of C_1 in $B_1 \cap B_2$; hence $B_1 \cap B_2 \subseteq C_1$ and $C_1 \cap C_2 = B_1 \cap C_2 \supseteq C_1' \cap C_2 \neq \varnothing$. The symmetric argument applies if $(B_1 \cap B_2)/\sigma_2$ has size 1.

So assume $(B_1 \cap B_2)/\sigma_1 = B_1/\sigma_1$ and $(B_1 \cap B_2)/\sigma_2 = B_2/\sigma_2$, and put $S = \{(a/\sigma_1, a/\sigma_2) : a \in B_1 \cap B_2\}$, which is then subdirect in $(B_1/\sigma_1) \times (B_2/\sigma_2)$. Apply Lemma 7.31(d) to each σ_1-class C of B_1 against B_2 with the congruence σ_2; the hypothesis $C \cap B_2 \neq \varnothing$ holds because S is subdirect. Each fibre of S is therefore a single point or the whole of B_2/σ_2.

If some fibre is full, then either S is full — and then the fibre of C_1/σ_1 contains C_2/σ_2, giving $C_1 \cap C_2 \neq \varnothing$ — or the centre of S is proper and nonempty, and Lemma 7.19 produces a proper nonempty binary absorbing or central subuniverse of B_1/σ_1 or B_2/σ_2, contradicting Definition 4.21(iii). If every fibre is a single point, then the fibre of C_1'/σ_1 is both C_2/σ_2 (witnessed by $C_1' \cap C_2$) and C_2'/σ_2 (witnessed by $C_1' \cap C_2'$), so $C_2 = C_2'$ and $C_1 \cap C_2 = C_1 \cap C_2' \neq \varnothing$. □

7.4 Bridges

Three of the statements below — Lemmas 7.36, 7.38 and 7.39 — require their bridge to be *pair-reflexive*, and the hypothesis is not cosmetic in any of them: Remark 7.37 exhibits a bridge satisfying every other hypothesis of the first and none of its conclusion. Lemma 3.32 is what makes the hypothesis free, since every bridge these statements are applied to below is a δ°.

Lemma 7.36. *Let σ be a congruence on $\mathbf{A}$ and δ a symmetric,* pair-reflexive *bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta) = \widetilde{\delta} = A^2$. Then there is an abelian group $(G; +, -)$ with $(A/\sigma; \delta/\sigma) \cong (G; x_1 - x_2 = x_3 - x_4)$.*

Proof. Factor by σ (Lemma 3.24) and assume $\sigma = 0_{\mathbf{A}}$. By Lemma 7.13 and Lemma 7.12, $\mathbf{A}$ is affine, hence polynomially equivalent to a $\mathbb{Z}_p$-module $\mathbf{G}$ and Maltsev via $x - y + z$.

View δ as a binary relation on $\mathbf{A}^2$ as in the proof of Lemma 7.13. It is reflexive, by pair-reflexivity together with $\mathrm{pr}_{1,2}(\delta) = A^2$, and symmetric by hypothesis. In a Maltsev algebra a reflexive compatible binary relation is a congruence: if η is reflexive and compatible then for $(x,y),(y,z) \in \eta$ the Maltsev term applied to $(x,y),(y,y),(y,z)$ gives $(x,z) \in \eta$. So δ is a congruence of $\mathbf{G}^2$.

Clause (d) of Definition 3.26 says that a pair $\big((x_1,x_2),(x_3,x_4)\big)$ of δ has $x_1 = x_2$ exactly when $x_3 = x_4$, so the diagonal of $\mathbf{G}^2$ is a δ-block. A congruence of $\mathbf{G}^2$ with the diagonal as a block is the kernel of $(x,y) \mapsto x - y$ composed with a quotient of $\mathbf{G}$; since the diagonal is exactly one block, no further collapsing occurs, and $\delta = \{x_1 - x_2 = x_3 - x_4\}$. Taking $G = \mathbf{G}$ finishes. ☐

Remark 7.37 (Pair-reflexivity is not removable)**.** Without pair-reflexivity the correct conclusion is $\delta = \{(x_1,x_2,x_3,x_4) : \varphi(x_1 - x_2) = x_3 - x_4\}$ for an automorphism φ of G: factoring out the diagonal exhibits δ as the preimage of a subgroup $K \le G \times G$ that meets $G \times 0$ and $0 \times G$ trivially and is subdirect, hence a graph. Symmetry says $\varphi^2 = \mathrm{id}$; pair-reflexivity says $\varphi = \mathrm{id}$.

Both φ occur. On $\mathbf{Z}_3 \in \mathcal{V}_4$ — the operation $x_1 + x_2 + x_3 + x_4$ is idempotent since $4 \equiv 1$, and is a special WNU — take $\sigma = 0$ and

$$\delta = \{(x_1,x_2,x_3,x_4) \in \mathbb{Z}_3^4 : x_1 - x_2 + x_3 - x_4 = 0\},$$

which is $\varphi = -\mathrm{id}$. It is a subuniverse of $\mathbf{A}^4$, being a subgroup; clause (d) holds because $x_1 = x_2$ iff $x_3 = x_4$; $\mathrm{pr}_{1,2}(\delta) = \widetilde{\delta} = A^2$; and it is symmetric. Yet no bijection $f \colon \mathbb{Z}_3 \to \mathbb{Z}_3$ carries δ to $\{x_1 - x_2 = x_3 - x_4\}$: such an f would have to satisfy $f(x_1) - f(x_2) = f(x_3) - f(x_4)$ whenever $x_1 - x_2 + x_3 - x_4 = 0$, and putting $(x_1,x_2,x_3,x_4) = (0,b,b,0)$ gives $2(f(0) - f(b)) = 0$, hence $f(0) = f(b)$ for every b since p is odd — so f is constant, not a bijection. Since every abelian group of order 3 is $\mathbb{Z}_3$, no G works, and Lemma 7.36 genuinely fails without its hypothesis.

Lemma 7.38. *Let δ be a bridge from $0_{\mathbf{A}}$ to $0_{\mathbf{A}}$ with $\mathrm{pr}_{1,2}(\delta) \subseteq \widetilde{\delta}$ and $\mathrm{pr}_{1,2}(\delta)$ linked. Then $\mathbf{A}$ is BA and centre free.*

Proof outline, with the delicate steps in full. Replace δ by $\sigma := \delta^\circ$ of Lemma 3.32 and put $\omega = \mathrm{pr}_{1,2}(\sigma) = \mathrm{pr}_{1,2}(\delta)$. Note first that $\widetilde{\delta}$ is *reflexive*: clause (c) of Definition 3.26 gives $0_A \subseteq \mathrm{pr}_{1,2}(\delta)$, and the hypothesis gives $\mathrm{pr}_{1,2}(\delta) \subseteq \widetilde{\delta}$. Hence δ is a reflexive bridge, so Lemma 3.32 applies and gives $\omega \subseteq \widetilde{\delta} \subseteq \widetilde{\sigma}$ with $\widetilde{\sigma}$ symmetric, which is what legitimises the "without loss of generality" below. Induct on $|A|$.

Suppose $B <_T \mathbf{A}$ with $T \in \{\textsc{ba}, \textsc{c}\}$. Since ω is linked, ω meets $(A \setminus B) \times B$ or $B \times (A \setminus B)$; say the first, and pick (a,b) there.

Case 1: some subalgebra $D \lneq A$ contains a and b. Put $\rho = (\omega \circ \omega^{-1}) \cap D^2$, which is a subalgebra of $\mathbf{D}^2$, reflexive because ω is reflexive and symmetric by construction. Its transitive closure is again a subalgebra, hence a congruence on $\mathbf{D}$; let E be the block containing a and b, a subuniverse. Because ω is reflexive, every undirected ω-edge is a ρ-edge and every ρ-edge is a two-edge undirected ω-path, so ρ and the undirected ω-graph have the same connected components; hence $\omega \cap E^2$ is linked, and a, b lie in one component. *Note that $(\omega \cup \omega^{-1}) \cap D^2$ would not serve: a union of two subuniverses need not be closed.* Moreover $\mathrm{pr}_{1,2}(\sigma \cap E^4) = \omega \cap E^2$: the inclusion $\subseteq$ is trivial and $\supseteq$ holds because pair-reflexivity puts (x_1, x_2, x_1, x_2) in σ for every $(x_1, x_2) \in \omega$. So the inductive hypothesis applies to $\sigma \cap E^4$ and makes E BA and centre free, while Lemma 7.14 makes $B \cap E$ a proper nonempty subuniverse of E of type T — proper because $a \in E \setminus B$, nonempty because $b \in B \cap E$. Contradiction.

Case 2: no such D. Then $\{a\} \circ \omega = A$, because $\{a\}$ is a subuniverse by idempotence, hence so is $\{a\} \circ \omega$, and it contains a and b. Put $\xi(x_1, x_2) = \sigma(a, x_1, x_2, b)$; from $(a, b, a, b), (a, a, b, b) \in \sigma$ both projections of ξ contain a and b, hence equal A. Put $C = B \circ \xi$, so $C \leq_T \mathbf{A}$ by Lemma 7.4, with $a \in C$ and $b \notin C$ — the latter because $b \in C$ would give $(a, c, b, b) \in \sigma$ with $c \in B$, and clause (d) would force $a = c \in B$.

If $T = \textsc{ba}$, applying the absorbing term to (a, b, a, b) and (a, a, b, b) gives $(a, b_1, b_2, b) \in \sigma$ with $b_1, b_2 \in B$, so $C \cap B \neq \varnothing$. If $T = \textsc{c}$, the three tuples (a, b, a, b), (a, a, a, a) and (a, a, b, b) witness

$$\sigma \cap (\{a\} \times A \times C \times B), \quad \sigma \cap (\{a\} \times C \times C \times A), \quad \sigma \cap (\{a\} \times C \times A \times B) \neq \varnothing,$$

which is hypothesis (4) of Corollary 6.15 for the ternary relation $\{(x_2, x_3, x_4) : \sigma(a, x_2, x_3, x_4)\}$ with $n = 3$, all types $\textsc{c}$ and all ambients A — legitimate because $\lhd\lhd\lhd$ is reflexive. *The three tight boxes of these witnesses — $\{a\} \times B \times C \times B$, $\{a\} \times C \times C \times C$, $\{a\} \times C \times B \times B$ — are subsets of the displayed ones and do not match the corollary's shape; the widening is what makes the appeal legitimate.* Since $n = 3$ and all types are $\textsc{c}$, every alternative of the corollary fails, so hypothesis (3) must fail: $\sigma \cap (\{a\} \times C \times C \times B) \neq \varnothing$. Put $C' = \mathrm{pr}_2$ of that set. Then $C' \leq_{\textsc{c}} A$ by Lemma 7.4, using $\{a\} \circ \omega = A$, and hence $C' \leq_{\textsc{c}} C$ by Lemma 7.14; and by clause (d) either $a \notin C'$ or $C \cap B \neq \varnothing$.

So in every case we have $\varnothing \neq C' <_T C <_T A$ — taking $C' = B \cap C$ when $C \cap B \neq \varnothing$. Then $|C| > 1$, and applying the inductive hypothesis to $\sigma \cap C^4$ contradicts $C' <_T C$: its hypotheses hold because $\{a\} \circ \omega = A$ makes $\omega \cap C^2$ a star centred at a, hence linked. □

Lemma 7.39. Let σ be an irreducible congruence on $\mathbf{A}$ and δ a *pair-reflexive* bridge from σ to σ with $\widetilde{\delta} \supsetneq \sigma$. Then

(a) σ^* is a congruence;
(b) B/σ is BA and centre free for every block B of σ^*;
(c) if δ is also symmetric then there is a prime p such that every block B of σ^* satisfies $(B/\sigma; (\delta \cap B^4)/\sigma) \cong (\mathbb{Z}_p^{n_B}; x_1 - x_2 = x_3 - x_4)$ for some $n_B \geq 0$.

Proof. By Lemma 3.24 we may assume $\sigma = 0_{\mathbf{A}}$. Since $\widetilde{\delta}$ is a subuniverse of $\mathbf{A}^2$ properly containing σ and stable under it, minimality gives $\sigma^* \subseteq \widetilde{\delta}$; the same argument applied to $\mathrm{pr}_{1,2}(\delta)$, which properly contains σ by clause (c), gives $\sigma^* \subseteq \mathrm{pr}_{1,2}(\delta)$.

Let B be a block of $\mathrm{LeftLinked}(\sigma^*)$ with $|B| > 1$ and put $\delta' = \delta \cap B^4 \cap (\sigma^* \times \sigma^*)$. Then $\mathrm{pr}_{1,2}(\delta') = \sigma^* \cap B^2$: for $(x_1, x_2) \in \sigma^* \cap B^2$ pair-reflexivity puts (x_1, x_2, x_1, x_2) in δ — legitimate because $\sigma^* \subseteq \mathrm{pr}_{1,2}(\delta)$ — and that tuple lies in B^4 and in $\sigma^* \times \sigma^*$ — *this is where pair-reflexivity is spent, and without it the projection can collapse to 0_B*. The resulting relation is linked precisely because B is a block, and is contained in $\widetilde{\delta'}$ because $\sigma^* \subseteq \widetilde{\delta}$; so Lemma 7.38 makes B BA and centre free. Then $\sigma^* \cap B^2$, being linked and subdirect on B, must be all of B^2 by Corollary 7.2. Blocks of size 1 satisfy $B^2 \subseteq \sigma^*$ trivially, so $\sigma^* = \mathrm{LeftLinked}(\sigma^*)$ is an equivalence relation and hence a congruence, giving (a); and (b) follows, one-element algebras being vacuously BA and centre free.

For (c), fix a block B of σ^* with $|B| \geq 2$ and apply Lemma 7.36 to $\delta \cap B^4$, whose hypotheses now hold: $\mathrm{pr}_{1,2}(\delta \cap B^4) = B^2$ by the same pair-reflexivity computation, and $\widetilde{\delta \cap B^4} = B^2$ because $B^2 \subseteq \sigma^* \subseteq \widetilde{\delta}$. This gives an abelian group G_B with $(B; \delta \cap B^4) \cong (G_B; x_1 - x_2 = x_3 - x_4)$.

It remains to make every G_B elementary abelian for one common prime. Put $\omega = \delta \cap (\sigma^* \times \sigma^*)$, which has the same properties. From $x_1 - x_2 = x_3 - x_4$ one pp-defines $(k+1)x_1 = k\,x_2 + x_3$ for every k, by

$$((k+1)x_1 = k\,x_2 + x_3) = \exists x_4\ (k\,x_1 = (k-1)x_2 + x_4) \wedge (x_1 - x_2 = x_3 - x_4),$$

and hence, identifying x_3 with x_1, the relation $q\,x_1 = q\,x_2$ for every q. Let S be what that pp-definition defines when $x_1 - x_2 = x_3 - x_4$ is replaced by ω. If some G_B had an element of order p_1^2, or elements of coprime orders, or if two blocks had elements of different prime orders, then for a suitable q the relation S would be a subuniverse of $\mathbf{A}^2$ with $S \supsetneq \sigma$ and $\sigma^* \not\subseteq S$, contradicting the minimality of σ^*. Hence every element of every G_B has one and the same prime order p, and $G_B \cong \mathbb{Z}_p^{n_B}$. ☐

Caveat 7.40 (It is minimality, not irreducibility)**.** The contradiction at the end of the proof is with the *minimality of* σ^*, which is a consequence of irreducibility and not the same statement. The distinction matters wherever σ^* is treated as data attached to a proof of irreducibility (Caveat 3.21).

Proof of Lemma 4.12. $(a) \Rightarrow (b)$: take $\delta = S$ of Definition 4.10, which by Lemma 4.11 is a bridge with $\widetilde{S} = \sigma^* \supsetneq \sigma$.

$(b) \Rightarrow (a)$: given a *reflexive* bridge δ with $\widetilde{\delta} \supsetneq \sigma$, form δ° of Lemma 3.32; it is symmetric and pair-reflexive, and $\widetilde{\delta^\circ} \supseteq \widetilde{\delta} \supsetneq \sigma$. Now Lemma 7.39 gives items (b) and (c) of Definition 4.10, with the witness $S = \delta^\circ \cap (\sigma^* \times \sigma^*)$. ☐

Caveat 7.41 (The reflexivisation is an open obligation)**.** The direction $(b) \Rightarrow (a)$ as just proved consumes a *reflexive* bridge, because Lemma 3.32 needs reflexivity of its input for two of its conclusions: that δ° is reflexive, and — what is actually used here — that $\widetilde{\delta^\circ} = \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ contains $\widetilde{\delta}$. Lemma 4.12(b) as stated supplies only a bridge.

Two routes close the gap and neither is taken here. Either prove a reflexivisation lemma — that a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$ may be replaced by a reflexive one with the same property — or restate (b) as "there is a *reflexive* bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$", which is what $(a) \Rightarrow (b)$ produces, since the witness of Lemma 4.11 *is* reflexive, and then check the two call sites. Until one is done, $(b) \Rightarrow (a)$ and the proof of Lemma 4.17, which makes the same move, are incomplete. See §11.

Lemma 7.42. Let σ be a congruence on $\mathbf{A}$ and δ a reflexive bridge from σ to σ with

- (i) $\delta(x_1, x_2, x_3, x_4) = \delta(x_3, x_4, x_1, x_2)$;
- (ii) $(a, b, a, b), (b, a, b, a) \in \delta$ for every $(a, b) \in \mathrm{pr}_{1,2}(\delta)$;
- (iii) $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\delta)) = A^2$.

Then $(a, a, b, b) \in \delta$ for some $a \neq b$.

Proof. Factor by σ and assume $\sigma = 0_{\mathbf{A}}$; clause (c) of Definition 3.26 then forces $|A| \geq 2$. Write $P = \mathrm{pr}_{1,2}(\delta)$, which is symmetric by (ii), and $W = \mathrm{pr}_{1,2,3}(\delta) \leq_{\mathrm{sd}} \mathbf{P} \times \mathbf{A}$. Induct on $|A|$.

Case 1: some $B <_T \mathbf{A}$ with $|B| > 1$, $T \in \{\textsc{ba}, \textsc{c}\}$. Put $W' = \mathrm{pr}_{1,2,3}(\delta \cap B^4)$ and write $W(x_1, x_2, x_3) = \exists x_4\, \delta(x_1, x_2, x_3, x_4) \wedge A(x_1) \wedge A(x_2) \wedge A(x_3) \wedge A(x_4)$ with A the full unary relation. Replacing A by B — a *single* application of Lemma 7.4, which replaces every occurrence, so no transitivity of $\leq_{\textsc{c}}$ is needed — gives $W' \leq_T W$. It is nonempty, containing

(b, b, b) for every $b \in B$, and its projections are $\mathrm{pr}_{1,2}(W') = P \cap B^2$, by (ii), and $\mathrm{pr}_3(W') = B$. So Lemma 7.16 makes W' linked, which is (iii) for $\delta \cap B^4$. Conditions (i) and (ii) are inherited and $\delta \cap B^4$ is reflexive. If it is a bridge, the inductive hypothesis applies and produces the required pair inside B. If it is not, then $\mathrm{pr}_{1,2}(\delta \cap B^4) = 0_B$, so by clause (d) every tuple of $\delta \cap B^4$ has the form (x, x, y, y); linkedness of W' on $|B| \geq 2$ points then forbids the perfect matching, so some $(a, a, b, b) \in \delta$ with $a \neq b$.

Case 2: some $\{a\} <_T \mathbf{A}$, $T \in \{\mathrm{BA}, \mathrm{C}\}$, *and no such* B *of size* > 1. By (i) the relation $\mathrm{pr}_{1,3}(\delta)$ is symmetric, and by (iii) it is linked, so $\{a\} \circ \mathrm{pr}_{1,3}(\delta) \neq \{a\}$; pick $(a, b, c, d) \in \delta$ with $c \neq a$. If $a = b$ then $c = d$ by clause (d) and we are done. If $\{a\} \circ \mathrm{pr}_{1,3}(\delta) \neq A$ then that set is a strong subuniverse containing a and c, so of size > 1, which is Case 1. Otherwise choose $(a, e, b, f) \in \delta$. By (ii), $(b, a, b, a) \in \delta$. Let g be a ternary absorbing operation for $\{a\}$, available for $T = \mathrm{C}$ by Lemma 4.4. Applying g to (a, e, b, f), (b, a, b, a) and (a, a, a, a) gives $(a, a, g(b, b, a), a) \in \delta$, whence $g(b, b, a) = a$ by clause (d); applying g to (b, a, b, a), (b, a, b, a) and (a, e, b, f) then gives $(a, a, b, a) \in \delta$, whence $b = a$, a contradiction.

Case 3: $\mathbf{A}$ *is BA and centre free.* Put $C = \{(a, b) \in P : \{(a, b)\} \times A \subseteq W\}$. By Lemma 7.21, applied to the transpose of W, $C \leq_{\mathrm{BA}} \mathbf{P}$. If $C = P$ then in particular $(a, a) \in C$ for any a; otherwise $C <_{\mathrm{BA}} P$ and Lemma 7.43 gives $C \cap 0_{\mathbf{A}} \neq \varnothing$. Either way $(a, a) \in C$ for some a, so for every c there is d with $\delta(a, a, c, d)$, and clause (d) forces $d = c$. Taking $c \neq a$ finishes the proof. ☐

Lemma 7.43 ([13, Lem. 7.2])**.** *If* $0_{\mathbf{A}} \subseteq \sigma \leq \mathbf{A}^2$ *and* $\omega <_{\mathrm{BA}} \sigma$, *then* $\omega \cap 0_{\mathbf{A}} \neq \varnothing$.

Proof of Lemma 4.15. Put $\xi = \delta * \delta^{-1}$, which is symmetric and pair-reflexive. Each of $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\xi))$ and $\mathrm{RightLinked}(\mathrm{pr}_{1,2,4}(\xi))$ is a congruence containing σ, so by minimality of σ^* each is σ or contains σ^*; this gives three cases.

If both equal σ, then for $(a, b, c, d) \in \xi$ the classes c/σ and d/σ are determined by $a/\sigma, b/\sigma$; since $(a, b, a, b) \in \xi$ we get $(a, c), (b, d) \in \sigma$, so ξ is trivial. *Transferring that determinacy from* $\xi = \delta * \delta^{-1}$ *to* δ *itself is a counting argument, and it is where the hypothesis* $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta)$ *is spent:* on σ^*/σ, which is finite, δ induces a bipartite relation whose left neighbourhoods are nonempty, pairwise disjoint — that is what the case gives — and cover the right side; equal finite cardinalities force every neighbourhood to be a singleton, so the induced relation is a bijection and determinacy holds in both directions.

With that, introduce

$$
\begin{aligned}
\zeta_1(x_1, x_2, x_3, x_4) = {} & \exists y \exists y' \exists z \exists z' \exists t_1 \exists t_2 \exists t_3 \exists t_4 \\
& \delta(x_1, y, z, t_1) \wedge \delta(x_2, y, z', t_2) \wedge \delta(x_3, y', z, t_3) \wedge \delta(x_4, y', z', t_4), \\
\zeta_2(x_1, x_2, x_3, x_4) = {} & \exists y \exists z \;\; \delta(x_1, y, z, x_3) \wedge \delta(x_2, y, z, x_4),
\end{aligned}
$$

and pick $(a, b, c, d) \in \delta$ with $(a, b) \in \sigma^* \setminus \sigma$. From $(a, b, c, d), (b, b, b, b) \in \delta$ one gets $(a, b, a, b) \in \zeta_1$ and hence $\mathrm{pr}_{1,2}(\zeta_1) \supseteq \sigma^*$. Split.

If ζ_1 *is a bridge*, then $\widetilde{\zeta_1} = \sigma$ — forced, since σ is PC — and $(a, a, c, c) \in \zeta_1$ for every $(a, b, c, d) \in \delta$, so $\mathrm{pr}_{1,3}(\delta) = \sigma$.

If ζ_1 *is not a bridge*, there is $(a, a, b, c) \in \zeta_1$ with $(b, c) \notin \sigma$. Take witnesses $y = d$, $y' = d'$, $z = e$, $z' = e'$, $t_i = f_i$. Determinacy makes $(e, e') \in \sigma$, so $(b, d', e, f_3), (c, d', e, f_4) \in \delta$ and therefore $\mathrm{pr}_{1,2}(\zeta_2) \supsetneq \sigma$; determinacy in both directions then makes ζ_2 a bridge, whence $\widetilde{\zeta_2} = \sigma$ and $\mathrm{pr}_{1,4}(\delta) = \sigma$.

So $\mathrm{pr}_{1,3}(\delta) = \sigma$ or $\mathrm{pr}_{1,4}(\delta) = \sigma$, and by the same argument with x_1, x_2 interchanged, $\mathrm{pr}_{2,4}(\delta) = \sigma$ or $\mathrm{pr}_{2,3}(\delta) = \sigma$. *Two of the four combinations must now be excluded:* $\mathrm{pr}_{1,3} = \mathrm{pr}_{2,3} = \sigma$ would give $\mathrm{pr}_{1,2}(\delta) \subseteq \sigma$, against $\mathrm{pr}_{1,2}(\delta) = \sigma^* \supsetneq \sigma$, and likewise for $\mathrm{pr}_{1,4} = \mathrm{pr}_{2,4}$. The two surviving combinations are the conclusion, the reverse inclusion following from stability under σ.

If $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\xi)) \supsetneq \sigma^*$, choose a block B of it that is not a σ-block. Then $a, b, d \in B$ whenever $c \in B$ and $(a, b, c, d) \in \xi$, so $\xi \cap (A^2 \times B \times A) \subseteq B^4$ and therefore $\mathrm{pr}_{1,2,3}(\xi \cap B^4) = \mathrm{pr}_{1,2,3}(\xi) \cap (A^2 \times B)$, which is linked. *If $\sigma^* \cap B^2 \supsetneq \sigma \cap B^2$ then $\xi \cap B^4$ is a bridge and Lemma 7.42 applies, contradicting PC-ness via Lemma 4.12. If not, the lemma cannot be cited — it is not a bridge — and the case must be inlined: clause (d) then puts both pairs of every tuple in σ, and linkedness on a B that is not a σ-block produces $(x_1, x_1, x_3, x_3) \in \xi$ with $(x_1, x_3) \notin \sigma$ directly.* The third case is the mirror image, using $\mathrm{pr}_{1,2,4}$. □

Caveat 7.44 (The third case)**.** The third case is genuinely the mirror of the second: $(a, b, a, b) \in \xi$ places b in the block, and $\sigma^* \subseteq \mathrm{RightLinked}$ places a and c. Writing it out is mechanical, and it is not written out here.

7.5 Types interaction

Lemma 7.45 ([13, Lem. 8.19])**.** *Let $\omega, \sigma_1, \sigma_2$ be congruences on $\mathbf{A}$ with $\omega \cap \sigma_1 = \omega \cap \sigma_2$ and $\omega \setminus \sigma_1 \neq \varnothing$. Then there is a bridge δ from σ_1 to σ_2 with $\widetilde{\delta} = \sigma_1 \circ \sigma_2$.*

Proof. Put $\delta(x_1, x_2, y_1, y_2) = \exists z_1 \exists z_2\ \sigma_1(x_1, z_1) \wedge \sigma_1(x_2, z_2) \wedge \omega(z_1, z_2) \wedge \sigma_2(z_1, y_1) \wedge \sigma_2(z_2, y_2)$. Clauses (a) and (b) are visible. For (d): if $(x_1, x_2) \in \sigma_1$ then $(z_1, z_2) \in \sigma_1 \cap \omega = \sigma_2 \cap \omega$, so $(y_1, y_2) \in \sigma_2$; *and conversely,* if $(y_1, y_2) \in \sigma_2$ then $(z_1, z_2) \in \omega \cap \sigma_2 = \omega \cap \sigma_1$, so $(x_1, x_2) \in \sigma_1$ — the direction the source omits, and the one that spends the other half of the hypothesis. For (c): $\sigma_1 \subseteq \mathrm{pr}_{1,2}(\delta)$ by taking $z_i = y_i = x_1$, and choosing $(a, b) \in \omega \setminus \sigma_1$ gives $(a, b, a, b) \in \delta$ with $(a, b) \notin \sigma_1$ and, since $\omega \setminus \sigma_1 = \omega \setminus \sigma_2$, also $(a, b) \notin \sigma_2$. Finally $\widetilde{\delta} = \sigma_1 \circ \sigma_2$: the inclusion $\supseteq$ by taking $z_1 = z_2$, using reflexivity of ω, and $\subseteq$ by reading off $(x, z_1) \in \sigma_1$, $(z_1, y) \in \sigma_2$. □

Lemma 7.46. Let $C_1 <^{A}_{T_1(\sigma_1)} B_1 \lhd\lhd\lhd A$ and $C_2 <^{A}_{T_2(\sigma_2)} B_2 \lhd\lhd\lhd A$ with $T_1, T_2 \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$, $C_1 \cap B_2 \neq \varnothing$, $B_1 \cap C_2 \neq \varnothing$ and $C_1 \cap C_2 = \varnothing$. Then $T_1 = T_2 \in \{\textsc{ba}, \textsc{c}, \textsc{d}\}$; and if $T_1 = T_2 = \textsc{d}$ there is a bridge δ from σ_1 to σ_2 with $\widetilde{\delta} = \sigma_1 \circ \sigma_2$.

Proof. No S. If $T_1 = \textsc{s}$, pick $S_1 \subseteq C_1$ with $S_1 <_{\textsc{ba},\textsc{c}} B_1$. Since $C_2 \lhd\lhd\lhd A$ — the chain for B_2 extended by one D step — and $B_1 \cap C_2 \neq \varnothing$, Lemma 7.31(s) gives $S_1 \cap C_2 \neq \varnothing$, hence $C_1 \cap C_2 \neq \varnothing$. Symmetrically for T_2.

Both absorbing. If $T_1, T_2 \in \{\textsc{ba}, \textsc{c}\}$ then by Lemma 7.14 $C_1 \cap B_2 \leq_{T_1} B_1 \cap B_2$ and $B_1 \cap C_2 \leq_{T_2} B_1 \cap B_2$; both are proper, since $C_1 \cap B_2 = B_1 \cap B_2$ would put $B_1 \cap C_2$ inside $C_1 \cap C_2 = \varnothing$. Their intersection is empty, so Lemma 7.23 gives $T_1 = T_2 \in \{\textsc{ba}, \textsc{c}\}$.

Mixed. Let $T_1 \in \{\textsc{ba}, \textsc{c}\}$ and $T_2 = \textsc{d}$; the mirror case $T_1 = \textsc{d}$, $T_2 \in \{\textsc{ba}, \textsc{c}\}$ is identical with the indices swapped, the hypotheses being symmetric. By Lemma 7.31(d), $(B_1 \cap B_2)/\sigma_2$ has size 1 or is all of B_2/σ_2. In the first case the single block is the one defining C_2, because $B_1 \cap C_2 \neq \varnothing$; then $B_1 \cap B_2 = B_1 \cap C_2$, so $C_1 \cap B_2 \subseteq C_1 \cap C_2 = \varnothing$, a contradiction. In the second, Lemmas 7.14 and 7.6 give $(C_1 \cap B_2)/\sigma_2 \leq_{T_1} B_2/\sigma_2$, and this is *proper*: equality would make the σ_2-block defining C_2 meet C_1, i.e. $C_1 \cap C_2 \neq \varnothing$. So B_2/σ_2 has a proper nonempty strong subuniverse, contradicting Definition 4.21(iii).

Both D. Let ω be the intersection of the dividing congruences of the chosen chains for $B_1 \lhd\lhd\lhd A$ and $B_2 \lhd\lhd\lhd A$. Apply Lemma 7.31(d) to B_1 and C_2 with the congruence σ_1 — *to C_2, not to B_2; against B_2 the step does not follow.* Its second alternative, $(B_1 \cap C_2)/\sigma_1 = B_1/\sigma_1$, would put a point of C_2 in the σ_1-block defining C_1; so $|(B_1 \cap C_2)/\sigma_1| = 1$ and the "moreover" clause gives $\sigma_1 \supseteq \omega \cap \sigma_2$, because the chain witnessing $C_2 \lhd\lhd\lhd A$ is the chain for B_2 extended by the step $C_2 <_{\textsc{d}(\sigma_2)} B_2$ (Caveat 4.25). Symmetrically $\sigma_2 \supseteq \omega \cap \sigma_1$, so $\omega \cap \sigma_1 = \omega \cap \sigma_2$. Finally, for $c_1 \in C_1 \cap B_2$ and $c_2 \in B_1 \cap C_2$ we have $(c_1, c_2) \in \omega$, since both lie in $B_1 \cap B_2$ and each B_i sits inside a single block of every dividing congruence of its own chain; and $(c_1, c_2) \notin \sigma_1$, since $c_2 \in B_1$ and $c_2\, \sigma_1\, c_1 \in C_1$ would give $c_2 \in C_1 \cap C_2$. Now Lemma 7.45 applies. □

7.6 Factorisation

Lemma 7.47. *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$ with $R \cap (B_1 \times B_2) \neq \varnothing$, $B_i \lhd\lhd\lhd A_i$, σ a congruence on $\mathbf{A}_1$ with B_1/σ BA and centre free. If some $c \in A_2$ has $(E \times \{c\}) \cap R \neq \varnothing$ for every $E \in B_1/\sigma$, then such a c may be chosen in B_2.*

Proof outline. If c may already be chosen in B_2 there is nothing to do; otherwise walk down a chain for $B_2 \lhd\lhd\lhd A_2$ and let $B_2'' <_{T(\delta)} B_2'$ be the first step at which the choice fails, so $B_2 \subseteq B_2''$. Put S' and S'' for the sets of σ-class tuples of length $|A_1|$ witnessed by some c in B_2', resp. B_2''. A single good $c \in B_2'$ makes $S' = (B_1/\sigma)^{|A_1|}$, and $S'' \neq \varnothing$ because $R \cap (B_1 \times B_2) \neq \varnothing$ and $B_2 \subseteq B_2''$ — *this is where that hypothesis is spent.* For $T \in \{\textsc{ba}, \textsc{c}\}$, Lemmas 7.4 and 7.8 then put a strong subuniverse on B_1/σ, a contradiction; for $T = \textsc{s}$ the same follows once $R \cap (B_1 \times D_2) \neq \varnothing$, which Lemma 7.31(s) supplies inside $\mathbf{R}$.

For $T = \textsc{d}$ one builds $S \leq (B_1/\sigma)^{|A_1|} \times B_2'/\delta$ and finds a full column over $d = c/\delta$, while $(B_1/\sigma)^{|A_1|} \times B_2''/\delta \nsubseteq S$ by minimality. Corollary 7.33 then computes $\mathrm{pr}_{|A_1|+1}(S)$. It offers three alternatives and only the third gives the contradiction. "Empty" is excluded by $R \cap (B_1 \times B_2) \neq \varnothing$. "Size one" is excluded thus: the single δ-class would have to be $[c]_\delta$, since c has an R-partner in B_1; and $B_2'' = B_2' \cap E$ for a δ-block E, so either $E = [c]_\delta$, putting c in B_2'' against the minimal choice of B_2', or $E \cap [c]_\delta = \varnothing$, making $R \cap (B_1 \times B_2'') = \varnothing$ against $B_2 \subseteq B_2''$. So S is a central relation, and Lemmas 7.19 and 7.8 contradict the hypotheses. □

Lemma 7.48. *Under the hypotheses of Lemma 7.47, if* $(\mathrm{LeftLinked}(R) \cap B_1^2)/\sigma = (B_1/\sigma)^2$ *then* $\mathrm{LeftLinked}\big(R \cap (B_1 \times B_2)\big)/\sigma = (B_1/\sigma)^2$.

Proof outline. Write $R_n = (R \circ R^{-1})^n$, which is reflexive and symmetric because R is subdirect, so $R_{2n} = R_n \circ R_n^{-1}$ — *this is why the source restricts to $n = 2^k$.* If $(R_1 \cap B_1^2)/\sigma$ is already full, Lemma 7.21 applied to $S = \{(a/\sigma, b) : a \in B_1, (a, b) \in R\}$, subdirect over $(B_1/\sigma) \times \mathrm{pr}_2(S)$ rather than over $(B_1/\sigma) \times A_2$, produces the required c, which Lemma 7.47 moves into B_2; two classes are then linked through c. Otherwise take the largest $n = 2^k$ at which fullness fails, apply Lemma 7.21 to R_n, push c into B_1, and conclude by Lemma 7.19 that $(R_n \cap B_1^2)/\sigma$ is full after all. □

Lemma 7.49. Let σ be a dividing congruence for $B \lhd\lhd\lhd A$, let δ be a congruence on $\mathbf{A}$ with $(\delta \cap B^2)/\sigma \neq B^2/\sigma$, and let ω be the intersection of the dividing congruences of the chosen chain for $B \lhd\lhd\lhd A$. Then (1) $\delta \cap B^2 \subseteq \sigma \cap B^2$; (2) $\sigma \supseteq \delta \cap \omega$; (3) $(\delta \vee (\sigma \cap \omega)) \cap B^2 = \sigma \cap B^2$; (4) $(\delta \vee (\sigma \cap \omega)) \cap \omega = \sigma \cap \omega$.

Proof. Throughout we use $B^2 \subseteq \omega$, which holds because each B_i in the chain lies inside a single block of its own dividing congruence, and B is contained in every B_i. *It is used at least four times below.*

(2). The hypothesis gives σ-classes C_1, C_2 meeting B with $\big((C_1 \cap B) \times (C_2 \cap B)\big) \cap \delta = \varnothing$. Then $C_1 \cap B <_{\textsc{d}(\sigma)} B$, and Lemma 7.30 with $m = k$ and $B_k = C_1 \cap B$ gives $\big((C_1 \cap B) \circ \delta \cap B\big)/\sigma = \{C_1\}$, since the value B/σ is excluded by the choice of C_2; its second clause is $\sigma \supseteq \delta \cap \omega$.

(1) follows, since $\delta \cap B^2 \subseteq \delta \cap \omega \subseteq \sigma$.

(4). Put $\delta' = \delta \vee (\sigma \cap \omega)$, which is $\mathrm{LeftLinked}(\delta \circ (\sigma \cap \omega))$. If $(\delta' \cap B^2)/\sigma$ were full then Lemma 7.48 would make LeftLinked of the restricted relation full, so some single linking step would cross σ-classes: there would be $a_1, a_2, c \in B$ and $b_1, b_2 \in A$ with $(a_i, b_i) \in \delta$, $(b_i, c) \in \sigma \cap \omega$ and $(a_1, a_2) \notin \sigma$. But $a_1, a_2, c \in B$ gives $(a_i, c) \in \omega$, hence $(a_i, b_i) \in \omega$, hence $(a_i, b_i) \in \sigma$ by (2), hence $(a_1, a_2) \in \sigma$ — a contradiction. So $(\delta' \cap B^2)/\sigma \neq (B/\sigma)^2$, and applying (2) to δ' gives $\sigma \supseteq \delta' \cap \omega$; with $\delta' \supseteq \sigma \cap \omega$ this is (4).

(3) follows from (4) by intersecting with $B^2 \subseteq \omega$. □

Lemma 7.50. *Let $B \lhd\lhd\lhd A$ and let σ be a congruence on $\mathbf{A}$ with $|B/\sigma| > 1$ and B/σ BA and centre free. Then there is a dividing congruence $\delta \supseteq \sigma$ for $B \leq A$.*

Proof. Choose $\delta \supseteq \sigma$ maximal with $|B/\delta| > 1$; it exists because σ qualifies. Then B/δ is BA and centre free, since the preimage under $B/\sigma \twoheadrightarrow B/\delta$ of a proper nonempty strong subuniverse would be one of B/σ.

It remains to see that δ is irreducible. Suppose $\delta = S_1 \cap \cdots \cap S_k$ with each $S_i \supsetneq \delta$ a subuniverse of $\mathbf{A}^2$ *stable under* δ. Some S_i has $B^2 \not\subseteq S_i$, else $B^2 \subseteq \delta$. Now LeftLinked(S_i) is a congruence containing $S_i \supsetneq \delta \supseteq \sigma$, so maximality of δ forces $B^2 \subseteq$ LeftLinked(S_i), i.e. $(\text{LeftLinked}(S_i) \cap B^2)/\delta = (B/\delta)^2$. By Lemma 7.48, applied *with the congruence* δ, LeftLinked$(S_i \cap B^2)/\delta = (B/\delta)^2$, so $(S_i \cap B^2)/\delta$ is linked. It is also *proper*: stability under δ makes S_i a union of products of δ-blocks, so $(S_i \cap B^2)/\delta = (B/\delta)^2$ would give $B^2 \subseteq S_i$. Corollary 7.2 then produces a proper nonempty strong subuniverse of B/δ, a contradiction. □

Caveat 7.51 (Where each hypothesis is spent)**.** Two things in that proof are easy to lose. The relation must be factored by δ and not by σ — $(S_i \cap B^2)/\sigma$ lives on B/σ and the conclusion is about B/δ, and it is B/δ that was shown BA and centre free. And properness needs the words *stable under* δ in the unfolding of irreducibility: that clause is what makes S_i a union of products of δ-blocks, and it is the only part of Definition 3.20 doing work here. Without it, $S_i \cap B^2 \subsetneq B^2$ survives factoring only by accident.

Lemma 7.52. *Let δ be a congruence on $\mathbf{A}$ and $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd A$ with $T \in \{\text{PC}, \text{L}\}$. Then $C/\delta = B/\delta$, or $C/\delta <_{\text{S}} B/\delta$, or $C/\delta <^{A/\delta}_{T} B/\delta$; and in the last case $\delta \cap B^2 \subseteq \sigma \cap B^2$.*

Proof outline. Put $R = \{(a/\sigma, a/\delta) : a \in B\}$ and note $R \circ R^{-1} = (\delta \cap B^2)/\sigma$, so LeftLinked$(R) = (B/\sigma)^2$ exactly when $(\delta \cap B^2)/\sigma = B^2/\sigma$. In that case Lemma 7.21 yields $U \leq_{\text{BA,C}} B/\delta$ with $(B/\sigma) \times U \subseteq R$, and $U \subseteq C/\delta$, giving the first two alternatives.

Otherwise Lemma 7.49(4) gives $\sigma \cap \omega = \delta' \cap \omega$ for $\delta' = \delta \vee (\sigma \cap \omega)$, and $B/\delta' \cong B/\sigma$ by (3), so B/δ' is BA and centre free and Lemma 7.50 produces a dividing congruence $\delta'' \supseteq \delta'$ for B. Applying Lemma 7.49(2) to δ'' — legitimate because $\sigma \cap B^2 \subseteq \delta''$ and $|B/\delta''| > 1$ give $(\delta'' \cap B^2)/\sigma \neq B^2/\sigma$ — yields $\delta'' \cap \omega = \sigma \cap \omega$, hence $\delta'' \cap B^2 = \sigma \cap B^2$, so the δ''-classes and σ-classes agree on B and $C/\delta <^{A/\delta}_{\mathcal{T}(\delta''/\delta)} B/\delta$. Finally $\mathcal{T} = T$: $\omega \setminus \delta'' \neq \varnothing$, since $\omega \supseteq B^2$ and $|B/\sigma| > 1$, so Lemma 7.45 gives a bridge from δ'' to σ, and Lemma 4.14 in both directions makes the two congruences of the same type. The last clause holds because $\delta \cap B^2 \subseteq \delta' \cap B^2 = \sigma \cap B^2$. □

7.7 The headline properties

Proof of Lemma 6.2. If $\mathbf{B}$ has a proper nonempty binary absorbing or central subuniverse, take it. Otherwise $\mathbf{B}$ is BA and centre free and $|B| > 1$, so Lemma 7.50 with $\sigma = 0_{\mathbf{A}}$ gives a dividing congruence δ for B; take $C = B \cap E$ for any block E *of* δ with $B \cap E \neq B$, which exists because $|B/\delta| > 1$. Its type is L or PC according to δ. □

Proof of Lemma 6.12 and Corollary 6.13. (t). For $T \in \{\text{BA}, \text{C}\}$ this is Lemma 7.14. For $T = \text{S}$, take $E \subseteq C$ with $E <_{\text{BA,C}} B$; if $C \cap D = \varnothing$ there is nothing to prove, and otherwise $B \cap D \neq \varnothing$, so Lemma 7.31(s) gives $E \cap D \neq \varnothing$ and Lemma 7.14 gives $E \cap D <_{\text{BA,C}} B \cap D$. For $T \in \{\text{PC}, \text{L}\}$, Lemma 7.31(d) makes $(B \cap D)/\sigma$ empty, a point, or all of B/σ; the three cases give $C \cap D = \varnothing$, $C \cap D \in \{\varnothing, B \cap D\}$, and $C \cap D <_{T(\sigma)} B \cap D$ respectively.

(i). Apply (t) to each step of a chain for $B \lhd\lhd\lhd^{A} A$, obtaining $B \cap D \ \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A} D$. The corollary follows by composing with $D \lhd\lhd\lhd A$. □

Caveat 7.53 (The superscript in (i) is needed)**.** Corollary 6.13 is weaker than Lemma 6.12(i), and three later arguments need the stronger form: Corollary 6.9(b) and (m), which conclude $R'' \mathrel{\dot{\lhd}\dot{\lhd}\dot{\lhd}}^R R'$, and Lemma 6.18. Stating only the corollary — as is natural, since it is the symmetric-looking form — makes all three unprovable.

Proof of Lemma 6.4. (bt) is the reverse-homomorphism lemma and (b) follows by iteration. (fs) is Lemma 7.6. (ft) splits by type. For $T \in \{\text{BA}, \text{C}, \text{S}\}$ it follows from (fs). For $T \in \{\text{PC}, \text{L}\}$ it is Lemma 7.52; and for $T = \text{D}$, which that lemma does not cover, split the witnessing congruence into its linear and PC cases and apply the lemma to each, the two conclusions being the same three alternatives. Then (f) follows by iteration, all three outcomes of (ft) staying inside the type list of Definition 4.24.

(bt) is the reverse-homomorphism step, and it is proved directly rather than through Corollary 6.7, which is derived from it. For $T \in \{\text{BA}, \text{C}\}$ the witnessing term transfers: f is a homomorphism, so $f\big(t(\bar{a})\big) = t\big(f(\bar{a})\big)$, and a tuple with all but one entry in $f^{-1}(C')$ and the remaining entry in $f^{-1}(B')$ maps to one with all but one entry in C' and the remaining entry in B'; the centrality clause pulls back the same way, since $f\big(\text{Sg}(X)\big) = \text{Sg}\big(f(X)\big)$ for surjective f. For $T = \text{S}$ the witness $D' \subseteq C'$ pulls back to $f^{-1}(D') \subseteq f^{-1}(C')$ and the previous case applies. For $T \in \{\text{D}, \text{L}, \text{PC}\}$, $C' = B' \cap E$ for a block E of σ, so $f^{-1}(C') = f^{-1}(B') \cap f^{-1}(E)$ with $f^{-1}(E)$ a block of $f^{-1}(\sigma)$; condition (i) of Definition 4.21 transfers because $f^{-1}(\sigma^*) = f^{-1}(\sigma)^*$ by Lemma 3.24, and condition (iii) because $f^{-1}(B')/f^{-1}(\sigma) \cong B'/\sigma$.

(fm). Let $C = C_1 \cap \cdots \cap C_t$ with $C_i <^A_{T(\sigma_i)} B$ and induct on t. Write δ for the kernel of f, so that $f(X)$ may be read as X/δ throughout. For $t = 1$ this is (ft), and *the hypothesis that $f(B)$ is* S*-free is exactly what kills its middle alternative*, which $\mathcal{M}T$ cannot absorb. If $C_i/\delta = B/\delta$ for some i, say $i = t$, put $D_j = C_j \cap C_t$; by Lemma 6.12(t) $D_j \leq^A_{T(\sigma_j)} C_t$, and the inductive hypothesis applied to $C \leq_{\mathcal{M}T} C_t$ gives the claim. Otherwise every $C_i/\delta <^{A/\delta}_T B/\delta$, and $\delta \cap B^2 \subseteq \sigma_i \cap B^2$ for every i by the last clause of Lemma 7.52. That is what makes the images meet correctly: $(C_i \circ \delta) \cap B = C_i$, so a δ-class E lying in every C_i/δ has $E \cap B \subseteq C_1 \cap \cdots \cap C_t$, whence

$$(C_1 \cap \cdots \cap C_t)/\delta = C_1/\delta \cap \cdots \cap C_t/\delta,$$

the inclusion $\subseteq$ being trivial. This is the one place where the failure of images to commute with intersections (§2.1, item viii) is repaired, and the saturation hypothesis $\delta \cap B^2 \subseteq \sigma_i \cap B^2$ is precisely what buys it. (bm) follows from (bt) by intersecting. $\square$

Theorem 7.54 (Stable intersection, restated from Theorem 6.14)**.** Let $C_i <^A_{T_i(\sigma_i)} B_i \lhd\lhd\lhd A$ with $T_i \in \{\text{BA}, \text{C}, \text{S}, \text{L}, \text{PC}\}$ for $i \in [n]$, $n \geq 2$, and suppose $\bigcap_i C_i = \varnothing$ while $B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing$ for every j. Then all T_i are BA; or all are L and for all k, ℓ there is a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \sigma_\ell$; or $n = 2$ and $T_1 = T_2 = \text{C}$; or $n = 2$, $T_1 = T_2 = \text{PC}$ and $\sigma_1 = \sigma_2$.

Proof. For $n = 2$ combine Lemmas 7.46 and 4.14. For general n put $B_2' = B_2 \cap C_3 \cap \cdots \cap C_n$ and $C_2' = C_2 \cap B_2'$; then $C_2' <_{T_2(\sigma_2)} B_2' \lhd\lhd\lhd A$ by Lemma 6.12, properly because $C_2' = B_2'$ would make $C_1 \cap B_2'$ empty. The three hypotheses of the pair case hold, so $T_1 = T_2$ and, for the dividing types, the bridge exists.

If $T_1 = \text{PC}$ then $\sigma_1 \circ \sigma_2 = \sigma_1 = \sigma_2$. Indeed, composing the bridge with its transpose gives a bridge from σ_1 to itself, and if $\sigma_1 \circ \sigma_2 \supsetneq \sigma_1$ its collapse properly contains σ_1, making σ_1 linear by Lemma 4.12. That leaves the possibility of proper nesting, $\sigma_1 \circ \sigma_2 = \sigma_1$ with $\sigma_1 \neq \sigma_2$, which is excluded by Lemma 4.16: a bridge between PC congruences makes $\mathbf{A}/\sigma_1 \cong \mathbf{A}/\sigma_2$ with the induced relation on blocks bijective, and two congruences on one finite algebra with equal quotient sizes and $\sigma_2 \subseteq \sigma_1$ are equal. With $\sigma_1 = \sigma_2 =: \sigma$, the blocks E_1, E_2 defining C_1, C_2 are *distinct* — equal blocks would put the point of $B_1 \cap C_2'$ in $C_1 \cap C_2'$ — so $C_1 \cap C_2 = \varnothing$; applying this to every pair and then the hypothesis at $j = 3$ forces $n = 2$.

For $T_1 = \textsc{c}$ the argument just given has no analogue: there is no congruence and no pair of blocks. Suppose $n \geq 3$, put $D = \bigcap_i B_i$ and $E_i = C_i \cap D$. Then D is a nonempty subuniverse, containing $B_1 \cap C_2 \cap \cdots \cap C_n$; each $E_i \leq_{\textsc{c}} D$ by Lemma 7.14, and properly, since $E_j = D$ would give $\bigcap_i E_i = \bigcap_{i \neq j} E_i \neq \varnothing$. Moreover $\bigcap_i E_i = \varnothing$ and, because $C_i \subseteq B_i$,

$$\textstyle\bigcap_{i \neq j} E_i = D \cap \bigcap_{i \neq j} C_i = B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing.$$

Now the diagonal $R = \{(a, \ldots, a) : a \in D\} \leq D^n$ is a subuniverse by idempotence, and the two displays say exactly that R is $(E_1, \ldots, E_n)$-essential. By [15, Lem. 6.11] no such relation exists for $n \geq 3$ when the E_i are central. Hence $n = 2$. □

Caveat 7.55 (What the type-$\textsc{c}$ case costs)**.** The type-$\textsc{c}$ argument is the one place in this section that reaches outside the material developed here, to [15, Lem. 6.11]: *for $n \geq 3$ and E_i central in $\mathbf{D}$, there is no $(E_1, \ldots, E_n)$-essential relation $R \leq D^n$.* Two notes for anyone checking it. Its printed proof cites *itself* three times where it means the preceding lemma, so a self-contained rendering must supply both. And its ingredients are exactly the Barto–Kazda criterion (Lemma 7.5) and "central implies ternary absorbing" (Lemma 4.4), both of which are already available above.

Nonemptiness and properness of each E_i are established before the essential-relation theorem is applied, and both are needed: essentiality is a statement about proper nonempty subsets.

Proof of Corollary 6.15. Pull back along the projections $f_i : \mathbf{R} \to \mathbf{A}_i$ using Lemma 6.4(b),(bt), apply Theorem 6.14 inside $\mathbf{R}$, and translate $\widetilde{\delta} = \sigma'_k \circ \sigma'_\ell$ back as $\sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$. □

Proof of Lemma 6.18. With $C = \bigcap_{i \leq n} C_i$ and $D_j = \bigcap_{i \leq j} C_i$, Lemma 6.12(t) gives $D_{j+1} \leq_T^A D_j$, using $D_j \lhd\lhd\lhd A$ from part (i); collapsing the equalities gives the chain. □

Proof of Lemma 4.17. By Lemma 4.12 there is a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$; take it reflexive (Caveat 7.41) and replace it by δ°, symmetric and pair-reflexive. Lemma 7.39(c) applies, and $\sigma^* = A^2$ is a congruence with the single block A, so $A/\sigma \cong \mathbb{Z}_p^m$ as a group. Now $m = 0$ gives $\sigma = A^2$, impossible for an irreducible congruence (§2.1, item iv). And $m \geq 2$ contradicts irreducibility: A/σ is affine, so the basic operation acts as $\sum_i c_i x_i$ with $\sum_i c_i = 1$, and the coordinate congruences $\theta_i = \{(x, y) : x_i = y_i\}$ are linear subspaces of $(\mathbb{Z}_p^m)^2$, hence subuniverses of $(\mathbf{A}/\sigma)^2$, each properly containing $0_{\mathbf{A}/\sigma}$, with $\bigcap_i \theta_i = 0_{\mathbf{A}/\sigma}$ — so $0_{\mathbf{A}/\sigma}$ is reducible, and by Lemma 3.24 so is σ. Hence $m = 1$, and the group isomorphism is an isomorphism of algebras by Proposition 2.7. □

Lemma 7.56. *Let $C_1 <_{\mathcal{M}T}^A B_1 \lhd\lhd\lhd A$ with $T \in \{\textsc{pc}, \textsc{l}, \textsc{d}\}$, let $B_2 \lhd\lhd\lhd A$, $C_1 \cap B_2 = \varnothing$ and $B_1 \cap B_2 \neq \varnothing$. Then $(C_1 \circ (\omega_1 \cap \cdots \cap \omega_s)) \cap B_2 = \varnothing$, where $\omega_1, \ldots, \omega_s$ are all congruences of type T on $\mathbf{A}$ with $\omega_i^* \supseteq B_1^2$.*

Proof. Downward induction on $|B_1|$, starting at $B_1 = A$. Write $C_1 = \bigcap_i C_1^i$ with $C_1^i <_{\textsc{d}(\sigma_i)}^A B_1$, so each σ_i is among the ω_j and $C_1^i = (C_1^i \circ \sigma_i) \cap B_1$; by Corollary 6.8(e),(f), $C_1^i \circ \sigma_i <_{\textsc{d}(\sigma_i)}^A B_1 \circ \sigma_i \lhd\lhd\lhd A$. Since $\bigcap_i (C_1^i \circ \sigma_i) \cap B_1 \cap B_2 = C_1 \cap B_2 = \varnothing$, walk down a chain from A to B_1: either the intersection with B_2 is already empty at A, and then $C_1 \circ \bigcap_j \omega_j \subseteq \bigcap_i (C_1^i \circ \sigma_i)$ finishes; or there is a first step $B_1' <_{\textsc{d}} B_1''$ at which it becomes empty, and the inductive hypothesis at B_1'', together with $C_1 \subseteq B_1'$ and $\bigcap_j \omega_j \subseteq \sigma_i$, finishes. □

Caveat 7.57 (This is not yet a proof)**.** The argument above is not complete, and it should not be read as though it were. It does not fix one induction statement — the components are rewritten as type $\textsc{d}$ although the lemma tracks a fixed $T \in \{\textsc{pc}, \textsc{l}, \textsc{d}\}$ — it does not construct the object to which the inductive hypothesis is applied at the first step, and the two inclusions at the end are asserted rather than derived. This needs a rewrite, not polishing. See §11.

One thing the argument does *not* need is a theorem: the Case 1/Case 2 dichotomy is elementary, since the intersection is empty at B_1 by hypothesis, so either it is empty at A or there is a first step where it becomes so.

Caveat 7.58 (Lemma 6.20 is open)**. Lemma 6.20 is stated above and is not proved.** It is the one statement of §6 in that position, and nothing in this document may be built on it until it is settled. What is known:

Write $\bar{X}$ for X/σ and $C_1 = \bigcap_{j\le t} C_1^j$ with $C_1^j <^A_{T(\sigma_j)} B_1$. By Lemma 6.4(ft) each $\bar{C}_1^j$ is $\bar{B}_1$, or $<_\mathrm{s} \bar{B}_1$, or $<_T \bar{B}_1$. The target is to show that the type-T factors alone already cut $\bar{B}_2$ out; given that, the saturation clause of Lemma 7.52 applies to them, the intersection of their images is the image of their intersection, and Lemma 7.56 finishes with the ambient still $\bar{B}_1$, which is what the conclusion's congruence family $\omega_i^* \supseteq B_1^2$ requires.

Two things are established. First, the $<_\mathrm{s}$ factors never help: a factor with $\bar{C}_1^j <_\mathrm{s} \bar{B}_1$ contains a subuniverse $\bar{D}$ of $\bar{B}_1$ that is simultaneously binary absorbing and central, and Lemma 7.22 makes $\bar{D}$ meet $\bar{B}_1 \cap \bar{B}_2$, so $\bar{C}_1^j \cap \bar{B}_2 \neq \varnothing$. Second, one cannot simply pass to a minimal subfamily F of the $\bar{C}_1^j$ with $\bigcap F \cap \bar{B}_2 = \varnothing$ and read off the types from Theorem 6.14: the hypothesis gives $\big(\bigcap_j C_1^j\big)/\sigma \cap \bar{B}_2 = \varnothing$, whereas $\big(\bigcap_j C_1^j\big)/\sigma \subseteq \bigcap_j \bar{C}_1^j$ with the inclusion generally strict (§2.1, item viii), so no such F need exist. The saturation identity that would close that gap is available for a factor precisely when the factor lands in the type-T alternative, which is what was to be proved.

Nor can the configuration be attacked by descent. Passing to a BA-and-central $\bar{D}$ reproduces it: $\bar{D} \lhd\lhd\lhd \bar{A}$, $\bar{C}_1 \cap \bar{D} \lhd\lhd\lhd \bar{D}$, $\bar{B}_2 \cap \bar{D} \lhd\lhd\lhd \bar{D}$, still disjoint, with $\bar{D} \cap \bar{B}_2 \neq \varnothing$. And the descent shrinks the ambient, while the conclusion's congruence family is indexed by the ambient. See §11.

8 The main statements

Everything converges here. Theorem 8.1 is the single induction that carries the whole proof; the three theorems the algorithm consumes are corollaries of it.

8.1 The three claims the algorithm needs

The algorithm of §9 is justified by exactly three facts. Informally:

(IC1) every domain of size ≥ 2 has a strong subset, or an equivalence σ with $D_x/\sigma \cong \mathbb{Z}_p$;
(IC2) reducing a variable's domain to a strong subset of it does not destroy solvability, for a consistent enough instance;
(IC3) for a linked, consistent enough, strong-subset-free instance, the set of "linear cosets" containing a solution is empty, full, or a codimension-one affine subspace.

(IC1) is Lemma 6.2 together with Lemma 4.17. (IC2) is Theorem 8.6. (IC3) is Theorem 8.8.

8.2 The main induction

Theorem 8.1 (Main inductive claim)**.** Let $\mathcal{I}$ be an irreducible cycle-consistent instance and let $D^{(1)}$ be a 1-consistent reduction for $\mathcal{I}$ with $D^{(1)} \lhd\!\lhd\!\lhd D$.

(1) If $\mathcal{I}$ is crucial in $D^{(1)}$ then (1a) holds, and (1b) or (1c) holds:
  (1a) every constraint of $\mathcal{I}$ has the parallelogram property;
  (1b) $\mathcal{I}$ is a connected linear-type instance whose solution set is subdirect;
  (1c) there is an expanded covering $\mathcal{J}$ of $\mathcal{I}$, crucial in $D^{(1)}$, with a linked connected subinstance Υ whose solution set is not subdirect.

(2) If $D^{(2)} \leq_T D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$ with $T \in \{\textsc{ba}, \textsc{c}\}$, and $\mathcal{I}^{(1)}$ has a solution, then $\mathcal{I}^{(2)}$ has a solution.

Caveat 8.2 (The induction, stated precisely)**.** "Induction on the size of $D^{(1)}$" hides the entire well-foundedness argument, and it must be made precise, because *the two halves of the conclusion are proved by mutual induction and each invokes the other.*

The intended measure is

$$\mu(D^{(1)}) = \sum_{x \in \mathrm{Var}(\mathcal{I})} |D_x^{(1)}| \in \mathbb{N},$$

and the statement being proved by strong induction on μ is the conjunction of (1) and (2), *universally quantified over* $\mathcal{I}$:

$$P(k): \quad \forall \mathcal{I},\ \forall D^{(1)} \text{ with } \mu(D^{(1)}) \leq k,\ \text{(1) and (2) hold.}$$

The quantifier over $\mathcal{I}$ must be *inside*, because every use of the inductive hypothesis changes the instance: (2) applies it to a weakening $\mathcal{I}'$ of $\mathcal{I}^{(2)}$; Case 1 of (1) applies it to $\mathcal{I}$ with $D^{(2)}$; Case 2 applies it to weakenings and to expanded coverings. An induction that fixed $\mathcal{I}$ and inducted on $D^{(1)}$ alone does not go through.

Three further points.

(a) *The two halves are established in a fixed order at each level.* Part (2) at a given measure uses (1) only at strictly smaller measure, whereas part (1) at that measure uses *(2) at the same measure* — Case 1 below does exactly that. So the scheme is: strong induction on $k = \mu(D^{(1)})$; at each level establish (2) for all instances, then (1) for all instances. That is acyclic, and it is why part (2) is proved first.

(b) *Which appeals decrease.* (2) invokes the hypothesis at $D^{(2)} \subsetneq D^{(1)}$: strict because $D^{(2)} <_T D^{(1)}$ has $C \lneq B$ at some variable. Case 1 of (1) likewise. Case 2 invokes it at $D^{(B,\perp)} \subseteq D^{(B,\top)}$, which is $D^{(1)}$ with the z-domain cut to $B \lneq D^{(1)}_z$. In every case the decrease is at a single variable and by at least one element. The uniform statement is: *if $D' \leq_T D$ with T any type and $D' \neq D$, then $\mu(D') < \mu(D)$.*

(c) *The measure does not survive expanded coverings, and this is open.* An expanded covering has *more* variables than $\mathcal{I}$, and the reduction of a covering is the extension of $D^{(1)}$ along the parent map (Proposition 5.15(h)), which repeats the parent domains; its μ is *larger*. Observing that the domain at one parent variable is properly cut does not compare the two sums, since the reductions live on different variable sets and arbitrarily many children can outweigh the decrease. So μ as displayed is not a well-founded measure for the appeals made at coverings, and putting the quantifier over instances inside does not repair it. A correct measure must be insensitive to duplicated covering variables — by measuring on the variables of a fixed base instance equipped with a map from the covering, or by ordering parent-domain profiles rather than summing over children. See §11.

Proof of (2). Suppose $\mathcal{I}^{(2)}$ has no solution. Weaken $\mathcal{I}^{(2)}$ to an instance $\mathcal{I}'$ crucial in $D^{(2)}$ (Remark 5.13). By the inductive hypothesis applied to $\mathcal{I}'$ and $D^{(2)}$, $\mathcal{I}'$ satisfies (1a) and (1b) or (1c).

Case (1c). There is an expanded covering $\mathcal{J}$ of $\mathcal{I}'$, crucial in $D^{(2)}$, with a linked connected subinstance Υ. Let x be a variable of a constraint $C \in \Upsilon$. By Lemma 8.14(p), $\mathrm{Con}_1(C,x)$ is a perfect linear congruence; choose $\zeta \subseteq \mathbf{D}_x \times \mathbf{D}_x \times \mathbf{Z}_p$ with $(y_1,y_2,0) \in \zeta \iff (y_1,y_2) \in \mathrm{Con}_1(C,x)$ and $\mathrm{pr}_{1,2}(\zeta) = \mathrm{Con}_1(C,x)^*$.

Replace the variable x of C in $\mathcal{J}$ by a fresh x' and add the constraint $\zeta(x,x',z)$ with z fresh of domain $\mathbf{Z}_p$; call the result Θ, and extend $D^{(1)}, D^{(2)}$ by $D^{(1)}_{x'} = D^{(2)}_{x'} = D_{x'}$. Let E_i be the set of $b \in \mathbf{Z}_p$ such that Θ has a solution in $D^{(i)}$ with $z = b$. By Lemma 6.10, $E_2 \dot{\leq}_T E_1$.

Since $\mathcal{J}$ is crucial in $D^{(2)}$, $\Theta^{(2)}$ has a solution but none with $z = 0$; since $\mathcal{I}^{(1)}$ has a solution, so does $\mathcal{J}^{(1)}$, hence $\Theta^{(1)}$ has one with $z = 0$. Thus $0 \in E_1$, $0 \notin E_2$, and E_1 has at least two elements. As $\mathbf{Z}_p$ has no proper subalgebra of size > 1, $E_1 = \mathbf{Z}_p$; then $E_2 <_T \mathbf{Z}_p$ with $T \in \{\textsc{ba}, \textsc{c}\}$, contradicting Lemma 6.25.

Case (1b). Corollary 8.21 gives a contradiction directly. □

Caveat 8.3 (Three gaps in the case (1c) argument)**.** (a) "E_1 contains at least two different elements, $0 \in E_1$ and $0 \notin E_2$, hence $E_1 = \mathbf{Z}_p$" uses that E_1 is a subuniverse of $\mathbf{Z}_p$ — true because $E_1 = \Theta^{(1)}(z)$ is a relation defined by an instance (Definition 5.3) — *and* that the subuniverses of $\mathbf{Z}_p$ are the singletons and the whole set. The latter needs $w^{\mathbf{Z}_p}$ to generate: from $a \neq b$ one reaches every element. Prove it; it is one line but it is the crux.

(b) $E_2 \dot{\leq}_T E_1$ is applied with the *dotted* conclusion of Lemma 6.10, and the argument then treats E_2 as a proper subuniverse. The case $E_2 = \varnothing$ must be handled: it is excluded because $\Theta^{(2)}$ has a solution. Say so.

(c) The choice of ζ is by Definition 3.34, which supplies ζ with $\mathrm{pr}_{1,2}\,\zeta = \sigma^*$ and $(a_1,a_2,b) \in \zeta \Rightarrow ((a_1,a_2) \in \sigma \iff b = 0)$. The proof uses the *biconditional* $(y_1,y_2,0) \in \zeta \iff (y_1,y_2) \in \mathrm{Con}_1(C,x)$, which is stronger: it needs, for each $(y_1,y_2) \in \sigma$, that $(y_1,y_2,0) \in \zeta$. This follows from $\mathrm{pr}_{1,2}\,\zeta = \sigma^* \supseteq \sigma$ plus the implication, but the step is suppressed.

Proof of (1)*, sketch of the structure.* First, $|D^{(1)}_x| > 1$ for some x: otherwise $\mathcal{I}^{(1)}$ has all domains singletons and 1-consistency makes it solvable, contradicting cruciality.

Case 1: some $D^{(1)}_x$ has a nontrivial BA *or central subuniverse.* Lemma 8.17 yields a 1-consistent reduction $D^{(2)} \leq_T D^{(1)}$, $T \in \{\textsc{ba}, \textsc{c}\}$. One checks $\mathcal{I}$ is crucial in $D^{(2)}$: for a weakening $\mathcal{J}$ of a constraint, $\mathcal{J}^{(1)}$ has a solution, so by the inductive hypothesis (part (2) at $D^{(2)}$) $\mathcal{J}^{(2)}$ has one. Applying the inductive hypothesis (part (1)) at $D^{(2)}$ gives the conclusion.

Case 2: otherwise. Lemma 6.2 gives $E <^{D_z}_{T(\sigma)} D^{(1)}_z$ for some z and $T \in \{\text{PC}, \text{L}\}$. Property (1a) is obtained by choosing, for each x, a minimal $D^{(2)}_x \leq_{\mathcal{M}T} D^{(1)}_x$ containing the value at x of a solution produced by weakening a fixed constraint C; Lemma 6.18 gives $D^{(2)} \lhd\lhd\lhd^D D^{(1)}$ and Lemma 8.11 splits into two subcases, in one of which the inductive hypothesis applies and in the other of which Lemma 8.18 applies directly.

For (1b)/(1c) one forms, for each B in the family $\mathcal{B} = \{B : B <^{D_z}_{T(\sigma)} D^{(1)}_z\}$, the reduction $D^{(B,\top)}$ that equals $D^{(1)}$ off z and B at z, and the maximal 1-consistent reduction $D^{(B,\perp)} \subseteq D^{(B,\top)}$ (possibly empty). Lemma 8.16 supplies tree-coverings $\Upsilon_{B,x}$ computing $D^{(B,\perp)}_x$, and their conjunction Υ_x works uniformly in B. Writing $\mathcal{B}_0$ for the B with $D^{(B,\perp)}$ nonempty, the argument splits:

- $\mathcal{B}_0 = \varnothing$: the type must be L, and Lemma 8.19 applied to a suitably weakened tree-covering produces, for any two constraints sharing a variable, a reflexive bridge between their Con_1's. Hence $\mathcal{I}$ is connected, and (1b) or (1c) holds according as its solution set is subdirect.
- $\mathcal{B}_0 \neq \varnothing$: one constructs a family Ω of weakenings of $\mathcal{I}$ that is *critically* unable to solve all of $\mathcal{B}_0$ (conditions 1–4 of Construction 8.5) and argues by whether some member fails (1b).

□

Caveat 8.4 (Construction 8.5 is the hardest object)**.** The set Ω is specified by four conditions, one of which (condition 3) is a minimality condition "if we replace any instance in Ω by all weaker instances then 2 is not satisfied". The paper obtains Ω by starting from $\{\mathcal{I}\}$ and repeatedly weakening, justified by "we cannot weaken forever". Condition 4 is then *derived* from condition 3.

This is the single most demanding step in §8, for three reasons. (i) The iteration weakens a *set* of instances, so the termination measure is a measure on finite sets of instances, not on instances — and Caveat 5.11 shows that even the single-instance measure is not settled. (ii) The derivation of condition 4 from condition 3 is a two-line argument that instantiates condition 3 at $\Omega \cup \{\mathcal{J}_C : C \in \mathcal{J}\} \setminus \{\mathcal{J}\}$ and reads off an existential; the existential witness B depends on $\mathcal{J}$, so condition 4 is a $\Pi\Sigma$ statement and must be stated as such. (iii) The subsequent Subcase 1 builds a *second* descending sequence $\mathcal{B}_1 \supseteq \mathcal{B}_2 \supseteq \dots$ inside the first, again terminating by finiteness. Nested well-founded constructions of this shape are where informal proofs most often hide a circularity, and this one is isolated here rather than left inside the proof.

Remark 8.5 (Construction of Ω)**.** Ω is a finite set of weakenings of $\mathcal{I}$ with:

1. every member is a weakening of $\mathcal{I}$;
2. $\bigcap_{\mathcal{J}\in\Omega} \text{Sol}(\mathcal{J}) = \varnothing$, where $\text{Sol}(\mathcal{J}) = \{B \in \mathcal{B}_0 : \mathcal{J}^{(B,\perp)}$ has a solution$\}$;
3. replacing any member by all weaker instances destroys 2;
4. for every $\mathcal{J} \in \Omega$ there is $B \in \mathcal{B}_0$ with (a) $\mathcal{J}$ crucial in $D^{(B,\perp)}$ and (b) $B \in \text{Sol}(\mathcal{J}')$ for every $\mathcal{J}' \in \Omega \setminus \{\mathcal{J}\}$.

8.3 The three consumed theorems

Theorem 8.6 (Reductions are safe)**.** *Let Θ be a cycle-consistent irreducible instance and $B <^{D_x}_T D_x$ with $T \in \{\text{BA}, \text{C}, \text{PC}\}$. Then Θ has a solution if and only if Θ has a solution with $x \in B$.*

Proof. For $T = \text{PC}$ this is Theorem 8.7. For $T \in \{\text{BA}, \text{C}\}$, Lemma 8.17 gives a 1-consistent reduction $D^{(1)} \leq_T D$ with $D^{(1)}_x \subseteq B$; by Theorem 8.1(2), $\Theta^{(1)}$ has a solution, hence Θ has one with $x \in B$. □

Theorem 8.7 (PC reductions do not kill all solutions)**.** *If $\mathcal{I}$ is cycle-consistent and irreducible, $B <^{D_y}_{\text{PC}(\sigma)} D_y$ for some $y \in \text{Var}(\mathcal{I})$, and $\mathcal{I}$ has a solution, then $\mathcal{I}$ has a solution with $y \in B$.*

Theorem 8.8 (Codimension one)**.** Suppose

(i) $\mathcal{I}$ is a linked cycle-consistent irreducible instance with $\text{Var}(\mathcal{I}) = \{x_1, \dots, x_m\}$;
(ii) $\mathbf{D}_{x_i}$ is S-free for every i;
(iii) weakening all constraints of $\mathcal{I}$ gives an instance with subdirect solution set;
(iv) σ_{x_i} is the intersection of all linear congruences σ on $\mathbf{D}_{x_i}$ with $\sigma^* = D^2_{x_i}$;
(v) $L_{x_i} = \mathbf{D}_{x_i}/\sigma_{x_i}$;
(vi) $\phi\colon \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \to L_{x_1} \times \cdots \times L_{x_m}$ is a homomorphism, q_i prime;
(vii) weakening any one constraint of $\mathcal{I}$ gives an instance with a solution in $\phi(\bar{a})$ for every $\bar{a} \in \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k}$.

Then $\Delta = \{\bar{a} : \mathcal{I} \text{ has a solution in } \phi(\bar{a})\}$ is empty, full, or an affine subspace of codimension 1 — the solution set of a single linear equation.

Proof. If Δ is full we are done. Otherwise pick $\bar{b} \notin \Delta$ and regard $\phi(\bar{b})$ as a reduction $D^{(1)}$ for $\mathcal{I}$; condition (vii) says $\mathcal{I}$ is crucial in $D^{(1)}$.

Step 1: some $\text{Con}_1(C, x)$ *is a perfect linear congruence.* By Lemma 8.11, either $C^{(1)}_0 = \varnothing$ for some constraint C_0, or $D^{(1)}$ is 1-consistent.

- If $C^{(1)}_0 = \varnothing$: cruciality forces $\mathcal{I} = \{C_0\}$, say $C_0 = R(y_1, \dots, y_t)$. By Lemma 8.18, R has the parallelogram property and $\text{Con}_1(R, 1)$ is linear with $\text{Con}_1(R, 1)^* = D^2_{y_1}$. By Lemma 4.17, $\mathbf{D}_{y_1}/\delta \cong \mathbf{Z}_p$; with ψ the quotient map, $\zeta = \{(a_1, a_2, b) : \psi(a_1) - \psi(a_2) = b\}$ witnesses perfection.
- If $D^{(1)}$ is 1-consistent: by Theorem 8.1(1) every constraint has the parallelogram property and (1b) or (1c) holds. In either case Lemma 8.14(p) applies — under (1c) to the linked connected subinstance, whose containing a constraint of $\mathcal{I}$ is forced by condition (iii); under (1b) to $\mathcal{I}$ itself.

Step 2: add a $\mathbf{Z}_p$ *coordinate.* Let ζ witness perfection of $\text{Con}_1(C, x)$. Add a variable z with domain $\mathbf{Z}_p$, replace x in C by a fresh x', and add $\zeta(x, x', z)$, giving $\mathcal{I}'$. Put

$$L = \{(\bar{a}, b) : \mathcal{I}' \text{ has a solution with } z = b \text{ in } \phi(\bar{a})\} \leq \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \times \mathbf{Z}_p.$$

By condition (vii), $\text{pr}_{1..k}(L)$ is full; and $(\bar{b}, 0) \notin L$. So L has dimension k and is the solution set of one linear equation. If that equation is $z = c$ with $c \neq 0$ then $\Delta = \varnothing$; otherwise setting $z = 0$ gives an equation defining Δ, of dimension $k - 1$. □

Caveat 8.9 (The linear algebra of Step 2)**.** Step 2 as written is not yet correct, and the repair is worth setting out.

L is a nonempty subuniverse of an idempotent affine algebra, hence in general an *affine coset* and not a subgroup. And a product of prime fields with different primes is not one vector space, so "dimension k" and "one linear equation" are not meaningful until the primary components are separated: such a group splits as a product over primes of $\mathbb{F}_q$-vector spaces, and only coordinates over $\mathbb{F}_p$ can carry a nonzero coefficient in an equation with target $\mathbf{Z}_p$. The footnote in [14] about "J may contain only variables on the same field" is the same issue surfacing in the algorithm.

A correct short argument runs as follows. Choose a point of L and translate, obtaining a subgroup. The kernel of the projection $L \to \prod_i \mathbf{Z}_{q_i}$ lies in $0 \times \mathbf{Z}_p$, hence is either trivial or all of $\mathbf{Z}_p$; a full kernel would put $(\bar{b}, 0)$ in the fibre over every $\bar{b}$, contradicting $(\bar{b}, 0) \notin L$. So the projection is bijective and L is the graph of an affine homomorphism $f\colon \prod_i \mathbf{Z}_{q_i} \to \mathbf{Z}_p$, whose linear part automatically ignores every q-primary factor with $q \neq p$. Then $\Delta = f^{-1}(0)$ is empty, full, or an affine hyperplane, which is the conclusion. Writing this into the proof is work not yet done; see §11.

8.4 The supporting lemmas

The lemmas invoked above, in dependency order.

Lemma 8.10 ([13, Lem. 6.1])**.** *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*

Lemma 8.11. *Suppose $D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$, every $D_x^{(1)}$ is* S*-free, $T \in \{\mathrm{PC}, \mathrm{L}, \mathrm{D}\}$, $D^{(1)} \lhd\lhd\lhd D$, and $D_x^{(2)} \leq_{\mathcal{M}T} D_x^{(1)}$ is a minimal $\mathcal{M}T$ subuniverse for every x. Then either $C^{(2)} = \varnothing$ for some constraint C, or $\mathcal{I}^{(2)}$ is 1-consistent.*

Lemma 8.12. *A constraint crucial in a reduction of a cycle-consistent irreducible instance is rectangular, and its* Con_1*'s are irreducible congruences.*

Lemma 8.13. *A relation whose restriction along strong subsets becomes empty yields a bridge between the witnessing congruences.*

Lemma 8.14 (Connected instances)**.** *Let $\mathcal{I}$ be a cycle-consistent connected instance. Then*

- (a) *any two constraints with a common variable are adjacent;*
- (b) *for constraints C_1, C_2, variables $x_i \in \mathrm{Var}(C_i)$, and any path from x_1 to x_2, there is a bridge δ from $\mathrm{Con}_1(C_1, x_1)$ to $\mathrm{Con}_1(C_2, x_2)$ with $\widetilde{\delta}$ containing all pairs connected by that path;*
- (p) *if $\mathcal{I}$ is linked then $\mathrm{Con}_1(C, x)$ is a perfect linear congruence for every $C \in \mathcal{I}$, $x \in \mathrm{Var}(C)$.*

Caveat 8.15. Clause (p) is the bridge between the bridge machinery and the linear algebra; it is what makes Theorem 8.8 possible. Its proof is (b) followed by Lemma 6.23: linkedness of the instance makes $\widetilde{\delta}$ linked as a relation, which upgrades the bridge. The two senses of "linked" (Caveat 3.12) meet exactly here, and the step where one becomes the other should be isolated.

Lemma 8.16 ([15, Lem. 5.6])**.** *For a 1-consistent instance and a reduction, there is a tree-covering whose projection onto a given variable computes the maximal 1-consistent subreduction at that variable.*

Lemma 8.17. *Let $D^{(1)}$ be a 1-consistent reduction of a cycle-consistent instance $\mathcal{I}$ with $D^{(1)} \lhd\lhd\lhd D$, and $B <_T^{D_x} D_x^{(1)}$ with $T \in \{\mathrm{BA}, \mathrm{C}, \mathrm{PC}\}$. Then there is a nonempty 1-consistent reduction $D^{(2)} \lhd\lhd\lhd D^{(1)}$ with $D_x^{(2)} \subseteq B$. Moreover, if $T \in \{\mathrm{BA}, \mathrm{C}\}$ then $D^{(2)} \leq_T D^{(1)}$; and if $T = \mathrm{PC}$ and every $D_y^{(1)}$ is* S*-free then $D^{(2)} \leq_{\mathcal{M}\mathrm{PC}} D^{(1)}$.*

Lemma 8.18. *A constraint that becomes empty under a minimal multi-type reduction, in an instance crucial there, has the parallelogram property; and in the linear case its first induced congruence σ satisfies $\sigma^* = D^2$.*

Lemma 8.19 (Bridges from a subdirect PC/linear instance)**.** Suppose

- (i) $\mathcal{I}$ has subdirect solution set;

(ii) $D^{(1)}$ is a reduction with $D_x^{(1)} \lhd\lhd\lhd D_x$ for every x;
(iii) $C \in \mathcal{I}$ is a constraint of type $T \in \{\textsc{pc}, \textsc{l}\}$;
(iv) $B <_{\mathcal{T}(\xi)}^{D_z} D_z^{(1)}$ for some variable z, with $\mathcal{T} \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{pc}, \textsc{l}\}$;
(v) if $T = \textsc{pc}$ then $\mathcal{T} \in \{\textsc{pc}, \textsc{l}\}$;
(vi) $\mathcal{I}^{(1)}$ has a solution;
(vii) $\mathcal{I}^{(1)}$ has no solution with $z \in B$;
(viii) weakening C in $\mathcal{I}$ gives an instance with a solution in $D^{(1)}$ having $z \in B$.

Then $\mathcal{T} = T$, and for every variable x of C there is a bridge δ from ξ to $\mathrm{Con}_1(C, x)$ with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$.

Caveat 8.20 (Stated only)**.** Lemma 8.19 carries eight numbered hypotheses and is *stated without proof here.* It is the only source of bridges in §8, and every use of connectedness routes through it; so are Lemmas 8.11, 8.12, 8.13, 8.17, 8.18, Corollary 8.21 and clauses (a) and (b) of Lemma 8.14. These are gaps in the logical chain of this document, not imported results, and §11 records them as such.

Its conclusion has two independent parts — "$\mathcal{T} = T$" and "there is a bridge with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$" — and different call sites use different parts. The dependency order is Theorem 6.14, which it consumes, then this lemma, then Theorem 8.1, which consumes it.

Corollary 8.21. *In the situation of Theorem 8.1(2) with $\mathcal{I}'$ satisfying (1b), the type $\mathcal{T}$ of the reduction and the type of the constraints must agree, which contradicts $\mathcal{T} \in \{\textsc{ba}, \textsc{c}\}$.*

9 The algorithm

The algorithm is [13]'s. It is stated here because target (T1) of §1 is a proposition *about* it and cannot be written down otherwise.

9.1 Solve

```
function Solve(I):
    repeat
        I := ForceConsistency(I)
        if I = false: return "No"
        if some D_x has a strong subset B:
            I := ReduceDomain(I, x, B)
    until nothing changed
    return SolveLinear(I)
```

ForceConsistency enforces cycle-consistency and irreducibility, returning `false` if some domain empties. The reduction step is licensed by Theorem 8.6: restricting to a strong subset cannot destroy solvability. When no strong subset remains, Lemma 6.2 together with Lemma 4.17 gives, for each domain of size at least 2, a congruence σ_x with $D_x/\sigma_x \cong \mathbb{Z}_{q_1} \times \cdots \times \mathbb{Z}_{q_{n_x}}$; choose σ_x minimal. That case is SolveLinear.

9.2 SolveLinear

Write $\phi\colon \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k} \to D_{x_1}/\sigma_{x_1} \times \cdots \times D_{x_m}/\sigma_{x_m}$ for a linear map and

$$\phi^{-1}(\mathcal{I}) = \{\bar{a} : \mathcal{I} \text{ has a solution in } \phi(\bar{a})\}.$$

Computing $\phi^{-1}(\mathcal{I})$ would solve $\mathcal{I}$; the algorithm cannot do that, but it can do four things:

- **(p0)** decide $\bar{a} \in \phi^{-1}(\mathcal{I})$, by reducing each domain to the corresponding block and recursing on a smaller domain;
- **(p1)** decide $\phi^{-1}(\mathcal{I}) = \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k}$, by testing $\bar{0}$ first and then the k unit vectors with (p0). The set $\phi^{-1}(\mathcal{I})$ is in general an affine *coset*, not a subgroup; testing $\bar{0}$ first is what makes it one, after which membership of the coordinate generators implies fullness;
- **(p2)** compute $\phi^{-1}(\mathcal{I})$ when $\mathcal{I}$ is not linked, by splitting into linked pieces, recursing, and taking the union;
- **(p3)** compute $\phi^{-1}(\mathcal{I})$ when it is empty or of codimension one, by finding $\bar{a} \notin \phi^{-1}(\mathcal{I})$, then for each i a b_i with $\bar{a}[i \mapsto b_i] \in \phi^{-1}(\mathcal{I})$ if one exists, and reading off the equation $\sum_{i \in J}(y_i - a_i)/(b_i - a_i) = 1$ over the i for which b_i exists. The displayed quotient is meaningful only within one prime: the codimension-one equation has a single target prime p, and only coordinates over $\mathbb{F}_p$ can carry a nonzero coefficient, so the computation is organised prime by prime (Caveat 8.9).

```
function SolveLinear(I):
    m := dim(D_{x_1}/s_1 x ... x D_{x_n}/s_n)
    phi := a bijective linear map Z_{p_1}x...xZ_{p_m} -> that product
    while phi^-1(I) != Z_{p_1}x...xZ_{p_m}:              // (p1)
        I' := I
        for C in I':                                 // drop what can be dropped
            if phi^-1(I' \ {C}) != Z_{p_1}x...xZ_{p_m}:   // (p1)
                I' := I' \ {C}
        F := phi^-1(I')                                // (p2) or (p3)
```

```
    if F = {}: return "No"
    m := dim F ;  psi := a bijective linear map onto F ;  phi := phi o psi
  return "Yes"
```

The inner loop makes $\mathcal{I}'$ minimal with $\phi^{-1}(\mathcal{I}')$ not full. At that point Theorem 8.8 applies — its hypothesis (vii) is exactly minimality — so $\phi^{-1}(\mathcal{I}')$ is empty or of codimension one, and (p3) computes it; or $\mathcal{I}'$ is not linked and (p2) does. Since $\phi^{-1}(\mathcal{I}) \subseteq \phi^{-1}(\mathcal{I}')$, the image of ϕ shrinks while still containing all solutions. The dimension strictly decreases, so the loop terminates.

9.3 What (T1) has to say

Caveat 9.1 (Two obstacles between the pseudocode and a function)**.** (a) *Termination is not structural.* Solve recurses through (p0) on strictly smaller domains, and SolveLinear loops on a strictly decreasing dimension. Neither is a structural recursion; both need an explicit well-founded measure, and the two interleave — SolveLinear calls Solve on a smaller domain, which calls SolveLinear again. The measure is lexicographic: $(\sum_x |D_x|,\ \dim\phi)$.

(b) *Several steps quantify over objects that are only finitely many by accident.* "If some D_x has a strong subset B" quantifies over subsets of D_x, over congruences on D_x, and — through absorption — over *term operations*, of which there are infinitely many. For the types the algorithm actually searches for, the search is finite for a reason that needs no general arity bound: BA has a binary witness by definition, and a central subuniverse has a ternary one by Lemma 4.4.

Theorem 9.2 (Target (T1))**.** *Let Solve be the function defined by the two displays above, recursing on the lexicographic measure of Caveat 9.1(a). Then for every instance $\mathcal{I}$ of* CSP(Γ)*,*

$$\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

Caveat 9.3 (The statement must name the function)**.** Theorem 9.2 is about *this* Solve, and that is the whole of its content. The weaker reading — "there exists a Boolean-valued function deciding satisfiability" — follows for a fixed finite template by exhaustive search and says nothing. Turning the two pseudocode displays into a function with a proved well-founded recursion, so that the theorem above has a subject, is work this document has not yet done; see §11.

Given that, Theorem 9.2 mentions no machine, is provable from §8, and is the honest content of "the algorithm works": that one particular, highly non-obvious algorithm — the one whose existence is the dichotomy — is correct.

Theorem 9.4 (Target (T2); stated, not proved)**.** *For each finite* Γ *there are a program computing Solve and a polynomial p_Γ bounding its running time on instances of* CSP(Γ) *in the size of the instance.*

Caveat 9.5 ((T2) is not a wrapper)**.** Theorem 9.4 is the only statement of the tractable half that needs a cost model, and it is not a corollary of Theorem 9.2: it requires Solve to be exhibited as a program with a proved step bound. Note also that the statement is non-uniform. Because Γ is fixed, the searches over subsets of A, over congruences on A, and over term operations of bounded arity are constant-time — but "constant" means compiled away into a program whose *size* depends on $|A|$. So Solve is a family $\textsc{Solve}_\Gamma$ with a polynomial for each Γ, and a bound uniform in Γ would be a different statement, and a false one.

10 The hard half

If no weak near-unanimity operation preserves Γ, then $\mathrm{CSP}(\Gamma)$ is NP-complete. This half is older, shorter, and better understood than the tractable half, and it factors cleanly into algebra and complexity.

10.1 The chain

(1) *Reduction to a core, and to an idempotent algebra.* One may assume Γ is a rigid core containing all singleton unary relations, so that $\mathrm{Pol}(\Gamma)$ is idempotent, at the cost of a logspace reduction [2, "Cores and Idempotent Reducts"]. This is Caveat 1.2.

(2) *Maróti–McKenzie.* A finite idempotent algebra has a Taylor term if and only if it has a weak near-unanimity term [10].

(3) *No Taylor term $\Rightarrow$ pp-interpretation.* If a finite idempotent algebra has no Taylor term, then the variety it generates contains a two-element algebra whose only operations are projections; and from there Γ primitively positively interprets every finite constraint language, in particular a template for NAE-3-SAT. The word "and" hides three steps — an HSP representation in a finite power, its translation into a pp-interpretation, and the finite Inv–Pol theorem (Theorem 10.6) — which are imported rather than proved here.

(4) *Bulatov–Jeavons–Krokhin.* A primitive positive interpretation of Δ in Γ yields a polynomial-time many-one reduction $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$ [1].

(5) *Cook–Levin.* NAE-3-SAT is NP-hard.

Steps (1)–(3) are algebra; step (4) is a syntactic construction plus a complexity-theoretic wrapper; step (5) is pure complexity theory.

10.2 The layered targets

Theorem 10.1 (Target (H0))**.** *Let* $\mathbf{A}$ *be a finite idempotent algebra with no weak near-unanimity term operation. Then* $\mathrm{Inv}(\mathbf{A})$ *primitively positively interprets the constraint language* $\{\textsc{Nae}\}$ *on* $\{0,1\}$*, where* $\textsc{Nae} = \{0,1\}^3 \setminus \{(0,0,0),(1,1,1)\}$.

Theorem 10.2 (Target (H1); stated, not proved)**.** *If* Γ *primitively positively interprets* Δ *then* $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$*. Consequently, under* (H0)*,* $\mathrm{CSP}(\Gamma)$ *is* NP*-hard.*

Caveat 10.3 (Hardness is not completeness)**.** Theorem 10.2 concludes NP-*hardness.* The word *complete* in Theorem 1.1 additionally needs that every fixed finite-template CSP lies in NP: a solution is a certificate of size linear in the number of variables, and checking it means evaluating each constraint, which for fixed Γ is a table lookup. Small, but it needs the cost model, so it belongs with (H1).

Theorem 10.1 mentions no machine. Theorem 10.2 is where the complexity layer enters, and it is small — the reduction is a syntactic substitution of formulas — but it cannot be stated at all without a precise notion of polynomial-time many-one reduction.

10.3 Definitions

Definition 10.4 (pp-formula, pp-definability)**.** A *primitive positive formula* over Γ is $\exists y_1 \ldots \exists y_s \bigwedge_j R_j(\bar{z}_j)$ with $R_j \in \Gamma \cup \{=\}$. A relation is *pp-definable* from Γ if some pp-formula defines it.

Definition 10.5 (pp-interpretation)**.** Γ on A *pp-interprets* Δ on B if there are $d \geq 1$, a pp-definable $F \subseteq A^d$, and a surjection $\pi\colon F \to B$ such that the π-preimage of every relation of Δ, and of the equality relation on B, is pp-definable from Γ.

Theorem 10.6 (Geiger; Bodnarchuk–Kalužnin–Kotov–Romov)**.** *For a finite A, a relation is pp-definable from Γ if and only if it is preserved by every polymorphism of Γ.*

Caveat 10.7 (The Galois connection is the expensive import)**.** Theorem 10.6 is what converts the algebraic hypothesis ("no WNU polymorphism") into a syntactic conclusion ("pp-defines a hard relation"), and it is the only genuinely substantial import of §10. The easy direction is a one-line calculation. The hard direction — every relation preserved by all polymorphisms is pp-definable — goes through the free algebra on $|A|^k$ generators and an indicator-instance construction; it is roughly four printed pages and is the single largest item in the (H0) budget.

Remark 10.8 (Why the hard half is not the hard half)**.** Despite its name, (H0) is far smaller than (T0): perhaps twenty printed pages against a hundred. The asymmetry is intrinsic. Showing a problem is hard requires exhibiting one construction; showing it is easy requires an algorithm that works on every instance, and the correctness proof must survive every configuration the instance can present.

11 Status and imports

This section says, for every statement in this document, whether it is proved here, imported, or not yet proved. It replaces the running commentary that a partial exposition invites, and it is the only place where completeness claims are made.

11.1 The five verdicts

proved A complete proof is given here, or the statement is an immediate specialisation of one that is.
imported Proved elsewhere and used as a black box. Every such statement carries an exact citation in §11.4, with the hypotheses in the form the source states them.
outlined An argument is given at the level of its case structure, or with steps suppressed. *This is not a proof* and the statement should not be relied on.
stated The statement appears and nothing is offered towards it. These are gaps in the logical chain, not imports.
open Confirmed to need an argument that is not available. There is one.

Nothing below is marked PROVED on the strength of a machine search. Two claims here rest on finite verification — the counterexamples of Remark 7.37 and of Caveat 6.11 — and both are written so that a reader can check them in a few lines by hand.

11.2 The table

Statement	Verdict	Note
§2–§4: vocabulary		
`prop:p-divides`, `cor:Zp-in-Vn`	PROVED	
`lem:sg-terms`	IMPORTED	standard; [5]
`lem:rect-basics`, `prop:cover`, `lem:irred-quotient`	PROVED	
`lem:bridge-comp`	IMPORTED	[13, Lem. 6.3]
`lem:symmetrise`	PROVED	
`lem:central-ternary`	IMPORTED	[15, Cor. 6.11.1]
`lem:sfree-equiv`, `lem:linear-bridge`	PROVED	
`lem:linear-equiv`	OUTLINED	$(b) \Rightarrow (a)$ needs the reflexivisation of Caveat 7.41
`lem:no-cross-bridge`, `lem:bridge-to-pc`	STATED	consumed by `thm:stable-intersection` and `lem:factor-by-delta`
`lem:pc-bridges-trivial`	PROVED	third case not written out
`lem:linear-on-top`	OUTLINED	same reflexivisation
`lem:pc-on-top`	STATED	consumed by nothing; Caveat 4.19
§5: instances		
`prop:covering-basics`, `lem:conone-basics`	STATED	routine, but not written
`rem:get-crucial`	STATED	termination is Caveat 5.11
§6–§7: strong subuniverses		
`thm:brady-absorption`	IMPORTED	[2, Thm. 3.11.1]
`lem:linked-implies-abs`	PROVED	a corollary of the above
`lem:pp-propagation`	IMPORTED	[4, Lem. 2.9]
`lem:barto-kazda`	IMPORTED	[4, Prop. 2.14]
`lem:factor-type`	IMPORTED	C only; [15, Lem. 6.8]
`lem:power-to-base`	IMPORTED	[15, Lem. 6.24]
`thm:central-relation-source`	IMPORTED	[15, Thm. 6.15]
`lem:hobby-mckenzie`	IMPORTED	[8]
`lem:possible-intersections`	IMPORTED	[15, Lem. 6.25]
`lem:absorbing-equality`	IMPORTED	[13, Lem. 7.2]

Statement	**Verdict**	**Note**
`lem:ba-center-implies`	IMPORTED	[15, Cor. 6.1.2, 6.9.2]; both hypotheses restored, Caveat 6.11
`lem:special-wnu`	IMPORTED	[10, Lem. 4.7]
`lem:building-perfect`	IMPORTED	[13, Cor. 8.17.1]
`lem:bridge-abelian`, `lem:ba-center-intersect`, `lem:no-absorbing-diagonal`, `lem:absorb-linked-sub`, `lem:no-unary-quotient`, `lem:central-relation`, `lem:strong-nonempty`, `lem:multiset-shift`, `lem:tot-sym-full`, `lem:tot-sym-irred`, `lem:full-fibre`, `cor:intersection-good`, `lem:parallelogram-D`, `lem:nice-bridge-abelian`, `lem:nontrivial-bridge`, `lem:pc-inductive-step`, `lem:bridge-between-congruences`, `lem:two-stable`, `lem:cut-or-nothing`, `lem:main-existence`, `lem:ubiquity`, `lem:intersect-all`, `cor:intersect-all-weak`, `lem:propagation`, `cor:prop-quotient`, `thm:stable-intersection`, `cor:stable-intersection`, `lem:multitype-chain`, `lem:perfect-from-linked`, `lem:no-abs-in-linear`	PROVED	
`lem:bacon-left`	OUTLINED	$\lvert A\rvert = 1$, the linkedness of W_k, and the nonemptiness and properness at the pp-propagation step are all suppressed
`lem:self-intersection-pc`	OUTLINED	
`lem:intersection-good`	OUTLINED	twelve subcases of (d) unexpanded; Caveat 7.32. **Much of §7.4 onwards depends on it**
`lem:block-good-bridge`, `lem:center-pushed-in`, `lem:leftlinked-stay-full`, `lem:factor-by-delta`	OUTLINED	
`lem:multiply-all-linear`	OUTLINED	not yet a proof; Caveat 7.57
`cor:prop-from-factor`, `cor:prop-times-cong`, `cor:prop-relations`, `lem:preserve-linked`	STATED	
`lem:maximal-mult`	OPEN	Caveat 7.58
§8: the main statements		
`thm:main-inductive`(2)	PROVED	modulo the common-term obligation of Caveat 6.11
`thm:main-inductive`(1)	OUTLINED	structure only; and the measure is not well founded across coverings, Caveat 8.2(c)
`thm:reductions-safe`	OUTLINED	the BA and C cases follow from `thm:main-inductive`(2); the PC case is `thm:pc-safe`, which is stated only
`thm:pc-safe`	STATED	
`thm:codim-one`	OUTLINED	Step 2's linear algebra is wrong as written; Caveat 8.9
`lem:expanded-consistency`	IMPORTED	[13, Lem. 6.1]
`lem:tree-coverings`	IMPORTED	[15, Lem. 5.6]

Statement	Verdict	Note
`lem:minimal-consistent`, `lem:crucial-irreducible`, `lem:bridge-from-relation`, `lem:connected`, `lem:find-consistent`, `lem:parallelogram-crucial`, `lem:bridge-from-instance`, `cor:same-type`	STATED	Caveat 8.20
§9–§10: the algorithm and the hard half		
`thm:t1`	STATED	needs SOLVE defined as a function; Caveat 9.3
`thm:t2`, `thm:h1`	STATED	need a cost model
`thm:h0`	STATED	
`thm:galois`	IMPORTED	Geiger; Bodnarchuk et al.
`thm:dichotomy`	STATED	and needs the core reduction, Caveat 1.2

11.3 What this adds up to

The algebraic core of the strong-subuniverse theory (§6–§7) is the part in the best shape: of its statements, thirty are proved here, eleven are imported with exact citations, seven are outlined, four are stated only, and one is open. §8 is in much worse shape: its spine — `lem:bridge-from-instance`, and with it most of Lemma 8.14 — is stated without proof, and part (1) of the main induction is an outline whose measure is not yet well founded.

Six items block the rest, and they are the honest reading of the table.

1. `lem:maximal-mult` (Caveat 7.58). OPEN.
2. The reflexivisation of a bridge, Caveat 7.41, which is used by both `lem:linear-equiv` and `lem:linear-on-top`.
3. The expansion of `lem:intersection-good`, on which much of §7.4 onwards rests, and the rewrite of `lem:multiply-all-linear`.
4. `lem:no-cross-bridge` and `lem:bridge-to-pc`, which `thm:stable-intersection` consumes.
5. The induction measure across expanded coverings (Caveat 8.2(c)) and the termination of weakening (Caveat 5.11).
6. The linear algebra of `thm:codim-one` (Caveat 8.9), and the common binary absorption witness at the call site of `lem:ba-center-implies` (Caveat 6.11).

The dependency order among the three large statements is

$$\texttt{thm:stable-intersection} \longrightarrow \texttt{lem:bridge-from-instance} \longrightarrow \texttt{thm:main-inductive}.$$

Stable intersection is the sole source of bridges; bridge-from-instance is the only thing that turns them into instance-level structure; the main induction consumes that. Nothing in the reverse direction is used.

11.4 The import ledger

Every statement used and not proved here, with its source and the hypotheses in the form the source states them. Where our statement differs from the printed one — because a hypothesis has been restored, or an alternative added — the difference is noted, and the reasons are in the companion audit document.

Here	Source	Hypotheses, and any difference
`thm:brady-absorption`	[2, Thm. 3.11.1]	$R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ subdirect and linked, $\mathbf{A}, \mathbf{B}$ finite idempotent Taylor. Stated here in the source's three-alternative form; `lem:linked-implies-abs` is our weaker symmetric *corollary* under properness, and is not Brady's theorem.
`lem:pp-propagation`	[4, Lem. 2.9], [15, Lem. 6.1, Thm. 6.9]	R pp-defined by Φ in which S occurs; $S' <_T S$ with $T \in \{\textsc{ba}, \textsc{c}\}$ replaces *every* occurrence. Note the conclusion is *dotted*: replacing a relation by a proper strong subrelation can make the pp-defined result empty, so $R' \dot{\leq}_T R$ unless nonemptiness of R' is separately known. Every use below is in dotted form.
`lem:barto-kazda`	[4, Prop. 2.14], [15, Lem. 3.2]	$B \leq \mathbf{A}$, $n \geq 2$. Stated as printed.
`lem:factor-type`	[15, Lem. 6.8]	$B \leq_{\textsc{c}} \mathbf{A}$, σ a congruence on $\mathbf{A}$. Only the $\textsc{c}$ case is imported; $\textsc{ba}$ and $\textsc{s}$ are proved here (Caveat 7.7).
`lem:power-to-base`	[15, Lem. 6.24]	R a nontrivial strong subuniverse of $\mathbf{A}_1 \times \cdots \times \mathbf{A}_n$ of type $T \neq \textsc{pc}$. Already general in T; the case $T = \textsc{s}$ needs no separate argument (Caveat 7.9).
`thm:central-relation-source`	[15, Thm. 6.15]	$R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$, $\mathbf{A}, \mathbf{B}$ finite idempotent. Three alternatives, the third being a nontrivial projective subuniverse of $\mathbf{B}$; `lem:central-relation` eliminates it under Convention 2.1, using also [15, Lem. 3.4].
`lem:possible-intersections`	[15, Lem. 6.25]	$B <_{T_1} \mathbf{A}$, $C <_{T_2} \mathbf{A}$, $B \cap C = \varnothing$, $T_1, T_2 \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$. Stated as printed.
`lem:ba-center-implies`	[15, Cor. 6.1.2, 6.9.2]	$R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ with $\mathrm{pr}_1(R) = A_1$, and $C_i \leq_{\textsc{ba}(t)} \mathbf{A}_i$ for a *single* binary t. Both hypotheses are as in [15] and both are needed (Caveat 6.11).
`lem:hobby-mckenzie`	[8]	Abelian iff some congruence of the square has the diagonal as a block; Abelian iff affine, for a finite algebra with a WNU term.
`lem:absorbing-equality`	[13, Lem. 7.2]	$0_{\mathbf{A}} \subseteq \sigma \leq \mathbf{A}^2$, $\omega <_{\textsc{ba}} \sigma$ *proper*. The equality case needs separate treatment and gets it at the one call site.
`lem:bridge-comp`	[13, Lem. 6.3]	$\sigma_0, \sigma_1, \sigma_2$ *all* irreducible. The hypothesis is easy to drop by accident; `lem:symmetrise` avoids the citation altogether.
`lem:building-perfect`	[13, Cor. 8.17.1]	σ irreducible on $\mathbf{A} \in \mathcal{V}_n$, δ a bridge from σ to σ with $\widetilde{\delta} = A^2$. Used only by `lem:perfect-from-linked`.
`lem:special-wnu`	[10, Lem. 4.7]	w an idempotent WNU of arity n on a finite A. We use only "Clo(w) contains a special idempotent WNU of some arity N", not the printed bound.
`lem:expanded-consistency`	[13, Lem. 6.1]	an expanded covering of a cycle-consistent irreducible instance.
`lem:tree-coverings`	[15, Lem. 5.6]	a 1-consistent instance and a reduction.
`lem:sg-terms`	[5]	standard; the "fixed generator list" phrasing is the one used (Caveat 3.9).
`lem:central-ternary`	[15, Cor. 6.11.1]	a central subuniverse is ternary absorbing.
`thm:galois`	Geiger; Bodnarchuk–Kalužnin–Kotov–Romov	A finite. See Caveat 10.7.

Here	Source	Hypotheses, and any difference
Essential relations for $n \geq 3$	[15, Lem. 6.11]	$C_i \leq_{\mathrm{C}} \mathbf{A}_i$, $n \geq 3$: no $(C_1, \ldots, C_n)$-essential R. Used once, in `thm:stable-intersection`; its printed proof cites itself where it means the preceding lemma (Caveat 7.55).
Cores and idempotent reducts	[2]	used by Caveat 1.2 and §10.

That is nineteen imported statements. The two that are real work are `thm:brady-absorption` and `thm:central-relation-source`; the rest are short or routine.

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